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Inventor(s)

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Surgical generator for ultrasonic and electrosurgical devices

Abstract

A method for determining motional branch current in an ultrasonic transducer of an ultrasonic surgical device over multiple frequencies of a transducer drive signal. The method may comprise, at each of a plurality of frequencies of the transducer drive signal, oversampling a current and voltage of the transducer drive signal, receiving, by a processor, the current and voltage samples, and determining, by the processor, the motional branch current based on the current and voltage samples, a static capacitance of the ultrasonic transducer and the frequency of the transducer drive signal.

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5484436	12/1995	Eggers et al.	N/A	N/A
5486162	12/1995	Brumbach	N/A	N/A
5486189	12/1995	Mudry et al.	N/A	N/A
5490860	12/1995	Middle et al.	N/A	N/A
5496317	12/1995	Goble et al.	N/A	N/A
5499992	12/1995	Meade et al.	N/A	N/A
5500216	12/1995	Julian et al.	N/A	N/A
5501654	12/1995	Failla et al.	N/A	N/A
5504650	12/1995	Katsui et al.	N/A	N/A
5505693	12/1995	Mackool	N/A	N/A
5507297	12/1995	Slater et al.	N/A	N/A
5507738	12/1995	Ciervo	N/A	N/A
5509922	12/1995	Aranyi et al.	N/A	N/A
5511556	12/1995	DeSantis	N/A	N/A
5520704	12/1995	Castro et al.	N/A	N/A
5522832	12/1995	Kugo et al.	N/A	N/A
5522839	12/1995	Pilling	N/A	N/A
5527331	12/1995	Kresch et al.	N/A	N/A
5531744	12/1995	Nardella et al.	N/A	N/A
5536267	12/1995	Edwards et al.	N/A	N/A
5540681	12/1995	Strul et al.	N/A	N/A
5540684	12/1995	Hassler, Jr.	N/A	N/A
5540693	12/1995	Fisher	N/A	N/A
5542916	12/1995	Hirsch et al.	N/A	N/A
5548286	12/1995	Craven	N/A	N/A
5549637	12/1995	Crainich	N/A	N/A
5553675	12/1995	Pitzen et al.	N/A	N/A
5558671	12/1995	Yates	N/A	N/A
5562609	12/1995	Brumbach	N/A	N/A
5562610	12/1995	Brumbach	N/A	N/A
5562659	12/1995	Morris	N/A	N/A
5562703	12/1995	Desai	N/A	N/A
5563179	12/1995	Stone et al.	N/A	N/A
5569164	12/1995	Lurz	N/A	N/A
5571121	12/1995	Heifetz	N/A	N/A
5573424	12/1995	Poppe	N/A	N/A
5573533	12/1995	Strul	N/A	N/A
5573534	12/1995	Stone	N/A	N/A
5577654	12/1995	Bishop	N/A	N/A
5584830	12/1995	Ladd et al.	N/A	N/A
5591187	12/1996	Dekel	N/A	N/A
5593414	12/1996	Shipp et al.	N/A	N/A

5599350	12/1996	Schulze et al.	N/A	N/A
5600526	12/1996	Russell et al.	N/A	N/A
5601601	12/1996	Tal et al.	N/A	N/A
5603773	12/1996	Campbell	N/A	N/A
5607436	12/1996	Pratt et al.	N/A	N/A
5607450	12/1996	Zvenyatsky et al.	N/A	N/A
5609573	12/1996	Sandock	N/A	N/A
5611813	12/1996	Lichtman	N/A	N/A
5618304	12/1996	Hart et al.	N/A	N/A
5618307	12/1996	Donlon et al.	N/A	N/A
5618492	12/1996	Auten et al.	N/A	N/A
5620447	12/1996	Smith et al.	N/A	N/A
5624452	12/1996	Yates	N/A	N/A
5626587	12/1996	Bishop et al.	N/A	N/A
5626595	12/1996	Sklar et al.	N/A	N/A
5626608	12/1996	Cuny et al.	N/A	N/A
5628760	12/1996	Knoepfler	N/A	N/A
5630420	12/1996	Vaitekunas	N/A	N/A
5632432	12/1996	Schulze et al.	N/A	N/A
5632717	12/1996	Yoon	N/A	N/A
5638827	12/1996	Palmer et al.	N/A	N/A
5640741	12/1996	Yano	N/A	N/A
D381077	12/1996	Hunt	N/A	N/A
5647871	12/1996	Levine et al.	N/A	N/A
5649937	12/1996	Bitto et al.	N/A	N/A
5649955	12/1996	Hashimoto et al.	N/A	N/A
5651780	12/1996	Jackson et al.	N/A	N/A
5653713	12/1996	Michelson	N/A	N/A
5655100	12/1996	Ebrahim et al.	N/A	N/A
5658281	12/1996	Heard	N/A	N/A
5662662	12/1996	Bishop et al.	N/A	N/A
5662667	12/1996	Knodel	N/A	N/A
5665085	12/1996	Nardella	N/A	N/A
5665100	12/1996	Yoon	N/A	N/A
5669922	12/1996	Hood	N/A	N/A
5674219	12/1996	Monson et al.	N/A	N/A
5674220	12/1996	Fox et al.	N/A	N/A
5674235	12/1996	Parisi	N/A	N/A
5678568	12/1996	Uchikubo et al.	N/A	N/A
5688270	12/1996	Yates et al.	N/A	N/A
5690269	12/1996	Bolanos et al.	N/A	N/A
5693042	12/1996	Boiarski et al.	N/A	N/A
5693051	12/1996	Schulze et al.	N/A	N/A
5694936	12/1996	Fujimoto et al.	N/A	N/A
5695510	12/1996	Hood	N/A	N/A
5700261	12/1996	Brinkerhoff	N/A	N/A
5704534	12/1997	Huitema et al.	N/A	N/A
5704791	12/1997	Gillio	N/A	N/A
5707369	12/1997	Vaitekunas et al.	N/A	N/A
5709680	12/1997	Yates et al.	N/A	N/A

5711472	12/1997	Bryan	N/A	N/A
5713896	12/1997	Nardella	N/A	N/A
5715817	12/1997	Stevens-Wright et al.	N/A	N/A
5716366	12/1997	Yates	N/A	N/A
5717306	12/1997	Shipp	N/A	N/A
5720742	12/1997	Zacharias	N/A	N/A
5720744	12/1997	Eggleston et al.	N/A	N/A
5722980	12/1997	Schulz et al.	N/A	N/A
5723970	12/1997	Bell	N/A	N/A
5728130	12/1997	Ishikawa et al.	N/A	N/A
5730752	12/1997	Alden et al.	N/A	N/A
5733074	12/1997	Stock et al.	N/A	N/A
5735848	12/1997	Yates et al.	N/A	N/A
5741226	12/1997	Strukel et al.	N/A	N/A
5743906	12/1997	Parins et al.	N/A	N/A
5752973	12/1997	Kieturakis	N/A	N/A
5755717	12/1997	Yates et al.	N/A	N/A
5759183	12/1997	VanDusseldorp	606/49	A61B 18/149
5762255	12/1997	Chrisman et al.	N/A	N/A
5766164	12/1997	Mueller et al.	N/A	N/A
5772659	12/1997	Becker et al.	N/A	N/A
5776130	12/1997	Buyse et al.	N/A	N/A
5776155	12/1997	Beaupre et al.	N/A	N/A
5779130	12/1997	Alesi et al.	N/A	N/A
5779701	12/1997	McBrayer et al.	N/A	N/A
5782834	12/1997	Lucey et al.	N/A	N/A
5792135	12/1997	Madhani et al.	N/A	N/A
5792138	12/1997	Shipp	N/A	N/A
5792165	12/1997	Klieman et al.	N/A	N/A
5796188	12/1997	Bays	N/A	N/A
5797941	12/1997	Schulze et al.	N/A	N/A
5797958	12/1997	Yoon	N/A	N/A
5797959	12/1997	Castro et al.	N/A	N/A
5800432	12/1997	Swanson	N/A	N/A
5800448	12/1997	Banko	N/A	N/A
5800449	12/1997	Wales	N/A	N/A
5805140	12/1997	Rosenberg et al.	N/A	N/A
5807393	12/1997	Williamson, IV et al.	N/A	N/A
5808396	12/1997	Boukhny	N/A	N/A
5810811	12/1997	Yates et al.	N/A	N/A
5810828	12/1997	Lightman et al.	N/A	N/A
5810859	12/1997	DiMatteo et al.	N/A	N/A
5817033	12/1997	DeSantis et al.	N/A	N/A
5817084	12/1997	Jensen	N/A	N/A
5817093	12/1997	Williamson, IV et al.	N/A	N/A
5817119	12/1997	Klieman et al.	N/A	N/A
5823197	12/1997	Edwards	N/A	N/A

5827271	12/1997	Buyse et al.	N/A	N/A
5827323	12/1997	Klieman et al.	N/A	N/A
5828160	12/1997	Sugishita	N/A	N/A
5833696	12/1997	Whitfield et al.	N/A	N/A
5836897	12/1997	Sakurai et al.	N/A	N/A
5836909	12/1997	Cosmescu	N/A	N/A
5836943	12/1997	Miller, III	N/A	N/A
5836957	12/1997	Schulz et al.	N/A	N/A
5836990	12/1997	Li	N/A	N/A
5843109	12/1997	Mehta et al.	N/A	N/A
5851212	12/1997	Zirps et al.	N/A	N/A
5853412	12/1997	Mayenberger	N/A	N/A
5854590	12/1997	Dalstein	N/A	N/A
5858018	12/1998	Shipp et al.	N/A	N/A
5865361	12/1998	Milliman et al.	N/A	N/A
5873873	12/1998	Smith et al.	N/A	N/A
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5876401	12/1998	Schulze et al.	N/A	N/A
5878193	12/1998	Wang et al.	N/A	N/A
5879364	12/1998	Bromfield et al.	N/A	N/A
5880668	12/1998	Hall	N/A	N/A
5883615	12/1998	Fago et al.	N/A	N/A
5891142	12/1998	Eggers et al.	N/A	N/A
5893835	12/1998	Witt et al.	N/A	N/A
5897523	12/1998	Wright et al.	N/A	N/A
5897569	12/1998	Kellogg et al.	N/A	N/A
5903607	12/1998	Tailliet	N/A	N/A
5904681	12/1998	West, Jr.	N/A	N/A
5906625	12/1998	Bitto et al.	N/A	N/A
5906627	12/1998	Spaulding	N/A	N/A
5906628	12/1998	Miyawaki et al.	N/A	N/A
5910129	12/1998	Koblish et al.	N/A	N/A
5911699	12/1998	Anis et al.	N/A	N/A
5913823	12/1998	Hedberg et al.	N/A	N/A
5916229	12/1998	Evans	N/A	N/A
5921956	12/1998	Grinberg et al.	N/A	N/A
5929846	12/1998	Rosenberg et al.	N/A	N/A
5935143	12/1998	Hood	N/A	N/A
5935144	12/1998	Estabrook	N/A	N/A
5938633	12/1998	Beaupre	N/A	N/A
5944718	12/1998	Austin et al.	N/A	N/A
5944737	12/1998	Tsonton et al.	N/A	N/A
5947984	12/1998	Whipple	N/A	N/A
5954717	12/1998	Behl et al.	N/A	N/A
5954736	12/1998	Bishop et al.	N/A	N/A
5954746	12/1998	Holthaus et al.	N/A	N/A
5957882	12/1998	Nita et al.	N/A	N/A
5957943	12/1998	Vaitekunas	N/A	N/A
5968007	12/1998	Simon et al.	N/A	N/A
5968060	12/1998	Kellogg	N/A	N/A

5974342	12/1998	Petrofsky	N/A	N/A
D416089	12/1998	Barton et al.	N/A	N/A
5980510	12/1998	Tsonton et al.	N/A	N/A
5980546	12/1998	Hood	N/A	N/A
5984938	12/1998	Yoon	N/A	N/A
5987344	12/1998	West	N/A	N/A
5989274	12/1998	Davison et al.	N/A	N/A
5989275	12/1998	Estabrook et al.	N/A	N/A
5993465	12/1998	Shipp et al.	N/A	N/A
5993972	12/1998	Reich et al.	N/A	N/A
5994855	12/1998	Lundell et al.	N/A	N/A
6003517	12/1998	Sheffield et al.	N/A	N/A
6004335	12/1998	Vaitekunas et al.	N/A	N/A
6011416	12/1999	Mizuno et al.	N/A	N/A
6013052	12/1999	Durman et al.	N/A	N/A
6024741	12/1999	Williamson, IV et al.	N/A	N/A
6024744	12/1999	Kese et al.	N/A	N/A
6024750	12/1999	Mastri et al.	N/A	N/A
6027515	12/1999	Cimino	N/A	N/A
6031526	12/1999	Shipp	N/A	N/A
6033375	12/1999	Brumbach	N/A	N/A
6033399	12/1999	Gines	N/A	N/A
6036667	12/1999	Manna et al.	N/A	N/A
6036707	12/1999	Spaulding	N/A	N/A
6039734	12/1999	Goble	N/A	N/A
6048224	12/1999	Kay	N/A	N/A
6050943	12/1999	Slayton et al.	N/A	N/A
6050996	12/1999	Schmaltz et al.	N/A	N/A
6051010	12/1999	DiMatteo et al.	N/A	N/A
6056735	12/1999	Okada et al.	N/A	N/A
6063098	12/1999	Houser et al.	N/A	N/A
6066132	12/1999	Chen et al.	N/A	N/A
6066151	12/1999	Miyawaki et al.	N/A	N/A
6068627	12/1999	Orszulak et al.	N/A	N/A
6068629	12/1999	Haissaguerre et al.	N/A	N/A
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6074389	12/1999	Levine et al.	N/A	N/A
6077285	12/1999	Boukhny	N/A	N/A
6080149	12/1999	Huang et al.	N/A	N/A
6083191	12/1999	Rose	N/A	N/A
6086584	12/1999	Miller	N/A	N/A
6090120	12/1999	Wright et al.	N/A	N/A
6091995	12/1999	Ingle et al.	N/A	N/A
6096033	12/1999	Tu et al.	N/A	N/A
6099483	12/1999	Palmer et al.	N/A	N/A
6099542	12/1999	Cohn et al.	N/A	N/A
6099550	12/1999	Yoon	N/A	N/A
6102909	12/1999	Chen et al.	N/A	N/A
6109500	12/1999	Alli et al.	N/A	N/A

6110127	12/1999	Suzuki	N/A	N/A
6113594	12/1999	Savage	N/A	N/A
6113598	12/1999	Baker	N/A	N/A
6117152	12/1999	Huitema	N/A	N/A
H1904	12/1999	Yates et al.	N/A	N/A
6126629	12/1999	Perkins	N/A	N/A
6126658	12/1999	Baker	N/A	N/A
6129735	12/1999	Okada et al.	N/A	N/A
6129740	12/1999	Michelson	N/A	N/A
6132368	12/1999	Cooper	N/A	N/A
6132427	12/1999	Jones et al.	N/A	N/A
6132429	12/1999	Baker	N/A	N/A
6132448	12/1999	Perez et al.	N/A	N/A
6139320	12/1999	Hahn	N/A	N/A
6139561	12/1999	Shibata et al.	N/A	N/A
6142615	12/1999	Qiu et al.	N/A	N/A
6142994	12/1999	Swanson et al.	N/A	N/A
6144402	12/1999	Norsworthy et al.	N/A	N/A
6147560	12/1999	Erhage et al.	N/A	N/A
6152902	12/1999	Christian et al.	N/A	N/A
6152923	12/1999	Ryan	N/A	N/A
6154198	12/1999	Rosenberg	N/A	N/A
6156029	12/1999	Mueller	N/A	N/A
6159160	12/1999	Hsei et al.	N/A	N/A
6159175	12/1999	Strukel et al.	N/A	N/A
6162194	12/1999	Shipp	N/A	N/A
6162208	12/1999	Hipps	N/A	N/A
6165150	12/1999	Banko	N/A	N/A
6174309	12/2000	Wrublewski et al.	N/A	N/A
6174310	12/2000	Kirwan, Jr.	N/A	N/A
6176857	12/2000	Ashley	N/A	N/A
6179853	12/2000	Sachse et al.	N/A	N/A
6183426	12/2000	Akisada et al.	N/A	N/A
6187003	12/2000	Buyse et al.	N/A	N/A
6190386	12/2000	Rydell	N/A	N/A
6193709	12/2000	Miyawaki et al.	N/A	N/A
6204592	12/2000	Hur	N/A	N/A
6205383	12/2000	Hermann	N/A	N/A
6205855	12/2000	Pfeiffer	N/A	N/A
6206844	12/2000	Reichel et al.	N/A	N/A
6206876	12/2000	Levine et al.	N/A	N/A
6210337	12/2000	Dunham et al.	N/A	N/A
6210402	12/2000	Olsen et al.	N/A	N/A
6210403	12/2000	Klicek	N/A	N/A
6214023	12/2000	Whipple et al.	N/A	N/A
6228080	12/2000	Gines	N/A	N/A
6231565	12/2000	Tovey et al.	N/A	N/A
6232899	12/2000	Craven	N/A	N/A
6233476	12/2000	Strommer et al.	N/A	N/A
6238366	12/2000	Savage et al.	N/A	N/A

6238384	12/2000	Peer	N/A	N/A
6241724	12/2000	Fleischman et al.	N/A	N/A
6245065	12/2000	Panescu	606/49	A61B 18/1206
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6252110	12/2000	Uemura et al.	N/A	N/A
D444365	12/2000	Bass et al.	N/A	N/A
D445092	12/2000	Lee	N/A	N/A
D445764	12/2000	Lee	N/A	N/A
6254623	12/2000	Haibel, Jr. et al.	N/A	N/A
6257241	12/2000	Wampler	N/A	N/A
6258034	12/2000	Hanafy	N/A	N/A
6259230	12/2000	Chou	N/A	N/A
6261286	12/2000	Goble et al.	N/A	N/A
6267761	12/2000	Ryan	N/A	N/A
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6274963	12/2000	Estabrook et al.	N/A	N/A
6277115	12/2000	Saadat	N/A	N/A
6277117	12/2000	Tetzlaff et al.	N/A	N/A
6278218	12/2000	Madan et al.	N/A	N/A
6280407	12/2000	Manna et al.	N/A	N/A
6283981	12/2000	Beaupre	N/A	N/A
6287344	12/2000	Wampler et al.	N/A	N/A
6290575	12/2000	Shipp	N/A	N/A
6292700	12/2000	Morrison et al.	N/A	N/A
6299591	12/2000	Banko	N/A	N/A
6306131	12/2000	Hareyama et al.	N/A	N/A
6306157	12/2000	Shchervinsky	N/A	N/A
6309400	12/2000	Beaupre	N/A	N/A
6311783	12/2000	Harpell	N/A	N/A
6319221	12/2000	Savage et al.	N/A	N/A
6325795	12/2000	Lindemann et al.	N/A	N/A
6325799	12/2000	Goble	N/A	N/A
6325811	12/2000	Messerly	N/A	N/A
6328751	12/2000	Beaupre	N/A	N/A
6332891	12/2000	Himes	N/A	N/A
6338657	12/2001	Harper et al.	N/A	N/A
6340352	12/2001	Okada et al.	N/A	N/A
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6356224	12/2001	Wohlfarth	N/A	N/A
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D457958	12/2001	Dycus et al.	N/A	N/A
6383194	12/2001	Pothula	N/A	N/A
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6387094	12/2001	Eitenmuller	N/A	N/A
6387109	12/2001	Davison et al.	N/A	N/A
6388657	12/2001	Natoli	N/A	N/A
6390973	12/2001	Ouchi	N/A	N/A
6391026	12/2001	Hung et al.	N/A	N/A
6391042	12/2001	Cimino	N/A	N/A
6398779	12/2001	Buysse et al.	N/A	N/A
6402743	12/2001	Orszulak et al.	N/A	N/A
6402748	12/2001	Schoenman et al.	N/A	N/A
6405184	12/2001	Bohme et al.	N/A	N/A
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H2037	12/2001	Yates et al.	N/A	N/A
6416469	12/2001	Phung et al.	N/A	N/A
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6417969	12/2001	DeLuca et al.	N/A	N/A
6419675	12/2001	Gallo, Sr.	N/A	N/A
6423073	12/2001	Bowman	N/A	N/A
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6428538	12/2001	Blewett et al.	N/A	N/A
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6430446	12/2001	Knowlton	N/A	N/A
6432118	12/2001	Messerly	N/A	N/A
6436114	12/2001	Novak et al.	N/A	N/A
6436115	12/2001	Beaupre	N/A	N/A
6436129	12/2001	Sharkey et al.	N/A	N/A
6440062	12/2001	Ouchi	N/A	N/A
6443968	12/2001	Holthaus et al.	N/A	N/A
6443969	12/2001	Novak et al.	N/A	N/A
6449006	12/2001	Shipp	N/A	N/A
6454781	12/2001	Witt et al.	N/A	N/A
6454782	12/2001	Schwemberger	N/A	N/A
6458128	12/2001	Schulze	N/A	N/A
6458130	12/2001	Frazier et al.	N/A	N/A
6458142	12/2001	Faller et al.	N/A	N/A
6459363	12/2001	Walker et al.	N/A	N/A
6461363	12/2001	Gadberry et al.	N/A	N/A
6464689	12/2001	Qin et al.	N/A	N/A
6464702	12/2001	Schulze et al.	N/A	N/A
6468270	12/2001	Hovda et al.	N/A	N/A
6475211	12/2001	Chess et al.	N/A	N/A
6475215	12/2001	Tanrisever	N/A	N/A
6480796	12/2001	Wiener	N/A	N/A
6485490	12/2001	Wampler et al.	N/A	N/A
6491690	12/2001	Goble et al.	N/A	N/A
6491701	12/2001	Tierney et al.	N/A	N/A
6491708	12/2001	Madan et al.	N/A	N/A
6497715	12/2001	Satou	N/A	N/A
6500112	12/2001	Khoury	N/A	N/A

6500176	12/2001	Truckai et al.	N/A	N/A
6500188	12/2001	Harper et al.	N/A	N/A
6500312	12/2001	Wedekamp	N/A	N/A
6503248	12/2002	Levine	N/A	N/A
6506208	12/2002	Hunt et al.	N/A	N/A
6511478	12/2002	Burnside et al.	N/A	N/A
6511480	12/2002	Tetzlaff et al.	N/A	N/A
6511493	12/2002	Moutafis et al.	N/A	N/A
6514252	12/2002	Nezhat et al.	N/A	N/A
6514267	12/2002	Jewett	N/A	N/A
6517565	12/2002	Whitman et al.	N/A	N/A
6524251	12/2002	Rabiner et al.	N/A	N/A
6524316	12/2002	Nicholson et al.	N/A	N/A
6527736	12/2002	Attinger et al.	N/A	N/A
6531846	12/2002	Smith	N/A	N/A
6533784	12/2002	Truckai et al.	N/A	N/A
6537272	12/2002	Christopherson et al.	N/A	N/A
6537291	12/2002	Friedman et al.	N/A	N/A
6543452	12/2002	Lavigne	N/A	N/A
6543456	12/2002	Freeman	N/A	N/A
6544260	12/2002	Markel et al.	N/A	N/A
6551309	12/2002	LePivert	N/A	N/A
6554829	12/2002	Schulze et al.	N/A	N/A
6558376	12/2002	Bishop	N/A	N/A
6558380	12/2002	Lingenfelder et al.	N/A	N/A
6561983	12/2002	Cronin et al.	N/A	N/A
6562035	12/2002	Levin	N/A	N/A
6562037	12/2002	Paton et al.	N/A	N/A
6565558	12/2002	Lindenmeier et al.	N/A	N/A
6572563	12/2002	Ouchi	N/A	N/A
6572632	12/2002	Zisterer et al.	N/A	N/A
6572639	12/2002	Ingle et al.	N/A	N/A
6575969	12/2002	Rittman, III et al.	N/A	N/A
6582427	12/2002	Goble et al.	N/A	N/A
6582451	12/2002	Marucci et al.	N/A	N/A
6584360	12/2002	Francischelli et al.	N/A	N/A
D477408	12/2002	Bromley	N/A	N/A
6585735	12/2002	Frazier et al.	N/A	N/A
6588277	12/2002	Giordano et al.	N/A	N/A
6589200	12/2002	Schwemberger et al.	N/A	N/A
6589239	12/2002	Khandkar et al.	N/A	N/A
6590733	12/2002	Wilson et al.	N/A	N/A
6599288	12/2002	Maguire et al.	N/A	N/A
6602252	12/2002	Mollenauer	N/A	N/A
6602262	12/2002	Griego et al.	N/A	N/A
6607540	12/2002	Shipp	N/A	N/A
6610059	12/2002	West, Jr.	N/A	N/A
6610060	12/2002	Mulier et al.	N/A	N/A

6611793	12/2002	Burnside et al.	N/A	N/A
6616450	12/2002	Mossle et al.	N/A	N/A
6619529	12/2002	Green et al.	N/A	N/A
6620161	12/2002	Schulze et al.	N/A	N/A
6622731	12/2002	Daniel et al.	N/A	N/A
6623482	12/2002	Pendekanti et al.	N/A	N/A
6623500	12/2002	Cook et al.	N/A	N/A
6623501	12/2002	Heller et al.	N/A	N/A
6626848	12/2002	Neuenfeldt	N/A	N/A
6626926	12/2002	Friedman et al.	N/A	N/A
6629974	12/2002	Penny et al.	N/A	N/A
6632221	12/2002	Edwards et al.	N/A	N/A
6633234	12/2002	Wiener et al.	N/A	N/A
6635057	12/2002	Harano et al.	N/A	N/A
6644532	12/2002	Green et al.	N/A	N/A
6651669	12/2002	Burnside	N/A	N/A
6652513	12/2002	Panescu	606/41	A61B 18/1206
6652539	12/2002	Shipp et al.	N/A	N/A
6652545	12/2002	Shipp et al.	N/A	N/A
6656132	12/2002	Ouchi	N/A	N/A
6656177	12/2002	Truckai et al.	N/A	N/A
6656198	12/2002	Tsonton et al.	N/A	N/A
6660017	12/2002	Beaupre	N/A	N/A
6662127	12/2002	Wiener et al.	N/A	N/A
6663941	12/2002	Brown et al.	N/A	N/A
6666860	12/2002	Takahashi	N/A	N/A
6666875	12/2002	Sakurai et al.	N/A	N/A
6669690	12/2002	Okada et al.	N/A	N/A
6669710	12/2002	Moutafis et al.	N/A	N/A
6673248	12/2003	Chowdhury	N/A	N/A
6676660	12/2003	Wampler et al.	N/A	N/A
6678621	12/2003	Wiener et al.	N/A	N/A
6679875	12/2003	Honda et al.	N/A	N/A
6679882	12/2003	Kornerup	N/A	N/A
6679899	12/2003	Wiener et al.	N/A	N/A
6682501	12/2003	Nelson et al.	N/A	N/A
6682544	12/2003	Mastri et al.	N/A	N/A
6685700	12/2003	Behl et al.	N/A	N/A
6685701	12/2003	Orszulak et al.	N/A	N/A
6685703	12/2003	Pearson et al.	N/A	N/A
6689145	12/2003	Lee et al.	N/A	N/A
6689146	12/2003	Himes	N/A	N/A
6690960	12/2003	Chen et al.	N/A	N/A
6695840	12/2003	Schulze	N/A	N/A
6702821	12/2003	Bonutti	N/A	N/A
6716215	12/2003	David et al.	N/A	N/A
6719692	12/2003	Kleffner et al.	N/A	N/A
6719765	12/2003	Bonutti	N/A	N/A
6719776	12/2003	Baxter et al.	N/A	N/A

6722552	12/2003	Fenton, Jr.	N/A	N/A
6723091	12/2003	Goble et al.	N/A	N/A
D490059	12/2003	Conway et al.	N/A	N/A
6730080	12/2003	Harano et al.	N/A	N/A
6731047	12/2003	Kauf et al.	N/A	N/A
6733498	12/2003	Paton et al.	N/A	N/A
6733506	12/2003	McDevitt et al.	N/A	N/A
6736813	12/2003	Yamauchi et al.	N/A	N/A
6739872	12/2003	Turri	N/A	N/A
6740079	12/2003	Eggers et al.	N/A	N/A
D491666	12/2003	Kimmell et al.	N/A	N/A
6743245	12/2003	Lobdell	N/A	N/A
6746284	12/2003	Spink, Jr.	N/A	N/A
6746443	12/2003	Morley et al.	N/A	N/A
6752815	12/2003	Beaupre	N/A	N/A
6755825	12/2003	Shoenman et al.	N/A	N/A
6761698	12/2003	Shibata et al.	N/A	N/A
6762535	12/2003	Take et al.	N/A	N/A
6766202	12/2003	Underwood et al.	N/A	N/A
6770072	12/2003	Truckai et al.	N/A	N/A
6773409	12/2003	Truckai et al.	N/A	N/A
6773434	12/2003	Ciarrocca	N/A	N/A
6773435	12/2003	Schulze et al.	N/A	N/A
6773443	12/2003	Truwit et al.	N/A	N/A
6773444	12/2003	Messerly	N/A	N/A
6775575	12/2003	Bommannan et al.	N/A	N/A
6778023	12/2003	Christensen	N/A	N/A
6783524	12/2003	Anderson et al.	N/A	N/A
6786382	12/2003	Hoffman	N/A	N/A
6786383	12/2003	Stegelmann	N/A	N/A
6789939	12/2003	Schrodinger et al.	N/A	N/A
6790173	12/2003	Saadat et al.	N/A	N/A
6790216	12/2003	Ishikawa	N/A	N/A
6794027	12/2003	Araki et al.	N/A	N/A
6796981	12/2003	Wham et al.	N/A	N/A
D496997	12/2003	Dycus et al.	N/A	N/A
6800085	12/2003	Selmon et al.	N/A	N/A
6802843	12/2003	Truckai et al.	N/A	N/A
6808525	12/2003	Latterell et al.	N/A	N/A
6809508	12/2003	Donofrio	N/A	N/A
6810281	12/2003	Brock et al.	N/A	N/A
6811842	12/2003	Ehrnsperger et al.	N/A	N/A
6814731	12/2003	Swanson	N/A	N/A
6819027	12/2003	Saraf	N/A	N/A
6821273	12/2003	Mollenauer	N/A	N/A
6827712	12/2003	Tovey et al.	N/A	N/A
6828712	12/2003	Battaglin et al.	N/A	N/A
6835082	12/2003	Gonnering	N/A	N/A
6835199	12/2003	McGuckin, Jr. et al.	N/A	N/A

6840938	12/2004	Morley et al.	N/A	N/A
6843789	12/2004	Goble	N/A	N/A
6849073	12/2004	Hoey et al.	N/A	N/A
6860878	12/2004	Brock	N/A	N/A
6860880	12/2004	Treat et al.	N/A	N/A
6863676	12/2004	Lee et al.	N/A	N/A
6866671	12/2004	Tierney et al.	N/A	N/A
6869439	12/2004	White et al.	N/A	N/A
6875220	12/2004	Du et al.	N/A	N/A
6877647	12/2004	Green et al.	N/A	N/A
6882439	12/2004	Ishijima	N/A	N/A
6887209	12/2004	Kadziauskas et al.	N/A	N/A
6887252	12/2004	Okada et al.	N/A	N/A
6893435	12/2004	Goble	N/A	N/A
6898536	12/2004	Wiener et al.	N/A	N/A
6899685	12/2004	Kermode et al.	N/A	N/A
6905497	12/2004	Truckai et al.	N/A	N/A
6908463	12/2004	Treat et al.	N/A	N/A
6908472	12/2004	Wiener et al.	N/A	N/A
6913579	12/2004	Truckai et al.	N/A	N/A
6915623	12/2004	Dey et al.	N/A	N/A
6923804	12/2004	Eggers et al.	N/A	N/A
6923806	12/2004	Hooven et al.	N/A	N/A
6926712	12/2004	Phan	N/A	N/A
6926716	12/2004	Baker et al.	N/A	N/A
6926717	12/2004	Garito et al.	N/A	N/A
6929602	12/2004	Hirakui et al.	N/A	N/A
6929622	12/2004	Chian	N/A	N/A
6929632	12/2004	Nita et al.	N/A	N/A
6929644	12/2004	Truckai et al.	N/A	N/A
6933656	12/2004	Matsushita et al.	N/A	N/A
D509589	12/2004	Wells	N/A	N/A
6942660	12/2004	Pantera et al.	N/A	N/A
6942677	12/2004	Nita et al.	N/A	N/A
6945981	12/2004	Donofrio et al.	N/A	N/A
6946779	12/2004	Birgel	N/A	N/A
6948503	12/2004	Refior et al.	N/A	N/A
6953461	12/2004	McClurken et al.	N/A	N/A
6958070	12/2004	Witt et al.	N/A	N/A
D511145	12/2004	Donofrio et al.	N/A	N/A
6974450	12/2004	Weber et al.	N/A	N/A
6976844	12/2004	Hickok et al.	N/A	N/A
6976969	12/2004	Messerly	N/A	N/A
6977495	12/2004	Donofrio	N/A	N/A
6979332	12/2004	Adams	N/A	N/A
6981628	12/2005	Wales	N/A	N/A
6984220	12/2005	Wuchinich	N/A	N/A
6984231	12/2005	Goble et al.	N/A	N/A
6988295	12/2005	Tillim	N/A	N/A
6988649	12/2005	Shelton, IV et al.	N/A	N/A

6994708	12/2005	Manzo	N/A	N/A
6994709	12/2005	Iida	N/A	N/A
7000818	12/2005	Shelton, IV et al.	N/A	N/A
7001335	12/2005	Adachi et al.	N/A	N/A
7001379	12/2005	Behl et al.	N/A	N/A
7001382	12/2005	Gallo, Sr.	N/A	N/A
7004951	12/2005	Gibbens, III	N/A	N/A
7011657	12/2005	Truckai et al.	N/A	N/A
7014638	12/2005	Michelson	N/A	N/A
7018389	12/2005	Camerlengo	N/A	N/A
7025732	12/2005	Thompson et al.	N/A	N/A
7033356	12/2005	Latterell et al.	N/A	N/A
7033357	12/2005	Baxter et al.	N/A	N/A
7037306	12/2005	Podany et al.	N/A	N/A
7041083	12/2005	Chu et al.	N/A	N/A
7041088	12/2005	Nawrocki et al.	N/A	N/A
7041102	12/2005	Truckai et al.	N/A	N/A
7044949	12/2005	Orszulak et al.	N/A	N/A
7052494	12/2005	Goble et al.	N/A	N/A
7052496	12/2005	Yamauchi	N/A	N/A
7055731	12/2005	Shelton, IV et al.	N/A	N/A
7063699	12/2005	Hess et al.	N/A	N/A
7066893	12/2005	Hibner et al.	N/A	N/A
7066895	12/2005	Podany	N/A	N/A
7066936	12/2005	Ryan	N/A	N/A
7070597	12/2005	Truckai et al.	N/A	N/A
7074218	12/2005	Washington et al.	N/A	N/A
7074219	12/2005	Levine et al.	N/A	N/A
7077039	12/2005	Gass et al.	N/A	N/A
7077845	12/2005	Hacker et al.	N/A	N/A
7077853	12/2005	Kramer et al.	N/A	N/A
7083075	12/2005	Swayze et al.	N/A	N/A
7083613	12/2005	Treat	N/A	N/A
7083618	12/2005	Couture et al.	N/A	N/A
7083619	12/2005	Truckai et al.	N/A	N/A
7087054	12/2005	Truckai et al.	N/A	N/A
7090637	12/2005	Danitz et al.	N/A	N/A
7090672	12/2005	Underwood et al.	N/A	N/A
7094235	12/2005	Francischelli	N/A	N/A
7101371	12/2005	Dycus et al.	N/A	N/A
7101372	12/2005	Dycus et al.	N/A	N/A
7101373	12/2005	Dycus et al.	N/A	N/A
7101378	12/2005	Salameh et al.	N/A	N/A
7104834	12/2005	Robinson et al.	N/A	N/A
7108695	12/2005	Witt et al.	N/A	N/A
7111769	12/2005	Wales et al.	N/A	N/A
7112201	12/2005	Truckai et al.	N/A	N/A
7113831	12/2005	Hooven	N/A	N/A
D531311	12/2005	Guerra et al.	N/A	N/A
7117034	12/2005	Kronberg	N/A	N/A

7118564	12/2005	Ritchie et al.	N/A	N/A
7118570	12/2005	Tetzlaff et al.	N/A	N/A
7118587	12/2005	Dycus et al.	N/A	N/A
7119516	12/2005	Denning	N/A	N/A
7124932	12/2005	Isaacson et al.	N/A	N/A
7125409	12/2005	Truckai et al.	N/A	N/A
7128720	12/2005	Podany	N/A	N/A
7131860	12/2005	Sartor et al.	N/A	N/A
7131970	12/2005	Moses et al.	N/A	N/A
7135018	12/2005	Ryan et al.	N/A	N/A
7135030	12/2005	Schwemberger et al.	N/A	N/A
7137980	12/2005	Buyse et al.	N/A	N/A
7143925	12/2005	Shelton, IV et al.	N/A	N/A
7144403	12/2005	Booth	N/A	N/A
7147138	12/2005	Shelton, IV	N/A	N/A
7153315	12/2005	Miller	N/A	N/A
D536093	12/2006	Nakajima et al.	N/A	N/A
7156189	12/2006	Bar-Cohen et al.	N/A	N/A
7156846	12/2006	Dycus et al.	N/A	N/A
7156853	12/2006	Muratsu	N/A	N/A
7157058	12/2006	Marhasin et al.	N/A	N/A
7159750	12/2006	Racenet et al.	N/A	N/A
7160259	12/2006	Tardy et al.	N/A	N/A
7160296	12/2006	Pearson et al.	N/A	N/A
7160298	12/2006	Lawes et al.	N/A	N/A
7160299	12/2006	Baily	N/A	N/A
7163548	12/2006	Stulen et al.	N/A	N/A
7166103	12/2006	Carmel et al.	N/A	N/A
7169144	12/2006	Hoey et al.	N/A	N/A
7169146	12/2006	Truckai et al.	N/A	N/A
7169156	12/2006	Hart	N/A	N/A
7179254	12/2006	Pendekanti et al.	N/A	N/A
7179271	12/2006	Friedman et al.	N/A	N/A
7186253	12/2006	Truckai et al.	N/A	N/A
7189233	12/2006	Truckai et al.	N/A	N/A
7195631	12/2006	Dumbauld	N/A	N/A
D541418	12/2006	Schechter et al.	N/A	N/A
7198635	12/2006	Danek et al.	N/A	N/A
7204820	12/2006	Akahoshi	N/A	N/A
7207471	12/2006	Heinrich et al.	N/A	N/A
7207997	12/2006	Shipp et al.	N/A	N/A
7208005	12/2006	Frecker et al.	N/A	N/A
7210881	12/2006	Greenberg	N/A	N/A
7211079	12/2006	Treat	N/A	N/A
7217128	12/2006	Atkin et al.	N/A	N/A
7217269	12/2006	El-Galley et al.	N/A	N/A
7220951	12/2006	Truckai et al.	N/A	N/A
7223229	12/2006	Inman et al.	N/A	N/A
7225964	12/2006	Mastri et al.	N/A	N/A

7226447	12/2006	Uchida et al.	N/A	N/A
7226448	12/2006	Bertolero et al.	N/A	N/A
7229455	12/2006	Sakurai et al.	N/A	N/A
7232440	12/2006	Dumbauld et al.	N/A	N/A
7235071	12/2006	Gonnering	N/A	N/A
7235073	12/2006	Levine et al.	N/A	N/A
7241294	12/2006	Reschke	N/A	N/A
7244262	12/2006	Wiener et al.	N/A	N/A
7251531	12/2006	Mosher et al.	N/A	N/A
7252641	12/2006	Thompson et al.	N/A	N/A
7252667	12/2006	Moses et al.	N/A	N/A
7258688	12/2006	Shah et al.	N/A	N/A
7264618	12/2006	Murakami et al.	N/A	N/A
7267677	12/2006	Johnson et al.	N/A	N/A
7267685	12/2006	Butaric et al.	N/A	N/A
7269873	12/2006	Brewer et al.	N/A	N/A
7273483	12/2006	Wiener et al.	N/A	N/A
D552241	12/2006	Bromley et al.	N/A	N/A
7282048	12/2006	Goble et al.	N/A	N/A
7285895	12/2006	Beaupre	N/A	N/A
7287682	12/2006	Ezzat et al.	N/A	N/A
7297149	12/2006	Vitali et al.	N/A	N/A
7300431	12/2006	Dubrovsky	N/A	N/A
7300435	12/2006	Wham et al.	N/A	N/A
7300446	12/2006	Beaupre	N/A	N/A
7300450	12/2006	Vleugels et al.	N/A	N/A
7303531	12/2006	Lee et al.	N/A	N/A
7303557	12/2006	Wham et al.	N/A	N/A
7306597	12/2006	Manzo	N/A	N/A
7307313	12/2006	Ohyanagi et al.	N/A	N/A
7309849	12/2006	Truckai et al.	N/A	N/A
7311706	12/2006	Schoenman et al.	N/A	N/A
7311709	12/2006	Truckai et al.	N/A	N/A
7317955	12/2007	McGreevy	N/A	N/A
7318831	12/2007	Alvarez et al.	N/A	N/A
7318832	12/2007	Young et al.	N/A	N/A
7326236	12/2007	Andreas et al.	N/A	N/A
7329257	12/2007	Kanehira et al.	N/A	N/A
7331410	12/2007	Yong et al.	N/A	N/A
7335165	12/2007	Truwit et al.	N/A	N/A
7335997	12/2007	Wiener	N/A	N/A
7337010	12/2007	Howard et al.	N/A	N/A
7353068	12/2007	Tanaka et al.	N/A	N/A
7354440	12/2007	Truckal et al.	N/A	N/A
7357287	12/2007	Shelton, IV et al.	N/A	N/A
7357802	12/2007	Palanker et al.	N/A	N/A
7361172	12/2007	Cimino	N/A	N/A
7364577	12/2007	Wham et al.	N/A	N/A
7367976	12/2007	Lawes et al.	N/A	N/A
7371227	12/2007	Zeiner	N/A	N/A

RE40388	12/2007	Gines	N/A	N/A
7380695	12/2007	Doll et al.	N/A	N/A
7380696	12/2007	Shelton, IV et al.	N/A	N/A
7381209	12/2007	Truckai et al.	N/A	N/A
7384420	12/2007	Dycus et al.	N/A	N/A
7390317	12/2007	Taylor et al.	N/A	N/A
7396356	12/2007	Mollenauer	N/A	N/A
7403224	12/2007	Fuller et al.	N/A	N/A
7404508	12/2007	Smith et al.	N/A	N/A
7407077	12/2007	Ortiz et al.	N/A	N/A
7408288	12/2007	Hara	N/A	N/A
7412008	12/2007	Lliev	N/A	N/A
7416101	12/2007	Shelton, IV et al.	N/A	N/A
7416437	12/2007	Sartor et al.	N/A	N/A
D576725	12/2007	Shumer et al.	N/A	N/A
7419490	12/2007	Falkenstein et al.	N/A	N/A
7422139	12/2007	Shelton, IV et al.	N/A	N/A
7422463	12/2007	Kuo	N/A	N/A
7422582	12/2007	Malackowski et al.	N/A	N/A
D578643	12/2007	Shumer et al.	N/A	N/A
D578644	12/2007	Shumer et al.	N/A	N/A
D578645	12/2007	Shumer et al.	N/A	N/A
7431694	12/2007	Stefanchik et al.	N/A	N/A
7431704	12/2007	Babaev	N/A	N/A
7431720	12/2007	Pendekanti et al.	N/A	N/A
7435582	12/2007	Zimmermann et al.	N/A	N/A
7441684	12/2007	Shelton, IV et al.	N/A	N/A
7442193	12/2007	Shields et al.	N/A	N/A
7445621	12/2007	Dumbauld et al.	N/A	N/A
7449004	12/2007	Yamada et al.	N/A	N/A
7451904	12/2007	Shelton, IV	N/A	N/A
7455208	12/2007	Wales et al.	N/A	N/A
7455641	12/2007	Yamada et al.	N/A	N/A
7462181	12/2007	Kraft et al.	N/A	N/A
7464846	12/2007	Shelton, IV et al.	N/A	N/A
7464849	12/2007	Shelton, IV et al.	N/A	N/A
7472815	12/2008	Shelton, IV et al.	N/A	N/A
7473145	12/2008	Ehr et al.	N/A	N/A
7473253	12/2008	Dycus et al.	N/A	N/A
7473263	12/2008	Johnston et al.	N/A	N/A
7479148	12/2008	Beaupre	N/A	N/A
7479160	12/2008	Branch et al.	N/A	N/A
7481775	12/2008	Weikel, Jr. et al.	N/A	N/A
7488285	12/2008	Honda et al.	N/A	N/A
7488319	12/2008	Yates	N/A	N/A
7491201	12/2008	Shields et al.	N/A	N/A
7491202	12/2008	Odom et al.	N/A	N/A
7494468	12/2008	Rabiner et al.	N/A	N/A
7494501	12/2008	Ahlberg et al.	N/A	N/A
7498080	12/2008	Tung et al.	N/A	N/A

7502234	12/2008	Goliszek et al.	N/A	N/A
7503893	12/2008	Kucklick	N/A	N/A
7503895	12/2008	Rabiner et al.	N/A	N/A
7506790	12/2008	Shelton, IV	N/A	N/A
7506791	12/2008	Omaits et al.	N/A	N/A
7507239	12/2008	Shadduck	N/A	N/A
7510107	12/2008	Timm et al.	N/A	N/A
7510556	12/2008	Nguyen et al.	N/A	N/A
7513025	12/2008	Fischer	N/A	N/A
7517349	12/2008	Truckai et al.	N/A	N/A
7520865	12/2008	Radley Young et al.	N/A	N/A
7524320	12/2008	Tierney et al.	N/A	N/A
7525309	12/2008	Sherman et al.	N/A	N/A
7530986	12/2008	Beaupre et al.	N/A	N/A
7534243	12/2008	Chin et al.	N/A	N/A
7535233	12/2008	Kojovic et al.	N/A	N/A
D594983	12/2008	Price et al.	N/A	N/A
7540871	12/2008	Gonnering	N/A	N/A
7540872	12/2008	Schechter et al.	N/A	N/A
7543730	12/2008	Marczyk	N/A	N/A
7544200	12/2008	Houser	N/A	N/A
7549564	12/2008	Boudreaux	N/A	N/A
7550216	12/2008	Ofer et al.	N/A	N/A
7553309	12/2008	Buysse et al.	N/A	N/A
7554343	12/2008	Bromfield	N/A	N/A
7559450	12/2008	Wales et al.	N/A	N/A
7559452	12/2008	Wales et al.	N/A	N/A
7563259	12/2008	Takahashi	N/A	N/A
7566318	12/2008	Haefner	N/A	N/A
7567012	12/2008	Namikawa	N/A	N/A
7568603	12/2008	Shelton, IV et al.	N/A	N/A
7569057	12/2008	Liu et al.	N/A	N/A
7572266	12/2008	Young et al.	N/A	N/A
7572268	12/2008	Babaev	N/A	N/A
7578820	12/2008	Moore et al.	N/A	N/A
7582084	12/2008	Swanson	606/49	A61B 18/1206
7582086	12/2008	Privitera et al.	N/A	N/A
7582087	12/2008	Tetzlaff et al.	N/A	N/A
7582095	12/2008	Shipp et al.	N/A	N/A
7585181	12/2008	Olsen	N/A	N/A
7586289	12/2008	Andruk et al.	N/A	N/A
7587536	12/2008	McLeod	N/A	N/A
7588176	12/2008	Timm et al.	N/A	N/A
7588177	12/2008	Racenet	N/A	N/A
7594925	12/2008	Danek et al.	N/A	N/A
7597693	12/2008	Garrison	N/A	N/A
7601119	12/2008	Shahinian	N/A	N/A
7601136	12/2008	Akahoshi	N/A	N/A

7604150	12/2008	Boudreaux	N/A	N/A
7607557	12/2008	Shelton, IV et al.	N/A	N/A
7617961	12/2008	Viola	N/A	N/A
7621930	12/2008	Houser	N/A	N/A
7625370	12/2008	Hart et al.	N/A	N/A
7628791	12/2008	Garrison et al.	N/A	N/A
7628792	12/2008	Guerra	N/A	N/A
7632267	12/2008	Dahla	N/A	N/A
7632269	12/2008	Truckai et al.	N/A	N/A
7637410	12/2008	Marczyk	N/A	N/A
7641653	12/2009	Dalla Betta et al.	N/A	N/A
7641671	12/2009	Crainich	N/A	N/A
7644848	12/2009	Swayze et al.	N/A	N/A
7645240	12/2009	Thompson et al.	N/A	N/A
7645277	12/2009	McClurken et al.	N/A	N/A
7645278	12/2009	Ichihashi et al.	N/A	N/A
7648499	12/2009	Orszulak et al.	N/A	N/A
7649410	12/2009	Andersen et al.	N/A	N/A
7654431	12/2009	Hueil et al.	N/A	N/A
7655003	12/2009	Lorang et al.	N/A	N/A
7658311	12/2009	Boudreaux	N/A	N/A
7659833	12/2009	Warner et al.	N/A	N/A
7662151	12/2009	Crompton, Jr. et al.	N/A	N/A
7665647	12/2009	Shelton, IV et al.	N/A	N/A
7666206	12/2009	Taniguchi et al.	N/A	N/A
7667592	12/2009	Ohyama et al.	N/A	N/A
7670334	12/2009	Hueil et al.	N/A	N/A
7670338	12/2009	Albrecht et al.	N/A	N/A
7674263	12/2009	Ryan	N/A	N/A
7678069	12/2009	Baker et al.	N/A	N/A
7678105	12/2009	McGreevy et al.	N/A	N/A
7678125	12/2009	Shipp	N/A	N/A
7682366	12/2009	Sakurai et al.	N/A	N/A
7686770	12/2009	Cohen	N/A	N/A
7686826	12/2009	Lee et al.	N/A	N/A
7688028	12/2009	Phillips et al.	N/A	N/A
7691095	12/2009	Bednarek et al.	N/A	N/A
7691098	12/2009	Wallace et al.	N/A	N/A
7696441	12/2009	Kataoka	N/A	N/A
7699846	12/2009	Ryan	N/A	N/A
7703459	12/2009	Saadat et al.	N/A	N/A
7703653	12/2009	Shah et al.	N/A	N/A
7708735	12/2009	Chapman et al.	N/A	N/A
7708751	12/2009	Hughes et al.	N/A	N/A
7708758	12/2009	Lee et al.	N/A	N/A
7708768	12/2009	Danek et al.	N/A	N/A
7713202	12/2009	Boukhny et al.	N/A	N/A
7713267	12/2009	Pozzato	N/A	N/A
7714481	12/2009	Sakai	N/A	N/A

7717312	12/2009	Beetel	N/A	N/A
7717914	12/2009	Kimura	N/A	N/A
7717915	12/2009	Miyazawa	N/A	N/A
7721935	12/2009	Racenet et al.	N/A	N/A
7722527	12/2009	Bouchier et al.	N/A	N/A
7722607	12/2009	Dumbauld et al.	N/A	N/A
D618797	12/2009	Price et al.	N/A	N/A
7726537	12/2009	Olson et al.	N/A	N/A
7727177	12/2009	Bayat	N/A	N/A
7731717	12/2009	Odom et al.	N/A	N/A
7738969	12/2009	Bleich	N/A	N/A
7740594	12/2009	Hibner	N/A	N/A
7744615	12/2009	Couture	N/A	N/A
7749240	12/2009	Takahashi et al.	N/A	N/A
7751115	12/2009	Song	N/A	N/A
7753245	12/2009	Boudreaux et al.	N/A	N/A
7753904	12/2009	Shelton, IV et al.	N/A	N/A
7753908	12/2009	Swanson	N/A	N/A
7762445	12/2009	Heinrich et al.	N/A	N/A
D621503	12/2009	Otten et al.	N/A	N/A
7766210	12/2009	Shelton, IV et al.	N/A	N/A
7766693	12/2009	Sartor et al.	N/A	N/A
7766910	12/2009	Hixson et al.	N/A	N/A
7768510	12/2009	Tsai et al.	N/A	N/A
7770774	12/2009	Mastri et al.	N/A	N/A
7770775	12/2009	Shelton, IV et al.	N/A	N/A
7771425	12/2009	Dycus et al.	N/A	N/A
7771444	12/2009	Patel et al.	N/A	N/A
7775972	12/2009	Brock et al.	N/A	N/A
7776036	12/2009	Schechter et al.	N/A	N/A
7776037	12/2009	Odom	N/A	N/A
7778733	12/2009	Nowlin et al.	N/A	N/A
7780054	12/2009	Wales	N/A	N/A
7780593	12/2009	Ueno et al.	N/A	N/A
7780651	12/2009	Madhani et al.	N/A	N/A
7780659	12/2009	Okada et al.	N/A	N/A
7780663	12/2009	Yates et al.	N/A	N/A
7784662	12/2009	Wales et al.	N/A	N/A
7784663	12/2009	Shelton, IV	N/A	N/A
7789883	12/2009	Takashino et al.	N/A	N/A
7793814	12/2009	Racenet et al.	N/A	N/A
7794475	12/2009	Hess et al.	N/A	N/A
7796969	12/2009	Kelly et al.	N/A	N/A
7798386	12/2009	Schall et al.	N/A	N/A
7799020	12/2009	Shores et al.	N/A	N/A
7799027	12/2009	Hafner	N/A	N/A
7799045	12/2009	Masuda	N/A	N/A
7803151	12/2009	Whitman	N/A	N/A
7803152	12/2009	Honda et al.	N/A	N/A
7803156	12/2009	Eder et al.	N/A	N/A

7803168	12/2009	Gifford et al.	N/A	N/A
7806891	12/2009	Nowlin et al.	N/A	N/A
7810693	12/2009	Broehl et al.	N/A	N/A
7811283	12/2009	Moses et al.	N/A	N/A
7815238	12/2009	Cao	N/A	N/A
7815641	12/2009	Dodde et al.	N/A	N/A
7819298	12/2009	Hall et al.	N/A	N/A
7819299	12/2009	Shelton, IV et al.	N/A	N/A
7819819	12/2009	Quick et al.	N/A	N/A
7819872	12/2009	Johnson et al.	N/A	N/A
7821143	12/2009	Wiener	N/A	N/A
D627066	12/2009	Romero	N/A	N/A
7824401	12/2009	Manzo et al.	N/A	N/A
7832408	12/2009	Shelton, IV et al.	N/A	N/A
7832611	12/2009	Boyden et al.	N/A	N/A
7832612	12/2009	Baxter, III et al.	N/A	N/A
7834484	12/2009	Sartor	N/A	N/A
7837699	12/2009	Yamada et al.	N/A	N/A
7845537	12/2009	Shelton, IV et al.	N/A	N/A
7846155	12/2009	Houser et al.	N/A	N/A
7846159	12/2009	Morrison et al.	N/A	N/A
7846160	12/2009	Payne et al.	N/A	N/A
7846161	12/2009	Dumbauld et al.	N/A	N/A
7854735	12/2009	Houser et al.	N/A	N/A
D631155	12/2010	Peine et al.	N/A	N/A
7861906	12/2010	Doll et al.	N/A	N/A
7862560	12/2010	Marion	N/A	N/A
7862561	12/2010	Swanson et al.	N/A	N/A
7867228	12/2010	Nobis et al.	N/A	N/A
7871392	12/2010	Sartor	N/A	N/A
7871423	12/2010	Livneh	N/A	N/A
7876030	12/2010	Taki et al.	N/A	N/A
D631965	12/2010	Price et al.	N/A	N/A
7877852	12/2010	Unger et al.	N/A	N/A
7878991	12/2010	Babaev	N/A	N/A
7879029	12/2010	Jimenez	N/A	N/A
7879033	12/2010	Sartor et al.	N/A	N/A
7879035	12/2010	Garrison et al.	N/A	N/A
7879070	12/2010	Ortiz et al.	N/A	N/A
7883475	12/2010	Dupont et al.	N/A	N/A
7892606	12/2010	Thies et al.	N/A	N/A
7896875	12/2010	Heim et al.	N/A	N/A
7897792	12/2010	Iikura et al.	N/A	N/A
7901400	12/2010	Wham et al.	N/A	N/A
7901423	12/2010	Stulen et al.	N/A	N/A
7905881	12/2010	Masuda et al.	N/A	N/A
7909220	12/2010	Viola	N/A	N/A
7909820	12/2010	Lipson et al.	N/A	N/A
7909824	12/2010	Masuda et al.	N/A	N/A
7918848	12/2010	Lau et al.	N/A	N/A

7919184	12/2010	Mohapatra et al.	N/A	N/A
7922061	12/2010	Shelton, IV et al.	N/A	N/A
7922651	12/2010	Yamada et al.	N/A	N/A
7931611	12/2010	Novak et al.	N/A	N/A
7931649	12/2010	Couture et al.	N/A	N/A
D637288	12/2010	Houghton	N/A	N/A
D638540	12/2010	Ijiri et al.	N/A	N/A
7935114	12/2010	Takashino et al.	N/A	N/A
7936203	12/2010	Zimlich	N/A	N/A
7951095	12/2010	Makin et al.	N/A	N/A
7951165	12/2010	Golden et al.	N/A	N/A
7954682	12/2010	Giordano et al.	N/A	N/A
7955331	12/2010	Truckai et al.	N/A	N/A
7956620	12/2010	Gilbert	N/A	N/A
7959050	12/2010	Smith et al.	N/A	N/A
7959626	12/2010	Hong et al.	N/A	N/A
7963963	12/2010	Francischelli et al.	N/A	N/A
7967602	12/2010	Lindquist	N/A	N/A
7972328	12/2010	Wham et al.	N/A	N/A
7972329	12/2010	Refior et al.	N/A	N/A
7975895	12/2010	Milliman	N/A	N/A
7976544	12/2010	McClurken et al.	N/A	N/A
7980443	12/2010	Scheib et al.	N/A	N/A
7981050	12/2010	Ritchart et al.	N/A	N/A
7981113	12/2010	Truckai et al.	N/A	N/A
7997278	12/2010	Utley et al.	N/A	N/A
7998157	12/2010	Culp et al.	N/A	N/A
8002732	12/2010	Visconti	N/A	N/A
8002770	12/2010	Swanson et al.	N/A	N/A
8020743	12/2010	Shelton, IV	N/A	N/A
8025672	12/2010	Novak et al.	N/A	N/A
8028885	12/2010	Smith et al.	N/A	N/A
8033173	12/2010	Ehlert et al.	N/A	N/A
8034049	12/2010	Odom et al.	N/A	N/A
8038693	12/2010	Allen	N/A	N/A
8048070	12/2010	O'Brien et al.	N/A	N/A
8048074	12/2010	Masuda	N/A	N/A
8052672	12/2010	Laufer et al.	N/A	N/A
8055208	12/2010	Lilla et al.	N/A	N/A
8056720	12/2010	Hawkes	N/A	N/A
8056787	12/2010	Boudreaux et al.	N/A	N/A
8057468	12/2010	Konesky	N/A	N/A
8057498	12/2010	Robertson	N/A	N/A
8058771	12/2010	Giordano et al.	N/A	N/A
8061014	12/2010	Smith et al.	N/A	N/A
8066167	12/2010	Measamer et al.	N/A	N/A
8070036	12/2010	Knodel	N/A	N/A
8070711	12/2010	Bassinger et al.	N/A	N/A
8070762	12/2010	Escudero et al.	N/A	N/A
8075555	12/2010	Truckai et al.	N/A	N/A

8075558	12/2010	Truckai et al.	N/A	N/A
8089197	12/2011	Rinner et al.	N/A	N/A
8092475	12/2011	Cotter et al.	N/A	N/A
8096459	12/2011	Ortiz et al.	N/A	N/A
8097012	12/2011	Kagarise	N/A	N/A
8100894	12/2011	Mucko et al.	N/A	N/A
8105230	12/2011	Honda et al.	N/A	N/A
8105323	12/2011	Buysse et al.	N/A	N/A
8105324	12/2011	Palanker et al.	N/A	N/A
8114104	12/2011	Young et al.	N/A	N/A
8118276	12/2011	Sanders et al.	N/A	N/A
8128624	12/2011	Couture et al.	N/A	N/A
8133218	12/2011	Daw et al.	N/A	N/A
8136712	12/2011	Zingman	N/A	N/A
8141762	12/2011	Bedi et al.	N/A	N/A
8142421	12/2011	Cooper et al.	N/A	N/A
8142461	12/2011	Houser et al.	N/A	N/A
8147485	12/2011	Wham et al.	N/A	N/A
8147488	12/2011	Masuda	N/A	N/A
8147508	12/2011	Madan et al.	N/A	N/A
8152801	12/2011	Goldberg et al.	N/A	N/A
8152825	12/2011	Madan et al.	N/A	N/A
8157145	12/2011	Shelton, IV et al.	N/A	N/A
8161977	12/2011	Shelton, IV et al.	N/A	N/A
8162966	12/2011	Connor et al.	N/A	N/A
8170717	12/2011	Sutherland et al.	N/A	N/A
8172846	12/2011	Brunnett et al.	N/A	N/A
8172870	12/2011	Shipp	N/A	N/A
8177800	12/2011	Spitz et al.	N/A	N/A
8182502	12/2011	Stulen et al.	N/A	N/A
8186560	12/2011	Hess et al.	N/A	N/A
8186877	12/2011	Klimovitch	374/170	A61B 18/1206
8187267	12/2011	Pappone et al.	N/A	N/A
D661801	12/2011	Price et al.	N/A	N/A
D661802	12/2011	Price et al.	N/A	N/A
D661803	12/2011	Price et al.	N/A	N/A
D661804	12/2011	Price et al.	N/A	N/A
8197472	12/2011	Lau et al.	N/A	N/A
8197479	12/2011	Olson et al.	N/A	N/A
8197502	12/2011	Smith et al.	N/A	N/A
8207651	12/2011	Gilbert	N/A	N/A
8210411	12/2011	Yates et al.	N/A	N/A
8211100	12/2011	Podhajsky et al.	N/A	N/A
8216223	12/2011	Wham et al.	N/A	N/A
8220688	12/2011	Laurent et al.	N/A	N/A
8221306	12/2011	Okada et al.	N/A	N/A
8221415	12/2011	Francischelli	N/A	N/A
8221418	12/2011	Prakash et al.	N/A	N/A
8226580	12/2011	Govari et al.	N/A	N/A

8226665	12/2011	Cohen	N/A	N/A
8226675	12/2011	Houser et al.	N/A	N/A
8231607	12/2011	Takuma	N/A	N/A
8235917	12/2011	Joseph et al.	N/A	N/A
8236018	12/2011	Yoshimine et al.	N/A	N/A
8236019	12/2011	Houser	N/A	N/A
8236020	12/2011	Smith et al.	N/A	N/A
8241235	12/2011	Kahler et al.	N/A	N/A
8241271	12/2011	Millman et al.	N/A	N/A
8241282	12/2011	Unger et al.	N/A	N/A
8241283	12/2011	Guerra et al.	N/A	N/A
8241284	12/2011	Dycus et al.	N/A	N/A
8241312	12/2011	Messerly	N/A	N/A
8246575	12/2011	Viola	N/A	N/A
8246615	12/2011	Behnke	N/A	N/A
8246616	12/2011	Amoah et al.	N/A	N/A
8246618	12/2011	Bucciaglia et al.	N/A	N/A
8246642	12/2011	Houser et al.	N/A	N/A
8251994	12/2011	McKenna et al.	N/A	N/A
8252012	12/2011	Stulen	N/A	N/A
8253303	12/2011	Giordano et al.	N/A	N/A
8257377	12/2011	Wiener et al.	N/A	N/A
8257387	12/2011	Cunningham	N/A	N/A
8262563	12/2011	Bakos et al.	N/A	N/A
8267300	12/2011	Boudreaux	N/A	N/A
8267935	12/2011	Couture et al.	N/A	N/A
8273087	12/2011	Kimura et al.	N/A	N/A
D669992	12/2011	Schafer et al.	N/A	N/A
D669993	12/2011	Merchant et al.	N/A	N/A
8277446	12/2011	Heard	N/A	N/A
8277447	12/2011	Garrison et al.	N/A	N/A
8277471	12/2011	Wiener et al.	N/A	N/A
8282581	12/2011	Zhao et al.	N/A	N/A
8282669	12/2011	Gerber et al.	N/A	N/A
8286846	12/2011	Smith et al.	N/A	N/A
8287485	12/2011	Kimura et al.	N/A	N/A
8287528	12/2011	Wham et al.	N/A	N/A
8287532	12/2011	Carroll et al.	N/A	N/A
8292886	12/2011	Kerr et al.	N/A	N/A
8292888	12/2011	Whitman	N/A	N/A
8292905	12/2011	Taylor et al.	N/A	N/A
8295902	12/2011	Salahieh et al.	N/A	N/A
8298223	12/2011	Wham et al.	N/A	N/A
8298225	12/2011	Gilbert	N/A	N/A
8298232	12/2011	Unger	N/A	N/A
8298233	12/2011	Mueller	N/A	N/A
8303576	12/2011	Brock	N/A	N/A
8303579	12/2011	Shibata	N/A	N/A
8303580	12/2011	Wham et al.	N/A	N/A
8303583	12/2011	Hosier et al.	N/A	N/A

8303613	12/2011	Crandall et al.	N/A	N/A
8306629	12/2011	Mioduski et al.	N/A	N/A
8308040	12/2011	Huang et al.	N/A	N/A
8308721	12/2011	Shibata et al.	N/A	N/A
8319400	12/2011	Houser et al.	N/A	N/A
8323302	12/2011	Robertson et al.	N/A	N/A
8323310	12/2011	Kingsley	N/A	N/A
8328061	12/2011	Kasvikis	N/A	N/A
8328761	12/2011	Widenhouse et al.	N/A	N/A
8328802	12/2011	Deville et al.	N/A	N/A
8328833	12/2011	Cuny	N/A	N/A
8328834	12/2011	Isaacs et al.	N/A	N/A
8333764	12/2011	Francischelli et al.	N/A	N/A
8333778	12/2011	Smith et al.	N/A	N/A
8333779	12/2011	Smith et al.	N/A	N/A
8334468	12/2011	Palmer et al.	N/A	N/A
8334635	12/2011	Voegele et al.	N/A	N/A
8337407	12/2011	Quistgaard et al.	N/A	N/A
8338726	12/2011	Palmer et al.	N/A	N/A
8343146	12/2012	Godara et al.	N/A	N/A
8344596	12/2012	Nield et al.	N/A	N/A
8348880	12/2012	Messerly et al.	N/A	N/A
8348947	12/2012	Takashino et al.	N/A	N/A
8348967	12/2012	Stulen	N/A	N/A
8353297	12/2012	Dacquay et al.	N/A	N/A
8357103	12/2012	Mark et al.	N/A	N/A
8357144	12/2012	Whitman et al.	N/A	N/A
8357149	12/2012	Govari et al.	N/A	N/A
8357158	12/2012	McKenna et al.	N/A	N/A
8360299	12/2012	Zemlok et al.	N/A	N/A
8361066	12/2012	Long et al.	N/A	N/A
8361072	12/2012	Dumbauld et al.	N/A	N/A
8361569	12/2012	Saito et al.	N/A	N/A
8366727	12/2012	Witt et al.	N/A	N/A
8372064	12/2012	Douglass et al.	N/A	N/A
8372099	12/2012	Deville et al.	N/A	N/A
8372101	12/2012	Smith et al.	N/A	N/A
8372102	12/2012	Stulen et al.	N/A	N/A
8374670	12/2012	Selkee	N/A	N/A
8377044	12/2012	Coe et al.	N/A	N/A
8377059	12/2012	Deville et al.	N/A	N/A
8377085	12/2012	Smith et al.	N/A	N/A
8382748	12/2012	Geisel	N/A	N/A
8382775	12/2012	Bender et al.	N/A	N/A
8382782	12/2012	Robertson et al.	N/A	N/A
8382792	12/2012	Chojin	N/A	N/A
8388646	12/2012	Chojin	N/A	N/A
8388647	12/2012	Nau, Jr. et al.	N/A	N/A
8393514	12/2012	Shelton, IV et al.	N/A	N/A
8394115	12/2012	Houser et al.	N/A	N/A

8397971	12/2012	Yates et al.	N/A	N/A
8398394	12/2012	Sauter et al.	N/A	N/A
8398674	12/2012	Prestel	N/A	N/A
8403926	12/2012	Nobis et al.	N/A	N/A
8403945	12/2012	Whitfield et al.	N/A	N/A
8403948	12/2012	Deville et al.	N/A	N/A
8403949	12/2012	Palmer et al.	N/A	N/A
8403950	12/2012	Palmer et al.	N/A	N/A
8409234	12/2012	Stahler et al.	N/A	N/A
8414577	12/2012	Boudreaux et al.	N/A	N/A
8418073	12/2012	Mohr et al.	N/A	N/A
8418349	12/2012	Smith et al.	N/A	N/A
8419757	12/2012	Smith et al.	N/A	N/A
8419758	12/2012	Smith et al.	N/A	N/A
8419759	12/2012	Dietz	N/A	N/A
8423182	12/2012	Robinson et al.	N/A	N/A
8425410	12/2012	Murray et al.	N/A	N/A
8425545	12/2012	Smith et al.	N/A	N/A
8430811	12/2012	Hess et al.	N/A	N/A
8430874	12/2012	Newton et al.	N/A	N/A
8430876	12/2012	Kappus et al.	N/A	N/A
8430897	12/2012	Novak et al.	N/A	N/A
8430898	12/2012	Wiener et al.	N/A	N/A
8435257	12/2012	Smith et al.	N/A	N/A
8437832	12/2012	Govari et al.	N/A	N/A
8439912	12/2012	Cunningham et al.	N/A	N/A
8439939	12/2012	Deville et al.	N/A	N/A
8444036	12/2012	Shelton, IV	N/A	N/A
8444637	12/2012	Podmore et al.	N/A	N/A
8444662	12/2012	Palmer et al.	N/A	N/A
8444663	12/2012	Houser et al.	N/A	N/A
8444664	12/2012	Balanev et al.	N/A	N/A
8453906	12/2012	Huang et al.	N/A	N/A
8454599	12/2012	Inagaki et al.	N/A	N/A
8454639	12/2012	Du et al.	N/A	N/A
8459525	12/2012	Yates et al.	N/A	N/A
8460284	12/2012	Aronow et al.	N/A	N/A
8460288	12/2012	Tamai et al.	N/A	N/A
8460292	12/2012	Truckai et al.	N/A	N/A
8461744	12/2012	Wiener et al.	N/A	N/A
8469981	12/2012	Robertson et al.	N/A	N/A
8471685	12/2012	Shingai	N/A	N/A
8479969	12/2012	Shelton, IV	N/A	N/A
8480703	12/2012	Nicholas et al.	N/A	N/A
8484833	12/2012	Cunningham et al.	N/A	N/A
8485413	12/2012	Scheib et al.	N/A	N/A
8485970	12/2012	Widenhouse et al.	N/A	N/A
8486057	12/2012	Behnke, II	N/A	N/A
8486096	12/2012	Robertson et al.	N/A	N/A
8491578	12/2012	Manwaring et al.	N/A	N/A

8491625	12/2012	Horner	N/A	N/A
8496682	12/2012	Guerra et al.	N/A	N/A
D687549	12/2012	Johnson et al.	N/A	N/A
8506555	12/2012	Ruiz Morales	N/A	N/A
8509318	12/2012	Tailliet	N/A	N/A
8512336	12/2012	Couture	N/A	N/A
8512337	12/2012	Francischelli et al.	N/A	N/A
8512359	12/2012	Whitman et al.	N/A	N/A
8512364	12/2012	Kowalski et al.	N/A	N/A
8512365	12/2012	Wiener et al.	N/A	N/A
8517239	12/2012	Scheib et al.	N/A	N/A
8518067	12/2012	Masuda et al.	N/A	N/A
8521331	12/2012	Itkowitz	N/A	N/A
8523043	12/2012	Ullrich et al.	N/A	N/A
8523882	12/2012	Huitema et al.	N/A	N/A
8523889	12/2012	Stulen et al.	N/A	N/A
8528563	12/2012	Gruber	N/A	N/A
8529437	12/2012	Taylor et al.	N/A	N/A
8529565	12/2012	Masuda et al.	N/A	N/A
8531064	12/2012	Robertson et al.	N/A	N/A
8535308	12/2012	Govari et al.	N/A	N/A
8535311	12/2012	Schall	N/A	N/A
8535340	12/2012	Allen	N/A	N/A
8535341	12/2012	Allen	N/A	N/A
8540128	12/2012	Shelton, IV et al.	N/A	N/A
8546996	12/2012	Messerly et al.	N/A	N/A
8546999	12/2012	Houser et al.	N/A	N/A
8551077	12/2012	Main et al.	N/A	N/A
8551086	12/2012	Kimura et al.	N/A	N/A
8556929	12/2012	Harper et al.	N/A	N/A
8561870	12/2012	Baxter, III et al.	N/A	N/A
8562592	12/2012	Conlon et al.	N/A	N/A
8562598	12/2012	Falkenstein et al.	N/A	N/A
8562600	12/2012	Kirkpatrick et al.	N/A	N/A
8562604	12/2012	Nishimura	N/A	N/A
8568390	12/2012	Mueller	N/A	N/A
8568397	12/2012	Horner et al.	N/A	N/A
8568400	12/2012	Gilbert	N/A	N/A
8568412	12/2012	Brandt et al.	N/A	N/A
8569997	12/2012	Lee	N/A	N/A
8573461	12/2012	Shelton, IV et al.	N/A	N/A
8573465	12/2012	Shelton, IV	N/A	N/A
8574231	12/2012	Boudreaux et al.	N/A	N/A
8574253	12/2012	Gruber et al.	N/A	N/A
8579176	12/2012	Smith et al.	N/A	N/A
8579897	12/2012	Vakharia et al.	N/A	N/A
8579928	12/2012	Robertson et al.	N/A	N/A
8579937	12/2012	Gresham	N/A	N/A
8585727	12/2012	Polo	N/A	N/A
8588371	12/2012	Ogawa et al.	N/A	N/A

8591459	12/2012	Clymer et al.	N/A	N/A
8591506	12/2012	Wham et al.	N/A	N/A
8591536	12/2012	Robertson	N/A	N/A
D695407	12/2012	Price et al.	N/A	N/A
D696631	12/2012	Price et al.	N/A	N/A
8596513	12/2012	Olson et al.	N/A	N/A
8597193	12/2012	Grunwald et al.	N/A	N/A
8597287	12/2012	Benamou et al.	N/A	N/A
8602031	12/2012	Reis et al.	N/A	N/A
8602288	12/2012	Shelton, IV et al.	N/A	N/A
8603085	12/2012	Jimenez	N/A	N/A
8603089	12/2012	Viola	N/A	N/A
8608044	12/2012	Hueil et al.	N/A	N/A
8608045	12/2012	Smith et al.	N/A	N/A
8608745	12/2012	Guzman et al.	N/A	N/A
8613383	12/2012	Beckman et al.	N/A	N/A
8616431	12/2012	Timm et al.	N/A	N/A
8617152	12/2012	Werneth et al.	N/A	N/A
8617194	12/2012	Beaupre	N/A	N/A
8622274	12/2013	Yates et al.	N/A	N/A
8623011	12/2013	Spivey	N/A	N/A
8623016	12/2013	Fischer	N/A	N/A
8623027	12/2013	Price et al.	N/A	N/A
8623040	12/2013	Artsyukhovich et al.	N/A	N/A
8623044	12/2013	Timm et al.	N/A	N/A
8628529	12/2013	Aldridge et al.	N/A	N/A
8628534	12/2013	Jones et al.	N/A	N/A
8632461	12/2013	Glossop	N/A	N/A
8636736	12/2013	Yates et al.	N/A	N/A
8638428	12/2013	Brown	N/A	N/A
8640788	12/2013	Dachs, II et al.	N/A	N/A
8641663	12/2013	Kirschenman et al.	N/A	N/A
8647350	12/2013	Mohan et al.	N/A	N/A
8650728	12/2013	Wan et al.	N/A	N/A
8652120	12/2013	Giordano et al.	N/A	N/A
8652132	12/2013	Tsuchiya et al.	N/A	N/A
8652155	12/2013	Houser et al.	N/A	N/A
8657489	12/2013	Ladurner et al.	N/A	N/A
8659208	12/2013	Rose et al.	N/A	N/A
8663214	12/2013	Weinberg et al.	N/A	N/A
8663220	12/2013	Wiener et al.	N/A	N/A
8663222	12/2013	Anderson et al.	N/A	N/A
8663223	12/2013	Masuda et al.	N/A	N/A
8663262	12/2013	Smith et al.	N/A	N/A
8668691	12/2013	Heard	N/A	N/A
8668710	12/2013	Slipszenko et al.	N/A	N/A
8684253	12/2013	Giordano et al.	N/A	N/A
8685016	12/2013	Wham et al.	N/A	N/A
8685020	12/2013	Weizman et al.	N/A	N/A

8690582	12/2013	Rohrbach et al.	N/A	N/A
8695866	12/2013	Leimbach et al.	N/A	N/A
8696366	12/2013	Chen et al.	N/A	N/A
8696665	12/2013	Hunt et al.	N/A	N/A
8696666	12/2013	Sanai et al.	N/A	N/A
8696917	12/2013	Petisce et al.	N/A	N/A
8702609	12/2013	Hadjicostis	N/A	N/A
8702702	12/2013	Edwards et al.	N/A	N/A
8702704	12/2013	Shelton, IV et al.	N/A	N/A
8704425	12/2013	Giordano et al.	N/A	N/A
8708213	12/2013	Shelton, IV et al.	N/A	N/A
8709008	12/2013	Willis et al.	N/A	N/A
8709031	12/2013	Stulen	N/A	N/A
8709035	12/2013	Johnson et al.	N/A	N/A
8715270	12/2013	Weitzner et al.	N/A	N/A
8715277	12/2013	Weizman	N/A	N/A
8721640	12/2013	Taylor et al.	N/A	N/A
8721657	12/2013	Kondoh et al.	N/A	N/A
8733613	12/2013	Huitema et al.	N/A	N/A
8733614	12/2013	Ross et al.	N/A	N/A
8734443	12/2013	Hixson et al.	N/A	N/A
8738110	12/2013	Tabada et al.	N/A	N/A
8747238	12/2013	Shelton, IV et al.	N/A	N/A
8747351	12/2013	Schultz	N/A	N/A
8747404	12/2013	Boudreaux et al.	N/A	N/A
8749116	12/2013	Messerly et al.	N/A	N/A
8752264	12/2013	Ackley et al.	N/A	N/A
8752749	12/2013	Moore et al.	N/A	N/A
8753338	12/2013	Widenhouse et al.	N/A	N/A
8754570	12/2013	Voegelé et al.	N/A	N/A
8758342	12/2013	Bales et al.	N/A	N/A
8758352	12/2013	Cooper et al.	N/A	N/A
8758391	12/2013	Swayze et al.	N/A	N/A
8764735	12/2013	Coe et al.	N/A	N/A
8764747	12/2013	Cummings et al.	N/A	N/A
8767970	12/2013	Eppolito	N/A	N/A
8770459	12/2013	Racenet et al.	N/A	N/A
8771269	12/2013	Sherman et al.	N/A	N/A
8771270	12/2013	Burbank	N/A	N/A
8771293	12/2013	Surti et al.	N/A	N/A
8773001	12/2013	Wiener et al.	N/A	N/A
8777944	12/2013	Frankhouser et al.	N/A	N/A
8777945	12/2013	Floume et al.	N/A	N/A
8779648	12/2013	Giordano et al.	N/A	N/A
8783541	12/2013	Shelton, IV et al.	N/A	N/A
8784415	12/2013	Malackowski et al.	N/A	N/A
8784418	12/2013	Romero	N/A	N/A
8790342	12/2013	Stulen et al.	N/A	N/A
8795274	12/2013	Hanna	N/A	N/A
8795275	12/2013	Hafner	N/A	N/A

8795276	12/2013	Dietz et al.	N/A	N/A
8795327	12/2013	Dietz et al.	N/A	N/A
8800838	12/2013	Shelton, IV	N/A	N/A
8801710	12/2013	Ullrich et al.	N/A	N/A
8801752	12/2013	Fortier et al.	N/A	N/A
8807414	12/2013	Ross et al.	N/A	N/A
8808204	12/2013	Irisawa et al.	N/A	N/A
8808319	12/2013	Houser et al.	N/A	N/A
8814856	12/2013	Elmouelhi et al.	N/A	N/A
8814870	12/2013	Paraschiv et al.	N/A	N/A
8820605	12/2013	Shelton, IV	N/A	N/A
8821388	12/2013	Naito et al.	N/A	N/A
8827992	12/2013	Koss et al.	N/A	N/A
8827995	12/2013	Schaller et al.	N/A	N/A
8831779	12/2013	Ortmaier et al.	N/A	N/A
8834466	12/2013	Cummings et al.	N/A	N/A
8834518	12/2013	Faller et al.	N/A	N/A
8844789	12/2013	Shelton, IV et al.	N/A	N/A
8845537	12/2013	Tanaka et al.	N/A	N/A
8845630	12/2013	Mehta et al.	N/A	N/A
8848808	12/2013	Dress	N/A	N/A
8851354	12/2013	Swensgard et al.	N/A	N/A
8852184	12/2013	Kucklick	N/A	N/A
8858547	12/2013	Brogna	N/A	N/A
8862955	12/2013	Cesari	N/A	N/A
8864749	12/2013	Okada	N/A	N/A
8864757	12/2013	Klimovitch et al.	N/A	N/A
8864761	12/2013	Johnson et al.	N/A	N/A
8870865	12/2013	Frankhouser et al.	N/A	N/A
8874220	12/2013	Draghici et al.	N/A	N/A
8876726	12/2013	Amit et al.	N/A	N/A
8876858	12/2013	Braun	N/A	N/A
8882766	12/2013	Couture et al.	N/A	N/A
8882791	12/2013	Stulen	N/A	N/A
8888776	12/2013	Dietz et al.	N/A	N/A
8888783	12/2013	Young	N/A	N/A
8888809	12/2013	Davison et al.	N/A	N/A
8899462	12/2013	Kostrzewski et al.	N/A	N/A
8900259	12/2013	Houser et al.	N/A	N/A
8906016	12/2013	Boudreaux et al.	N/A	N/A
8906017	12/2013	Rioux et al.	N/A	N/A
8911438	12/2013	Swoyer et al.	N/A	N/A
8911460	12/2013	Neurohr et al.	N/A	N/A
8920412	12/2013	Fritz et al.	N/A	N/A
8920414	12/2013	Stone et al.	N/A	N/A
8920421	12/2013	Rupp	N/A	N/A
8926607	12/2014	Norvell et al.	N/A	N/A
8926608	12/2014	Bacher et al.	N/A	N/A
8926620	12/2014	Chasmawala et al.	N/A	N/A
8931682	12/2014	Timm et al.	N/A	N/A

8932282	12/2014	Gilbert	N/A	N/A
8932299	12/2014	Bono et al.	N/A	N/A
8936614	12/2014	Allen, IV	N/A	N/A
8939974	12/2014	Boudreaux et al.	N/A	N/A
8945126	12/2014	Garrison et al.	N/A	N/A
8951248	12/2014	Messerly	439/39	A61B 18/1206
8951272	12/2014	Robertson et al.	N/A	N/A
8956349	12/2014	Aldridge et al.	N/A	N/A
8960520	12/2014	McCuen	N/A	N/A
8961515	12/2014	Twomey et al.	N/A	N/A
8961547	12/2014	Dietz et al.	N/A	N/A
8967443	12/2014	McCuen	N/A	N/A
8968283	12/2014	Kharin	N/A	N/A
8968294	12/2014	Maass et al.	N/A	N/A
8968296	12/2014	McPherson	N/A	N/A
8968355	12/2014	Malkowski et al.	N/A	N/A
8974447	12/2014	Kimball et al.	N/A	N/A
8974477	12/2014	Yamada	N/A	N/A
8974479	12/2014	Ross et al.	N/A	N/A
8974932	12/2014	McGahan et al.	N/A	N/A
8979843	12/2014	Timm et al.	N/A	N/A
8979844	12/2014	White et al.	N/A	N/A
8979890	12/2014	Boudreaux	N/A	N/A
8986287	12/2014	Park et al.	N/A	N/A
8986297	12/2014	Daniel et al.	N/A	N/A
8986302	12/2014	Aldridge et al.	N/A	N/A
8989855	12/2014	Murphy et al.	N/A	N/A
8989903	12/2014	Weir et al.	N/A	N/A
8991678	12/2014	Wellman et al.	N/A	N/A
8992422	12/2014	Spivey et al.	N/A	N/A
8992526	12/2014	Brodbeck et al.	N/A	N/A
8998891	12/2014	Garito et al.	N/A	N/A
9005199	12/2014	Beckman et al.	N/A	N/A
9011437	12/2014	Woodruff et al.	N/A	N/A
9011471	12/2014	Timm et al.	N/A	N/A
9017326	12/2014	DiNardo et al.	N/A	N/A
9017355	12/2014	Smith et al.	N/A	N/A
9017370	12/2014	Reschke et al.	N/A	N/A
9017372	12/2014	Artale et al.	N/A	N/A
9023035	12/2014	Allen, IV et al.	N/A	N/A
9023070	12/2014	Levine et al.	N/A	N/A
9023071	12/2014	Miller et al.	N/A	N/A
9028397	12/2014	Naito	N/A	N/A
9028476	12/2014	Bonn	N/A	N/A
9028478	12/2014	Mueller	N/A	N/A
9028481	12/2014	Behnke, II	N/A	N/A
9028494	12/2014	Shelton, IV et al.	N/A	N/A
9028519	12/2014	Yates et al.	N/A	N/A
9031667	12/2014	Williams	N/A	N/A

9033973	12/2014	Krapohl et al.	N/A	N/A
9035741	12/2014	Hamel et al.	N/A	N/A
9037259	12/2014	Mathur	N/A	N/A
9039690	12/2014	Kersten et al.	N/A	N/A
9039691	12/2014	Moua et al.	N/A	N/A
9039692	12/2014	Behnke, II et al.	N/A	N/A
9039695	12/2014	Giordano et al.	N/A	N/A
9039696	12/2014	Assmus et al.	N/A	N/A
9039705	12/2014	Takashino	N/A	N/A
9039731	12/2014	Joseph	N/A	N/A
9043018	12/2014	Mohr	N/A	N/A
9044227	12/2014	Shelton, IV et al.	N/A	N/A
9044230	12/2014	Morgan et al.	N/A	N/A
9044238	12/2014	Orszulak	N/A	N/A
9044243	12/2014	Johnson et al.	N/A	N/A
9044245	12/2014	Condie et al.	N/A	N/A
9044256	12/2014	Cadeddu et al.	N/A	N/A
9044261	12/2014	Houser	N/A	N/A
9050083	12/2014	Yates et al.	N/A	N/A
9050093	12/2014	Aldridge et al.	N/A	N/A
9050098	12/2014	Deville et al.	N/A	N/A
9050123	12/2014	Krause et al.	N/A	N/A
9050124	12/2014	Houser	N/A	N/A
9055961	12/2014	Manzo et al.	N/A	N/A
9059547	12/2014	McLawhorn	N/A	N/A
9060770	12/2014	Shelton, IV et al.	N/A	N/A
9060775	12/2014	Wiener et al.	N/A	N/A
9060776	12/2014	Yates et al.	N/A	N/A
9060778	12/2014	Condie et al.	N/A	N/A
9066720	12/2014	Ballakur et al.	N/A	N/A
9066723	12/2014	Beller et al.	N/A	N/A
9066747	12/2014	Robertson	N/A	N/A
9072523	12/2014	Houser et al.	N/A	N/A
9072535	12/2014	Shelton, IV et al.	N/A	N/A
9072536	12/2014	Shelton, IV et al.	N/A	N/A
9072538	12/2014	Suzuki et al.	N/A	N/A
9072539	12/2014	Messerly et al.	N/A	N/A
9084624	12/2014	Larkin et al.	N/A	N/A
9089327	12/2014	Worrell et al.	N/A	N/A
9089360	12/2014	Messerly et al.	N/A	N/A
9095333	12/2014	Konesky et al.	N/A	N/A
9095362	12/2014	Dachs, II et al.	N/A	N/A
9095367	12/2014	Olson et al.	N/A	N/A
9099863	12/2014	Smith et al.	N/A	N/A
9101358	12/2014	Kerr et al.	N/A	N/A
9101385	12/2014	Shelton, IV et al.	N/A	N/A
9107684	12/2014	Ma	N/A	N/A
9107689	12/2014	Robertson et al.	N/A	N/A
9107690	12/2014	Bales, Jr. et al.	N/A	N/A
9113900	12/2014	Buyse et al.	N/A	N/A

9113907	12/2014	Allen, IV et al.	N/A	N/A
9113940	12/2014	Twomey	N/A	N/A
9119657	12/2014	Shelton, IV et al.	N/A	N/A
9119957	12/2014	Gantz et al.	N/A	N/A
9125662	12/2014	Shelton, IV	N/A	N/A
9125667	12/2014	Stone et al.	N/A	N/A
9144453	12/2014	Rencher et al.	N/A	N/A
9147965	12/2014	Lee	N/A	N/A
9149324	12/2014	Huang et al.	N/A	N/A
9149325	12/2014	Worrell et al.	N/A	N/A
9161803	12/2014	Yates et al.	N/A	N/A
9165114	12/2014	Jain et al.	N/A	N/A
9168054	12/2014	Turner	N/A	A61N 7/00
9168085	12/2014	Juzkiw et al.	N/A	N/A
9168089	12/2014	Buyse et al.	N/A	N/A
9173656	12/2014	Schurr et al.	N/A	N/A
9179912	12/2014	Yates et al.	N/A	N/A
9186199	12/2014	Strauss et al.	N/A	N/A
9186204	12/2014	Nishimura et al.	N/A	N/A
9186796	12/2014	Ogawa	N/A	N/A
9192380	12/2014	(Tarinelli) Racenet et al.	N/A	N/A
9192421	12/2014	Garrison	N/A	N/A
9192428	12/2014	Houser et al.	N/A	N/A
9192431	12/2014	Woodruff et al.	N/A	N/A
9198714	12/2014	Worrell et al.	N/A	N/A
9198715	12/2014	Livneh	N/A	N/A
9198718	12/2014	Marczyk et al.	N/A	N/A
9198776	12/2014	Young	N/A	N/A
9204879	12/2014	Shelton, IV	N/A	N/A
9204891	12/2014	Weitzman	N/A	N/A
9204918	12/2014	Germain et al.	N/A	N/A
9204923	12/2014	Manzo et al.	N/A	N/A
9216050	12/2014	Condie et al.	N/A	N/A
9216051	12/2014	Fischer et al.	N/A	N/A
9216062	12/2014	Duque et al.	N/A	N/A
9220483	12/2014	Frankhouser et al.	N/A	N/A
9220527	12/2014	Houser et al.	N/A	N/A
9220559	12/2014	Worrell et al.	N/A	N/A
9226750	12/2015	Weir et al.	N/A	N/A
9226751	12/2015	Shelton, IV et al.	N/A	N/A
9226766	12/2015	Aldridge et al.	N/A	N/A
9226767	12/2015	Stulen et al.	N/A	N/A
9232979	12/2015	Parihar et al.	N/A	N/A
9237891	12/2015	Shelton, IV	N/A	N/A
9237921	12/2015	Messerly et al.	N/A	N/A
9241060	12/2015	Fujisaki	N/A	N/A
9241692	12/2015	Gunday et al.	N/A	N/A
9241728	12/2015	Price et al.	N/A	N/A
9241730	12/2015	Babaev	N/A	N/A

9241731	12/2015	Boudreaux et al.	N/A	N/A
9241768	12/2015	Sandhu et al.	N/A	N/A
9247953	12/2015	Palmer et al.	N/A	N/A
9254165	12/2015	Aronow et al.	N/A	N/A
9259234	12/2015	Robertson et al.	N/A	N/A
9259265	12/2015	Harris et al.	N/A	N/A
9265567	12/2015	Orban, III et al.	N/A	N/A
9265926	12/2015	Strobl et al.	N/A	N/A
9265973	12/2015	Akagane	N/A	N/A
9266310	12/2015	Krogdahl et al.	N/A	N/A
9277962	12/2015	Koss et al.	N/A	N/A
9282974	12/2015	Shelton, IV	N/A	N/A
9283027	12/2015	Monson et al.	N/A	N/A
9283045	12/2015	Rhee et al.	N/A	N/A
9283054	12/2015	Morgan et al.	N/A	N/A
9289256	12/2015	Shelton, IV et al.	N/A	N/A
9295514	12/2015	Shelton, IV et al.	N/A	N/A
9301759	12/2015	Spivey et al.	N/A	N/A
9305497	12/2015	Seo et al.	N/A	N/A
9307388	12/2015	Liang et al.	N/A	N/A
9307986	12/2015	Hall et al.	N/A	N/A
9308009	12/2015	Madan et al.	N/A	N/A
9308014	12/2015	Fischer	N/A	N/A
9314261	12/2015	Bales, Jr. et al.	N/A	N/A
9314292	12/2015	Trees et al.	N/A	N/A
9314301	12/2015	Ben-Haim et al.	N/A	N/A
9326754	12/2015	Polster	N/A	N/A
9326767	12/2015	Koch, Jr. et al.	N/A	N/A
9326787	12/2015	Sanai et al.	N/A	N/A
9326788	12/2015	Batross et al.	N/A	N/A
9332987	12/2015	Leimbach et al.	N/A	N/A
9333025	12/2015	Monson et al.	N/A	N/A
9333034	12/2015	Hancock	N/A	N/A
9339289	12/2015	Robertson	N/A	N/A
9339323	12/2015	Eder et al.	N/A	N/A
9339326	12/2015	McCullagh et al.	N/A	N/A
9345481	12/2015	Hall et al.	N/A	N/A
9345534	12/2015	Artale et al.	N/A	N/A
9345900	12/2015	Wu et al.	N/A	N/A
9351642	12/2015	Nadkarni et al.	N/A	N/A
9351726	12/2015	Leimbach et al.	N/A	N/A
9351727	12/2015	Leimbach et al.	N/A	N/A
9351754	12/2015	Vakharia et al.	N/A	N/A
9352173	12/2015	Yamada et al.	N/A	N/A
9358003	12/2015	Hall et al.	N/A	N/A
9358065	12/2015	Ladtkow et al.	N/A	N/A
9364171	12/2015	Harris et al.	N/A	N/A
9364230	12/2015	Shelton, IV et al.	N/A	N/A
9364279	12/2015	Houser et al.	N/A	N/A
9370364	12/2015	Smith et al.	N/A	N/A

9370400	12/2015	Parihar	N/A	N/A
9370611	12/2015	Ross et al.	N/A	N/A
9375206	12/2015	Vidal et al.	N/A	N/A
9375230	12/2015	Ross et al.	N/A	N/A
9375232	12/2015	Hunt et al.	N/A	N/A
9375256	12/2015	Cunningham et al.	N/A	N/A
9375264	12/2015	Horner et al.	N/A	N/A
9375267	12/2015	Kerr et al.	N/A	N/A
9385831	12/2015	Marr et al.	N/A	N/A
9386983	12/2015	Swensgard et al.	N/A	N/A
9393037	12/2015	Olson et al.	N/A	N/A
9393070	12/2015	Gelfand et al.	N/A	N/A
9398911	12/2015	Auld	N/A	N/A
9402680	12/2015	Ginnebaugh et al.	N/A	N/A
9402682	12/2015	Worrell et al.	N/A	N/A
9408606	12/2015	Shelton, IV	N/A	N/A
9408622	12/2015	Stulen et al.	N/A	N/A
9408660	12/2015	Strobl et al.	N/A	N/A
9414853	12/2015	Stulen et al.	N/A	N/A
9414880	12/2015	Monson et al.	N/A	N/A
9421014	12/2015	Ingmanson et al.	N/A	N/A
9421060	12/2015	Monson et al.	N/A	N/A
9427249	12/2015	Robertson et al.	N/A	N/A
9427279	12/2015	Muniz-Medina et al.	N/A	N/A
9439668	12/2015	Timm et al.	N/A	N/A
9439669	12/2015	Wiener et al.	N/A	N/A
9439671	12/2015	Akagane	N/A	N/A
9442288	12/2015	Tanimura	N/A	N/A
9445784	12/2015	O'Keeffe	N/A	N/A
9445832	12/2015	Wiener et al.	N/A	N/A
9451967	12/2015	Jordan et al.	N/A	N/A
9456863	12/2015	Moua	N/A	N/A
9456864	12/2015	Witt et al.	N/A	N/A
9468438	12/2015	Baber et al.	N/A	N/A
9468498	12/2015	Sigmon, Jr.	N/A	N/A
9474542	12/2015	Slipszenko et al.	N/A	N/A
9474568	12/2015	Akagane	N/A	N/A
9486236	12/2015	Price et al.	N/A	N/A
9492146	12/2015	Kostrzewski et al.	N/A	N/A
9492224	12/2015	Boudreaux et al.	N/A	N/A
9498245	12/2015	Voegelé et al.	N/A	N/A
9498275	12/2015	Wham et al.	N/A	N/A
9504483	12/2015	Houser et al.	N/A	N/A
9504520	12/2015	Worrell et al.	N/A	N/A
9504524	12/2015	Behnke, II	N/A	N/A
9504855	12/2015	Messerly et al.	N/A	N/A
9510850	12/2015	Robertson et al.	N/A	N/A
9510906	12/2015	Boudreaux et al.	N/A	N/A
9522029	12/2015	Yates et al.	N/A	N/A

9522032	12/2015	Behnke	N/A	N/A
9526564	12/2015	Rusin	N/A	N/A
9526565	12/2015	Strobl	N/A	N/A
9545253	12/2016	Worrell et al.	N/A	N/A
9545497	12/2016	Wenderow et al.	N/A	N/A
9554465	12/2016	Liu et al.	N/A	N/A
9554794	12/2016	Baber et al.	N/A	N/A
9554846	12/2016	Boudreaux	N/A	N/A
9554854	12/2016	Yates et al.	N/A	N/A
9560995	12/2016	Addison et al.	N/A	N/A
9561038	12/2016	Shelton, IV et al.	N/A	N/A
9572592	12/2016	Price et al.	N/A	N/A
9574644	12/2016	Parihar	N/A	N/A
9585714	12/2016	Livneh	N/A	N/A
9592056	12/2016	Mozdzierz et al.	N/A	N/A
9592072	12/2016	Akagane	N/A	N/A
9597143	12/2016	Madan et al.	N/A	N/A
9603669	12/2016	Govari et al.	N/A	N/A
9610091	12/2016	Johnson et al.	N/A	N/A
9610114	12/2016	Baxter, III et al.	N/A	N/A
9615877	12/2016	Tyrrell et al.	N/A	N/A
9623237	12/2016	Turner et al.	N/A	N/A
9629623	12/2016	Lytle, IV et al.	N/A	N/A
9629629	12/2016	Leimbach et al.	N/A	N/A
9632573	12/2016	Ogawa et al.	N/A	N/A
9636135	12/2016	Stulen	N/A	N/A
9636165	12/2016	Larson et al.	N/A	N/A
9636167	12/2016	Gregg	N/A	N/A
9638770	12/2016	Dietz et al.	N/A	N/A
9642644	12/2016	Houser et al.	N/A	N/A
9642669	12/2016	Takashino et al.	N/A	N/A
9643052	12/2016	Tchao et al.	N/A	N/A
9649110	12/2016	Parihar et al.	N/A	N/A
9649111	12/2016	Shelton, IV et al.	N/A	N/A
9649126	12/2016	Robertson et al.	N/A	N/A
9649173	12/2016	Choi et al.	N/A	N/A
9655670	12/2016	Larson et al.	N/A	N/A
9662131	12/2016	Omori et al.	N/A	N/A
9668806	12/2016	Unger et al.	N/A	N/A
9671860	12/2016	Ogawa et al.	N/A	N/A
9674949	12/2016	Liu et al.	N/A	N/A
9675374	12/2016	Stulen et al.	N/A	N/A
9675375	12/2016	Houser et al.	N/A	N/A
9681884	12/2016	Clem et al.	N/A	N/A
9687230	12/2016	Leimbach et al.	N/A	N/A
9687290	12/2016	Keller	N/A	N/A
9690362	12/2016	Leimbach et al.	N/A	N/A
9693817	12/2016	Mehta et al.	N/A	N/A
9700309	12/2016	Jaworek et al.	N/A	N/A
9700339	12/2016	Nield	N/A	N/A

9700343	12/2016	Messerly et al.	N/A	N/A
9705456	12/2016	Gilbert	N/A	N/A
9707004	12/2016	Houser et al.	N/A	N/A
9707027	12/2016	Ruddenklau et al.	N/A	N/A
9707030	12/2016	Davison et al.	N/A	N/A
9713507	12/2016	Stulen et al.	N/A	N/A
9717548	12/2016	Couture	N/A	N/A
9717552	12/2016	Cosman et al.	N/A	N/A
9724094	12/2016	Baber et al.	N/A	N/A
9724118	12/2016	Schulte et al.	N/A	N/A
9724120	12/2016	Faller et al.	N/A	N/A
9724152	12/2016	Horlle et al.	N/A	N/A
9730695	12/2016	Leimbach et al.	N/A	N/A
9733663	12/2016	Leimbach et al.	N/A	N/A
9737301	12/2016	Baber et al.	N/A	N/A
9737326	12/2016	Worrell et al.	N/A	N/A
9737355	12/2016	Yates et al.	N/A	N/A
9737358	12/2016	Beckman et al.	N/A	N/A
9743929	12/2016	Leimbach et al.	N/A	N/A
9743946	12/2016	Faller et al.	N/A	N/A
9743947	12/2016	Price et al.	N/A	N/A
9750499	12/2016	Leimbach et al.	N/A	N/A
9757128	12/2016	Baber et al.	N/A	N/A
9757142	12/2016	Shimizu	N/A	N/A
9757150	12/2016	Alexander et al.	N/A	N/A
9757186	12/2016	Boudreaux et al.	N/A	N/A
9764164	12/2016	Wiener et al.	N/A	N/A
9770285	12/2016	Zoran et al.	N/A	N/A
9782169	12/2016	Kimsey et al.	N/A	N/A
9782214	12/2016	Houser et al.	N/A	N/A
9788836	12/2016	Overmyer et al.	N/A	N/A
9788851	12/2016	Dannaher et al.	N/A	N/A
9795405	12/2016	Price et al.	N/A	N/A
9795436	12/2016	Yates et al.	N/A	N/A
9795808	12/2016	Messerly et al.	N/A	N/A
9801626	12/2016	Parihar et al.	N/A	N/A
9801648	12/2016	Houser et al.	N/A	N/A
9802033	12/2016	Hibner et al.	N/A	N/A
9804618	12/2016	Leimbach et al.	N/A	N/A
9808244	12/2016	Leimbach et al.	N/A	N/A
9808246	12/2016	Shelton, IV et al.	N/A	N/A
9808308	12/2016	Faller et al.	N/A	N/A
9814460	12/2016	Kimsey et al.	N/A	N/A
9814514	12/2016	Shelton, IV et al.	N/A	N/A
9815211	12/2016	Cao et al.	N/A	N/A
9820738	12/2016	Lytle, IV et al.	N/A	N/A
9820768	12/2016	Gee et al.	N/A	N/A
9820771	12/2016	Norton et al.	N/A	N/A
9820806	12/2016	Lee et al.	N/A	N/A
9826976	12/2016	Parihar et al.	N/A	N/A

9826977	12/2016	Leimbach et al.	N/A	N/A
9839443	12/2016	Brockman et al.	N/A	N/A
9844368	12/2016	Boudreaux et al.	N/A	N/A
9844374	12/2016	Lytle, IV et al.	N/A	N/A
9844375	12/2016	Overmyer et al.	N/A	N/A
9848901	12/2016	Robertson et al.	N/A	N/A
9848902	12/2016	Price et al.	N/A	N/A
9848937	12/2016	Trees et al.	N/A	N/A
9861381	12/2017	Johnson	N/A	N/A
9861428	12/2017	Trees et al.	N/A	N/A
9867612	12/2017	Parihar et al.	N/A	N/A
9867651	12/2017	Wham	N/A	N/A
9867670	12/2017	Brannan et al.	N/A	N/A
9872722	12/2017	Lech	N/A	N/A
9872725	12/2017	Worrell et al.	N/A	N/A
9872726	12/2017	Morisaki	N/A	N/A
9877720	12/2017	Worrell et al.	N/A	N/A
9877776	12/2017	Boudreaux	N/A	N/A
9877782	12/2017	Voegele et al.	N/A	N/A
9878184	12/2017	Beaupre	N/A	N/A
9883860	12/2017	Leimbach et al.	N/A	N/A
9883884	12/2017	Neurohr et al.	N/A	N/A
9888919	12/2017	Leimbach et al.	N/A	N/A
9888958	12/2017	Evans et al.	N/A	N/A
9895148	12/2017	Shelton, IV et al.	N/A	N/A
9895160	12/2017	Fan et al.	N/A	N/A
9901321	12/2017	Harks et al.	N/A	N/A
9901342	12/2017	Shelton, IV et al.	N/A	N/A
9901383	12/2017	Hassler, Jr.	N/A	N/A
9901754	12/2017	Yamada	N/A	N/A
9907563	12/2017	Germain et al.	N/A	N/A
9913642	12/2017	Leimbach et al.	N/A	N/A
9913656	12/2017	Stulen	N/A	N/A
9913680	12/2017	Voegele et al.	N/A	N/A
9918730	12/2017	Trees et al.	N/A	N/A
9924961	12/2017	Shelton, IV et al.	N/A	N/A
9925003	12/2017	Parihar et al.	N/A	N/A
9931118	12/2017	Shelton, IV et al.	N/A	N/A
9937001	12/2017	Nakamura	N/A	N/A
9943309	12/2017	Shelton, IV et al.	N/A	N/A
9949785	12/2017	Price et al.	N/A	N/A
9949788	12/2017	Boudreaux	N/A	N/A
9962182	12/2017	Dietz et al.	N/A	N/A
9968355	12/2017	Shelton, IV et al.	N/A	N/A
9974539	12/2017	Yates et al.	N/A	N/A
9987000	12/2017	Shelton, IV et al.	N/A	N/A
9987033	12/2017	Neurohr et al.	N/A	N/A
9993248	12/2017	Shelton, IV et al.	N/A	N/A
9993258	12/2017	Shelton, IV et al.	N/A	N/A
9993289	12/2017	Sobajima et al.	N/A	N/A

10004497	12/2017	Overmyer et al.	N/A	N/A
10004501	12/2017	Shelton, IV et al.	N/A	N/A
10004526	12/2017	Dycus et al.	N/A	N/A
10004527	12/2017	Gee et al.	N/A	N/A
D822206	12/2017	Shelton, IV et al.	N/A	N/A
10010339	12/2017	Witt et al.	N/A	N/A
10010341	12/2017	Houser et al.	N/A	N/A
10013049	12/2017	Leimbach et al.	N/A	N/A
10016199	12/2017	Baber et al.	N/A	N/A
10016207	12/2017	Suzuki et al.	N/A	N/A
10022142	12/2017	Aranyi et al.	N/A	N/A
10022567	12/2017	Messerly et al.	N/A	N/A
10022568	12/2017	Messerly et al.	N/A	N/A
10028761	12/2017	Leimbach et al.	N/A	N/A
10028786	12/2017	Mucilli et al.	N/A	N/A
10034684	12/2017	Weisenburgh, II et al.	N/A	N/A
10034704	12/2017	Asher et al.	N/A	N/A
D826405	12/2017	Shelton, IV et al.	N/A	N/A
10039588	12/2017	Harper et al.	N/A	N/A
10041822	12/2017	Zemlok	N/A	N/A
10045776	12/2017	Shelton, IV et al.	N/A	N/A
10045779	12/2017	Savage et al.	N/A	N/A
10045794	12/2017	Witt et al.	N/A	N/A
10045810	12/2017	Schall et al.	N/A	N/A
10045819	12/2017	Jensen et al.	N/A	N/A
10052044	12/2017	Shelton, IV et al.	N/A	N/A
10052102	12/2017	Baxter, III et al.	N/A	N/A
10070916	12/2017	Artale	N/A	N/A
10080609	12/2017	Hancock et al.	N/A	N/A
10085748	12/2017	Morgan et al.	N/A	N/A
10085762	12/2017	Timm et al.	N/A	N/A
10085792	12/2017	Johnson et al.	N/A	N/A
10092310	12/2017	Boudreaux et al.	N/A	N/A
10092344	12/2017	Mohr et al.	N/A	N/A
10092347	12/2017	Weisshaupt et al.	N/A	N/A
10092348	12/2017	Boudreaux	N/A	N/A
10092350	12/2017	Rothweiler et al.	N/A	N/A
10105140	12/2017	Malinouskas et al.	N/A	N/A
10111679	12/2017	Baber et al.	N/A	N/A
10111699	12/2017	Boudreaux	N/A	N/A
10111703	12/2017	Cosman, Jr. et al.	N/A	N/A
10117649	12/2017	Baxter, III et al.	N/A	N/A
10117667	12/2017	Robertson et al.	N/A	N/A
10117702	12/2017	Danziger et al.	N/A	N/A
10123835	12/2017	Keller et al.	N/A	N/A
10130367	12/2017	Cappola et al.	N/A	N/A
10130410	12/2017	Strobl et al.	N/A	N/A
10130412	12/2017	Wham	N/A	N/A
10135242	12/2017	Baber et al.	N/A	N/A

10136887	12/2017	Shelton, IV et al.	N/A	N/A
10149680	12/2017	Parihar et al.	N/A	N/A
10154848	12/2017	Chernov et al.	N/A	N/A
10154852	12/2017	Conlon et al.	N/A	N/A
10159483	12/2017	Beckman et al.	N/A	N/A
10159524	12/2017	Yates et al.	N/A	N/A
10166060	12/2018	Johnson et al.	N/A	N/A
10172665	12/2018	Heckel et al.	N/A	N/A
10172669	12/2018	Felder et al.	N/A	N/A
10178992	12/2018	Wise et al.	N/A	N/A
10179022	12/2018	Yates et al.	N/A	N/A
10180463	12/2018	Beckman et al.	N/A	N/A
10182816	12/2018	Shelton, IV et al.	N/A	N/A
10182818	12/2018	Hensel et al.	N/A	N/A
10188385	12/2018	Kerr et al.	N/A	N/A
10188455	12/2018	Hancock et al.	N/A	N/A
10194907	12/2018	Marczyk et al.	N/A	N/A
10194972	12/2018	Yates et al.	N/A	N/A
10194973	12/2018	Wiener et al.	N/A	N/A
10194976	12/2018	Boudreaux	N/A	N/A
10194977	12/2018	Yang	N/A	N/A
10194999	12/2018	Bacher et al.	N/A	N/A
10201364	12/2018	Leimbach et al.	N/A	N/A
10201365	12/2018	Boudreaux et al.	N/A	N/A
10201382	12/2018	Wiener et al.	N/A	N/A
10206676	12/2018	Shelton, IV	N/A	N/A
10226250	12/2018	Beckman et al.	N/A	N/A
10226273	12/2018	Messerly et al.	N/A	N/A
10231747	12/2018	Stulen et al.	N/A	N/A
10238385	12/2018	Yates et al.	N/A	N/A
10238391	12/2018	Leimbach et al.	N/A	N/A
10245027	12/2018	Shelton, IV et al.	N/A	N/A
10245028	12/2018	Shelton, IV et al.	N/A	N/A
10245029	12/2018	Hunter et al.	N/A	N/A
10245030	12/2018	Hunter et al.	N/A	N/A
10245033	12/2018	Overmyer et al.	N/A	N/A
10245095	12/2018	Boudreaux	N/A	N/A
10245097	12/2018	Honda et al.	N/A	N/A
10245104	12/2018	Mckenna et al.	N/A	N/A
10251664	12/2018	Shelton, IV et al.	N/A	N/A
10258331	12/2018	Shelton, IV et al.	N/A	N/A
10258505	12/2018	Ovchinnikov	N/A	N/A
10263171	12/2018	Wiener	N/A	H10N 30/802
10265068	12/2018	Harris et al.	N/A	N/A
10265117	12/2018	Wiener et al.	N/A	N/A
10265118	12/2018	Gerhardt	N/A	N/A
10271840	12/2018	Sapre	N/A	N/A
10271851	12/2018	Shelton, IV et al.	N/A	N/A
D847989	12/2018	Shelton, IV et al.	N/A	N/A
10278721	12/2018	Dietz et al.	N/A	N/A

10285705	12/2018	Shelton, IV et al.	N/A	N/A
10285724	12/2018	Faller et al.	N/A	N/A
10285750	12/2018	Coulson et al.	N/A	N/A
10292704	12/2018	Harris et al.	N/A	N/A
10299810	12/2018	Robertson et al.	N/A	N/A
10299821	12/2018	Shelton, IV et al.	N/A	N/A
D850617	12/2018	Shelton, IV et al.	N/A	N/A
D851762	12/2018	Shelton, IV et al.	N/A	N/A
10307159	12/2018	Harris et al.	N/A	N/A
10314579	12/2018	Chowaniec et al.	N/A	N/A
10314582	12/2018	Shelton, IV et al.	N/A	N/A
10314638	12/2018	Gee et al.	N/A	N/A
10321907	12/2018	Shelton, IV et al.	N/A	N/A
10321950	12/2018	Yates et al.	N/A	N/A
D854151	12/2018	Shelton, IV et al.	N/A	N/A
10335149	12/2018	Baxter, III et al.	N/A	N/A
10335182	12/2018	Stulen et al.	N/A	N/A
10335183	12/2018	Worrell et al.	N/A	N/A
10335614	12/2018	Messerly et al.	N/A	N/A
10342543	12/2018	Shelton, IV et al.	N/A	N/A
10342602	12/2018	Strobl et al.	N/A	N/A
10342606	12/2018	Cosman et al.	N/A	N/A
10342623	12/2018	Huelman et al.	N/A	N/A
10348941	12/2018	Elliot, Jr. et al.	N/A	N/A
10349999	12/2018	Yates et al.	N/A	N/A
10350016	12/2018	Burbank et al.	N/A	N/A
10350025	12/2018	Loyd et al.	N/A	N/A
10357246	12/2018	Shelton, IV et al.	N/A	N/A
10357247	12/2018	Shelton, IV et al.	N/A	N/A
10357303	12/2018	Conlon et al.	N/A	N/A
10363084	12/2018	Friedrichs	N/A	N/A
10368861	12/2018	Baxter, III et al.	N/A	N/A
10368865	12/2018	Harris et al.	N/A	N/A
10376263	12/2018	Morgan et al.	N/A	N/A
10376305	12/2018	Yates et al.	N/A	N/A
10390841	12/2018	Shelton, IV et al.	N/A	N/A
10398439	12/2018	Cabrera et al.	N/A	N/A
10398466	12/2018	Stulen et al.	N/A	N/A
10398497	12/2018	Batross et al.	N/A	N/A
10405857	12/2018	Shelton, IV et al.	N/A	N/A
10405863	12/2018	Wise et al.	N/A	N/A
10413291	12/2018	Worthington et al.	N/A	N/A
10413293	12/2018	Shelton, IV et al.	N/A	N/A
10413297	12/2018	Harris et al.	N/A	N/A
10413352	12/2018	Thomas et al.	N/A	N/A
10413353	12/2018	Kerr et al.	N/A	N/A
10420552	12/2018	Shelton, IV et al.	N/A	N/A
10420579	12/2018	Wiener et al.	N/A	N/A
10420607	12/2018	Woloszko et al.	N/A	N/A
D865175	12/2018	Widenhouse et al.	N/A	N/A

10426471	12/2018	Shelton, IV et al.	N/A	N/A
10426507	12/2018	Wiener et al.	N/A	N/A
10426546	12/2018	Graham et al.	N/A	N/A
10426978	12/2018	Akagane	N/A	N/A
10433837	12/2018	Worthington et al.	N/A	N/A
10433849	12/2018	Shelton, IV et al.	N/A	N/A
10433865	12/2018	Witt et al.	N/A	N/A
10433866	12/2018	Witt et al.	N/A	N/A
10433900	12/2018	Harris et al.	N/A	N/A
10441279	12/2018	Shelton, IV et al.	N/A	N/A
10441308	12/2018	Robertson	N/A	N/A
10441310	12/2018	Olson et al.	N/A	N/A
10441345	12/2018	Aldridge et al.	N/A	N/A
10448948	12/2018	Shelton, IV et al.	N/A	N/A
10448950	12/2018	Shelton, IV et al.	N/A	N/A
10448986	12/2018	Zikorus et al.	N/A	N/A
10456140	12/2018	Shelton, IV et al.	N/A	N/A
10456193	12/2018	Yates et al.	N/A	N/A
10463421	12/2018	Boudreaux et al.	N/A	N/A
10463887	12/2018	Witt et al.	N/A	N/A
10470762	12/2018	Leimbach et al.	N/A	N/A
10470764	12/2018	Baxter, III et al.	N/A	N/A
10478182	12/2018	Taylor	N/A	N/A
10478190	12/2018	Miller et al.	N/A	N/A
10485542	12/2018	Shelton, IV et al.	N/A	N/A
10485543	12/2018	Shelton, IV et al.	N/A	N/A
10485607	12/2018	Strobl et al.	N/A	N/A
D869655	12/2018	Shelton, IV et al.	N/A	N/A
10492785	12/2018	Overmyer et al.	N/A	N/A
10492849	12/2018	Juergens et al.	N/A	N/A
10499914	12/2018	Huang et al.	N/A	N/A
10507033	12/2018	Dickerson et al.	N/A	N/A
10512795	12/2018	Voegele et al.	N/A	N/A
10517595	12/2018	Hunter et al.	N/A	N/A
10517596	12/2018	Hunter et al.	N/A	N/A
10517627	12/2018	Timm et al.	N/A	N/A
10524787	12/2019	Shelton, IV et al.	N/A	N/A
10524789	12/2019	Swayze et al.	N/A	N/A
10524854	12/2019	Woodruff et al.	N/A	N/A
10524872	12/2019	Stewart et al.	N/A	N/A
10531874	12/2019	Morgan et al.	N/A	N/A
10537324	12/2019	Shelton, IV et al.	N/A	N/A
10537325	12/2019	Bakos et al.	N/A	N/A
10537351	12/2019	Shelton, IV et al.	N/A	N/A
10542979	12/2019	Shelton, IV et al.	N/A	N/A
10542982	12/2019	Beckman et al.	N/A	N/A
10542991	12/2019	Shelton, IV et al.	N/A	N/A
10543008	12/2019	Vakharia et al.	N/A	N/A
10548504	12/2019	Shelton, IV et al.	N/A	N/A
10548655	12/2019	Scheib et al.	N/A	N/A

10555769	12/2019	Worrell et al.	N/A	N/A
10561560	12/2019	Boutoussov et al.	N/A	N/A
10568624	12/2019	Shelton, IV et al.	N/A	N/A
10568625	12/2019	Harris et al.	N/A	N/A
10568626	12/2019	Shelton, IV et al.	N/A	N/A
10568632	12/2019	Miller et al.	N/A	N/A
10575892	12/2019	Danziger et al.	N/A	N/A
10582928	12/2019	Hunter et al.	N/A	N/A
10588625	12/2019	Weaner et al.	N/A	N/A
10588630	12/2019	Shelton, IV et al.	N/A	N/A
10588631	12/2019	Shelton, IV et al.	N/A	N/A
10588632	12/2019	Shelton, IV et al.	N/A	N/A
10588633	12/2019	Shelton, IV et al.	N/A	N/A
10595929	12/2019	Boudreaux et al.	N/A	N/A
10595930	12/2019	Scheib et al.	N/A	N/A
10603036	12/2019	Hunter et al.	N/A	N/A
10610224	12/2019	Shelton, IV et al.	N/A	N/A
10610286	12/2019	Wiener et al.	N/A	N/A
10610313	12/2019	Bailey et al.	N/A	N/A
10617412	12/2019	Shelton, IV et al.	N/A	N/A
10617420	12/2019	Shelton, IV et al.	N/A	N/A
10617464	12/2019	Duppuis	N/A	N/A
10624635	12/2019	Harris et al.	N/A	N/A
10624691	12/2019	Wiener et al.	N/A	N/A
10631858	12/2019	Burbank	N/A	N/A
10631859	12/2019	Shelton, IV et al.	N/A	N/A
10631928	12/2019	Basu et al.	N/A	N/A
10632630	12/2019	Cao et al.	N/A	N/A
RE47996	12/2019	Turner et al.	N/A	N/A
10639034	12/2019	Harris et al.	N/A	N/A
10639035	12/2019	Shelton, IV et al.	N/A	N/A
10639037	12/2019	Shelton, IV et al.	N/A	N/A
10639092	12/2019	Corbett et al.	N/A	N/A
10639098	12/2019	Cosman et al.	N/A	N/A
10646269	12/2019	Worrell et al.	N/A	N/A
10646292	12/2019	Solomon et al.	N/A	N/A
10653413	12/2019	Worthington et al.	N/A	N/A
10660692	12/2019	Lesko et al.	N/A	N/A
10667809	12/2019	Bakos et al.	N/A	N/A
10667810	12/2019	Shelton, IV et al.	N/A	N/A
10667811	12/2019	Harris et al.	N/A	N/A
10675021	12/2019	Harris et al.	N/A	N/A
10675024	12/2019	Shelton, IV et al.	N/A	N/A
10675025	12/2019	Swayze et al.	N/A	N/A
10675026	12/2019	Harris et al.	N/A	N/A
10677764	12/2019	Ross et al.	N/A	N/A
10682136	12/2019	Harris et al.	N/A	N/A
10682138	12/2019	Shelton, IV et al.	N/A	N/A
10687806	12/2019	Shelton, IV et al.	N/A	N/A
10687809	12/2019	Shelton, IV et al.	N/A	N/A

10687810	12/2019	Shelton, IV et al.	N/A	N/A
10687884	12/2019	Wiener et al.	N/A	N/A
10688321	12/2019	Wiener et al.	N/A	N/A
10695055	12/2019	Shelton, IV et al.	N/A	N/A
10695057	12/2019	Shelton, IV et al.	N/A	N/A
10695058	12/2019	Lytle, IV et al.	N/A	N/A
10695119	12/2019	Smith	N/A	N/A
10702270	12/2019	Shelton, IV et al.	N/A	N/A
10702329	12/2019	Strobl et al.	N/A	N/A
10709446	12/2019	Harris et al.	N/A	N/A
10709469	12/2019	Shelton, IV et al.	N/A	N/A
10709906	12/2019	Nield	N/A	N/A
10716615	12/2019	Shelton, IV et al.	N/A	N/A
10722233	12/2019	Wellman	N/A	N/A
D893717	12/2019	Messerly et al.	N/A	N/A
10729458	12/2019	Stoddard et al.	N/A	N/A
10729494	12/2019	Parihar et al.	N/A	N/A
10736629	12/2019	Shelton, IV et al.	N/A	N/A
10736685	12/2019	Wiener et al.	N/A	N/A
10751108	12/2019	Yates et al.	N/A	N/A
10751138	12/2019	Giordano et al.	N/A	N/A
10758229	12/2019	Shelton, IV et al.	N/A	N/A
10758230	12/2019	Shelton, IV et al.	N/A	N/A
10758232	12/2019	Shelton, IV et al.	N/A	N/A
10758294	12/2019	Jones	N/A	N/A
10765427	12/2019	Shelton, IV et al.	N/A	N/A
10765470	12/2019	Yates et al.	N/A	N/A
10772629	12/2019	Shelton, IV et al.	N/A	N/A
10772630	12/2019	Wixey	N/A	N/A
10779821	12/2019	Harris et al.	N/A	N/A
10779823	12/2019	Shelton, IV et al.	N/A	N/A
10779824	12/2019	Shelton, IV et al.	N/A	N/A
10779825	12/2019	Shelton, IV et al.	N/A	N/A
10779845	12/2019	Timm et al.	N/A	N/A
10779849	12/2019	Shelton, IV et al.	N/A	N/A
10779879	12/2019	Yates et al.	N/A	N/A
10786253	12/2019	Shelton, IV et al.	N/A	N/A
10786276	12/2019	Hirai et al.	N/A	N/A
10806454	12/2019	Kopp	N/A	N/A
10813638	12/2019	Shelton, IV et al.	N/A	N/A
10820938	12/2019	Fischer et al.	N/A	N/A
10828032	12/2019	Leimbach et al.	N/A	N/A
10828058	12/2019	Shelton, IV et al.	N/A	N/A
10835245	12/2019	Swayze et al.	N/A	N/A
10835246	12/2019	Shelton, IV et al.	N/A	N/A
10835247	12/2019	Shelton, IV et al.	N/A	N/A
10835307	12/2019	Shelton, IV et al.	N/A	N/A
10842492	12/2019	Shelton, IV et al.	N/A	N/A
10842523	12/2019	Shelton, IV et al.	N/A	N/A
10842563	12/2019	Gilbert et al.	N/A	N/A

D906355	12/2019	Messerly et al.	N/A	N/A
10856867	12/2019	Shelton, IV et al.	N/A	N/A
10856868	12/2019	Shelton, IV et al.	N/A	N/A
10856869	12/2019	Shelton, IV et al.	N/A	N/A
10856870	12/2019	Harris et al.	N/A	N/A
10856896	12/2019	Eichmann et al.	N/A	N/A
10856929	12/2019	Yates et al.	N/A	N/A
10856934	12/2019	Trees et al.	N/A	N/A
10874465	12/2019	Weir et al.	N/A	N/A
D908216	12/2020	Messerly et al.	N/A	N/A
10881399	12/2020	Shelton, IV et al.	N/A	N/A
10881401	12/2020	Baber et al.	N/A	N/A
10881409	12/2020	Cabrera	N/A	N/A
10881449	12/2020	Boudreaux et al.	N/A	N/A
10888322	12/2020	Morgan et al.	N/A	N/A
10888347	12/2020	Witt et al.	N/A	N/A
10893863	12/2020	Shelton, IV et al.	N/A	N/A
10893864	12/2020	Harris et al.	N/A	N/A
10893883	12/2020	Dannaher	N/A	N/A
10898186	12/2020	Bakos et al.	N/A	N/A
10898256	12/2020	Yates et al.	N/A	N/A
10912559	12/2020	Harris et al.	N/A	N/A
10912580	12/2020	Green et al.	N/A	N/A
10912603	12/2020	Boudreaux et al.	N/A	N/A
10918385	12/2020	Overmyer et al.	N/A	N/A
10925659	12/2020	Shelton, IV et al.	N/A	N/A
10926022	12/2020	Hickey et al.	N/A	N/A
D914878	12/2020	Shelton, IV et al.	N/A	N/A
10932766	12/2020	Tesar et al.	N/A	N/A
10932847	12/2020	Yates et al.	N/A	N/A
10945727	12/2020	Shelton, IV et al.	N/A	N/A
10952788	12/2020	Asher et al.	N/A	N/A
10959727	12/2020	Hunter et al.	N/A	N/A
10966741	12/2020	Illizaliturri-Sanchez et al.	N/A	N/A
10966747	12/2020	Worrell et al.	N/A	N/A
10973516	12/2020	Shelton, IV et al.	N/A	N/A
10973517	12/2020	Wixey	N/A	N/A
10973520	12/2020	Shelton, IV et al.	N/A	N/A
10980536	12/2020	Weaner et al.	N/A	N/A
10987105	12/2020	Cappola et al.	N/A	N/A
10987123	12/2020	Weir et al.	N/A	N/A
10987156	12/2020	Trees et al.	N/A	N/A
10993715	12/2020	Shelton, IV et al.	N/A	N/A
10993716	12/2020	Shelton, IV et al.	N/A	N/A
10993763	12/2020	Batross et al.	N/A	N/A
11000278	12/2020	Shelton, IV et al.	N/A	N/A
11000279	12/2020	Shelton, IV et al.	N/A	N/A
11020114	12/2020	Shelton, IV et al.	N/A	N/A
11020140	12/2020	Gee et al.	N/A	N/A

11033322	12/2020	Wiener et al.	N/A	N/A
11039834	12/2020	Harris et al.	N/A	N/A
11045191	12/2020	Shelton, IV et al.	N/A	N/A
11045192	12/2020	Harris et al.	N/A	N/A
11045275	12/2020	Boudreaux et al.	N/A	N/A
11051840	12/2020	Shelton, IV et al.	N/A	N/A
11051873	12/2020	Wiener et al.	N/A	N/A
11058424	12/2020	Shelton, IV et al.	N/A	N/A
11058447	12/2020	Houser	N/A	N/A
11058448	12/2020	Shelton, IV et al.	N/A	N/A
11058475	12/2020	Wiener et al.	N/A	N/A
11064997	12/2020	Shelton, IV et al.	N/A	N/A
11065048	12/2020	Messerly et al.	N/A	N/A
11083455	12/2020	Shelton, IV et al.	N/A	N/A
11083458	12/2020	Harris et al.	N/A	N/A
11090048	12/2020	Fanelli et al.	N/A	N/A
11090049	12/2020	Bakos et al.	N/A	N/A
11090104	12/2020	Wiener	N/A	A61B 17/320092
11096688	12/2020	Shelton, IV et al.	N/A	N/A
11096752	12/2020	Stulen et al.	N/A	N/A
11109866	12/2020	Shelton, IV et al.	N/A	N/A
11129611	12/2020	Shelton, IV et al.	N/A	N/A
11129666	12/2020	Messerly et al.	N/A	N/A
11129669	12/2020	Stulen et al.	N/A	N/A
11129670	12/2020	Shelton, IV et al.	N/A	N/A
11134942	12/2020	Harris et al.	N/A	N/A
11134978	12/2020	Shelton, IV et al.	N/A	N/A
11141154	12/2020	Shelton, IV et al.	N/A	N/A
11141213	12/2020	Yates et al.	N/A	N/A
11147551	12/2020	Shelton, IV	N/A	N/A
11147553	12/2020	Shelton, IV	N/A	N/A
11160551	12/2020	Shelton, IV et al.	N/A	N/A
11166716	12/2020	Shelton, IV et al.	N/A	N/A
11172929	12/2020	Shelton, IV	N/A	N/A
11179155	12/2020	Shelton, IV et al.	N/A	N/A
11179173	12/2020	Price et al.	N/A	N/A
11191539	12/2020	Overmyer et al.	N/A	N/A
11191540	12/2020	Aronhalt et al.	N/A	N/A
11197668	12/2020	Shelton, IV et al.	N/A	N/A
11202670	12/2020	Worrell et al.	N/A	N/A
11207065	12/2020	Harris et al.	N/A	N/A
11207067	12/2020	Shelton, IV et al.	N/A	N/A
11213293	12/2021	Worthington et al.	N/A	N/A
11213294	12/2021	Shelton, IV et al.	N/A	N/A
11219453	12/2021	Shelton, IV et al.	N/A	N/A
11224426	12/2021	Shelton, IV et al.	N/A	N/A
11224497	12/2021	Shelton, IV et al.	N/A	N/A
11229437	12/2021	Shelton, IV et al.	N/A	N/A
11229450	12/2021	Shelton, IV et al.	N/A	N/A

11229471	12/2021	Shelton, IV et al.	N/A	N/A
11229472	12/2021	Shelton, IV et al.	N/A	N/A
11234698	12/2021	Shelton, IV et al.	N/A	N/A
11241235	12/2021	Shelton, IV et al.	N/A	N/A
11246592	12/2021	Shelton, IV et al.	N/A	N/A
11246625	12/2021	Kane et al.	N/A	N/A
11246678	12/2021	Shelton, IV et al.	N/A	N/A
11253256	12/2021	Harris et al.	N/A	N/A
11259803	12/2021	Shelton, IV et al.	N/A	N/A
11259805	12/2021	Shelton, IV et al.	N/A	N/A
11259806	12/2021	Shelton, IV et al.	N/A	N/A
11259807	12/2021	Shelton, IV et al.	N/A	N/A
11266405	12/2021	Shelton, IV et al.	N/A	N/A
11266430	12/2021	Clauda et al.	N/A	N/A
11272931	12/2021	Boudreaux et al.	N/A	N/A
11278280	12/2021	Shelton, IV et al.	N/A	N/A
11284890	12/2021	Nalagatla et al.	N/A	N/A
11291440	12/2021	Harris et al.	N/A	N/A
11291444	12/2021	Boudreaux et al.	N/A	N/A
11291445	12/2021	Shelton, IV et al.	N/A	N/A
11291447	12/2021	Shelton, IV et al.	N/A	N/A
11291451	12/2021	Shelton, IV	N/A	N/A
11298127	12/2021	Shelton, IV	N/A	N/A
11298129	12/2021	Bakos et al.	N/A	N/A
11298130	12/2021	Bakos et al.	N/A	N/A
11304695	12/2021	Shelton, IV et al.	N/A	N/A
11304696	12/2021	Shelton, IV et al.	N/A	N/A
11304699	12/2021	Shelton, IV et al.	N/A	N/A
11311306	12/2021	Shelton, IV et al.	N/A	N/A
11311326	12/2021	Boudreaux	N/A	N/A
11311342	12/2021	Parihar et al.	N/A	N/A
D950728	12/2021	Bakos et al.	N/A	N/A
D952144	12/2021	Boudreaux	N/A	N/A
11317915	12/2021	Boudreaux et al.	N/A	N/A
11324503	12/2021	Shelton, IV et al.	N/A	N/A
11324527	12/2021	Aldridge et al.	N/A	N/A
11324557	12/2021	Shelton, IV et al.	N/A	N/A
11331100	12/2021	Boudreaux et al.	N/A	N/A
11331101	12/2021	Harris et al.	N/A	N/A
11337747	12/2021	Voegelé et al.	N/A	N/A
11344362	12/2021	Yates et al.	N/A	N/A
11350938	12/2021	Shelton, IV et al.	N/A	N/A
11357503	12/2021	Bakos et al.	N/A	N/A
11361176	12/2021	Shelton, IV et al.	N/A	N/A
11369377	12/2021	Boudreaux et al.	N/A	N/A
11376098	12/2021	Shelton, IV et al.	N/A	N/A
11382642	12/2021	Robertson et al.	N/A	N/A
11389161	12/2021	Shelton, IV et al.	N/A	N/A
11389164	12/2021	Yates et al.	N/A	N/A
11399837	12/2021	Shelton, IV et al.	N/A	N/A

11399855	12/2021	Boudreaux et al.	N/A	N/A
11406382	12/2021	Shelton, IV et al.	N/A	N/A
11406386	12/2021	Baber et al.	N/A	N/A
11413060	12/2021	Faller et al.	N/A	N/A
11419606	12/2021	Overmyer et al.	N/A	N/A
11419626	12/2021	Timm et al.	N/A	N/A
11424027	12/2021	Shelton, IV	N/A	N/A
11426167	12/2021	Shelton, IV et al.	N/A	N/A
11426191	12/2021	Vakharia et al.	N/A	N/A
D964564	12/2021	Boudreaux	N/A	N/A
11446029	12/2021	Shelton, IV et al.	N/A	N/A
11452525	12/2021	Shelton, IV et al.	N/A	N/A
11464511	12/2021	Timm et al.	N/A	N/A
11464512	12/2021	Shelton, IV et al.	N/A	N/A
11464601	12/2021	Shelton, IV et al.	N/A	N/A
11471155	12/2021	Shelton, IV et al.	N/A	N/A
11471156	12/2021	Shelton, IV et al.	N/A	N/A
11471206	12/2021	Henderson et al.	N/A	N/A
11471209	12/2021	Yates et al.	N/A	N/A
11478242	12/2021	Shelton, IV et al.	N/A	N/A
11484310	12/2021	Shelton, IV et al.	N/A	N/A
11497547	12/2021	McKenna et al.	N/A	N/A
11504122	12/2021	Shelton, IV et al.	N/A	N/A
11517309	12/2021	Bakos et al.	N/A	N/A
11529137	12/2021	Shelton, IV et al.	N/A	N/A
11529139	12/2021	Shelton, IV et al.	N/A	N/A
11553971	12/2022	Shelton, IV et al.	N/A	N/A
11559304	12/2022	Boudreaux et al.	N/A	N/A
11559307	12/2022	Shelton, IV et al.	N/A	N/A
11559308	12/2022	Yates et al.	N/A	N/A
11559347	12/2022	Wiener et al.	N/A	N/A
11571210	12/2022	Shelton, IV et al.	N/A	N/A
11576672	12/2022	Shelton, IV et al.	N/A	N/A
11576677	12/2022	Shelton, IV et al.	N/A	N/A
11583306	12/2022	Olson et al.	N/A	N/A
11589865	12/2022	Shelton, IV et al.	N/A	N/A
11589888	12/2022	Shelton, IV et al.	N/A	N/A
11589916	12/2022	Shelton, IV et al.	N/A	N/A
11607219	12/2022	Shelton, IV et al.	N/A	N/A
11638587	12/2022	Shelton, IV et al.	N/A	N/A
11653920	12/2022	Shelton, IV et al.	N/A	N/A
11659023	12/2022	Shelton, IV et al.	N/A	N/A
11660089	12/2022	Shelton, IV et al.	N/A	N/A
11660163	12/2022	Shelton, IV et al.	N/A	N/A
11666368	12/2022	Henderson et al.	N/A	N/A
11666375	12/2022	Scheib et al.	N/A	N/A
11678880	12/2022	Shelton, IV et al.	N/A	N/A
11678925	12/2022	Henderson et al.	N/A	N/A
11684402	12/2022	Shelton, IV et al.	N/A	N/A
11684412	12/2022	Shelton, IV et al.	N/A	N/A

11684434	12/2022	Shelton, IV	N/A	N/A
11696759	12/2022	Shelton, IV et al.	N/A	N/A
11696761	12/2022	Shelton, IV	N/A	N/A
11696776	12/2022	Shelton, IV et al.	N/A	N/A
11701111	12/2022	Shelton, IV et al.	N/A	N/A
11707318	12/2022	Shelton, IV et al.	N/A	N/A
11717311	12/2022	Weir et al.	N/A	N/A
11717706	12/2022	Wiener et al.	N/A	N/A
11723716	12/2022	Fiebig et al.	N/A	N/A
11744636	12/2022	Shelton, IV et al.	N/A	N/A
11751872	12/2022	Zeiner et al.	N/A	N/A
11751929	12/2022	Shelton, IV et al.	N/A	N/A
11759251	12/2022	Shelton, IV et al.	N/A	N/A
11766287	12/2022	Wiener et al.	N/A	N/A
11771419	12/2022	Shelton, IV et al.	N/A	N/A
11779329	12/2022	Shelton, IV et al.	N/A	N/A
11779387	12/2022	Salguero et al.	N/A	N/A
11786291	12/2022	Sarley et al.	N/A	N/A
11786294	12/2022	Shelton, IV et al.	N/A	N/A
11812957	12/2022	Shelton, IV et al.	N/A	N/A
11853835	12/2022	Shelton, IV	N/A	N/A
11857152	12/2023	Shelton, IV et al.	N/A	N/A
11871901	12/2023	Shelton, IV et al.	N/A	N/A
11871955	12/2023	Stulen et al.	N/A	N/A
11871982	12/2023	Wiener	N/A	H10N 30/802
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2002/0002377	12/2001	Cimino	N/A	N/A
2002/0002380	12/2001	Bishop	N/A	N/A
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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application claiming priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 17/370,128, entitled SURGICAL GENERATOR FOR ULTRASONIC AND ELECTROSURGICAL DEVICES, filed Jul. 8, 2021, which issued on Jan. 16, 2024 as U.S. Pat. No. 11,871,982, which is a continuation application claiming priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 16/248,160, entitled SURGICAL GENERATOR FOR ULTRASONIC AND ELECTROSURGICAL DEVICES, filed Jan. 15, 2019, which issued on Aug. 17, 2021 as U.S. Pat. No. 11,090,104, which is a continuation application claiming priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 14/657,876, entitled SURGICAL GENERATOR FOR ULTRASONIC AND ELECTROSURGICAL DEVICES, filed Mar. 13, 2015, which issued on Apr. 16, 2019 as U.S. Pat. No. 10,263,171, which is a continuation application claiming priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 13/448,175, entitled SURGICAL GENERATOR FOR ULTRASONIC AND ELECTROSURGICAL DEVICES, filed Apr. 16, 2012, which issued on Oct. 27, 2015 as U.S. Pat. No. 9,168,054, which is a continuation-in-part application claiming priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 13/251,766 entitled SURGICAL GENERATOR FOR ULTRASONIC AND ELECTROSURGICAL DEVICES,

filed Oct. 3, 2011, which issued on Oct. 15, 2019 as U.S. Pat. No. 10,441,345, which is a continuation-in-part application claiming priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 12/896,360, entitled SURGICAL GENERATOR FOR ULTRASONIC AND ELECTROSURGICAL DEVICES, filed Oct. 1, 2010, which issued on Jun. 23, 2015 as U.S. Pat. No. 9,060,775, which claims the benefit under Title 35, United States Code § 119 (e), of U.S. Provisional Patent Application Ser. No. 61/250,217, filed Oct. 9, 2009 and entitled A DUAL BIPOLAR AND ULTRASONIC GENERATOR FOR ELECTRO-SURGICAL INSTRUMENTS. Each of the applications is hereby incorporated by reference in their entireties. (2) The present application also is related to the following U.S. Patent Applications, each of which is incorporated herein by reference in its entirety: (1) U.S. patent application Ser. No. 12/896,351, entitled DEVICES AND TECHNIQUES FOR CUTTING AND COAGULATING TISSUE, now U.S. Pat. No. 9,089,360; (2) U.S. patent application Ser. No. 12/896,479, entitled SURGICAL GENERATOR FOR ULTRASONIC AND ELECTROSURGICAL DEVICES, now U.S. Pat. No. 8,956,349; (3) U.S. patent application Ser. No. 12/896,345, entitled SURGICAL GENERATOR FOR ULTRASONIC AND ELECTROSURGICAL DEVICES, now U.S. Pat. No. 8,986,302; (4) U.S. patent application Ser. No. 12/896,384, entitled SURGICAL GENERATOR FOR ULTRASONIC AND ELECTROSURGICAL DEVICES, now U.S. Pat. No. 8,951,248; (5) U.S. patent application Ser. No. 12/896,467, entitled SURGICAL GENERATOR FOR ULTRASONIC AND ELECTROSURGICAL DEVICES, now U.S. Pat. No. 9,050,093; (6) U.S. patent application Ser. No. 12/896,451, entitled SURGICAL GENERATOR FOR ULTRASONIC AND ELECTROSURGICAL DEVICES, now U.S. Pat. No. 9,039,695; (7) U.S. patent application Ser. No. 12/896,470, entitled SURGICAL GENERATOR FOR ULTRASONIC AND ELECTROSURGICAL DEVICES, now U.S. Pat. No. 9,060,776; and (8) U.S. patent application Ser. No. 14/657,850, entitled SURGICAL GENERATOR FOR ULTRASONIC AND ELECTROSURGICAL DEVICES, now U.S. Patent Publication No. 2015/0182276.

BACKGROUND

(1) Various embodiments are directed to surgical devices, and generators for supplying energy to surgical devices, for use in open or minimally invasive surgical environments.

(2) Ultrasonic surgical devices, such as ultrasonic scalpels, are finding increasingly widespread applications in surgical procedures by virtue of their unique performance characteristics. Depending upon specific device configurations and operational parameters, ultrasonic surgical devices can provide substantially simultaneous transection of tissue and homeostasis by coagulation, desirably minimizing patient trauma. An ultrasonic surgical device may comprise a handpiece containing an ultrasonic transducer, and an instrument coupled to the ultrasonic transducer having a distally-mounted end effector (e.g., a blade tip) to cut and seal tissue. In some cases, the instrument may be permanently affixed to the handpiece. In other cases, the instrument may be detachable from the handpiece, as in the case of a disposable instrument or an instrument that is interchangeable between different handpieces. The end effector transmits ultrasonic energy to tissue brought into contact with the end effector to realize cutting and sealing action. Ultrasonic surgical devices of this nature can be configured for open surgical use, laparoscopic, or endoscopic surgical procedures including robotic-assisted procedures.

(3) Ultrasonic energy cuts and coagulates tissue using temperatures lower than those used in electrosurgical procedures and can be transmitted to the end effector by an ultrasonic generator in communication with the handpiece. Vibrating at high frequencies (e.g., 55,500 times per second), the ultrasonic blade denatures protein in the tissue to form a sticky coagulum. Pressure exerted on tissue by the blade surface collapses blood vessels and allows the coagulum to form a haemostatic seal. A surgeon can control the cutting speed and coagulation by the force applied to the tissue by the end effector, the time over which the force is applied and the selected excursion level of the end effector.

- (4) The ultrasonic transducer may be modeled as an equivalent circuit comprising a first branch having a static capacitance and a second “motional” branch having a serially connected inductance, resistance and capacitance that define the electromechanical properties of a resonator. Known ultrasonic generators may include a tuning inductor for tuning out the static capacitance at a resonant frequency so that substantially all of generator's drive signal current flows into the motional branch. Accordingly, by using a tuning inductor, the generator's drive signal current represents the motional branch current, and the generator is thus able to control its drive signal to maintain the ultrasonic transducer's resonant frequency. The tuning inductor may also transform the phase impedance plot of the ultrasonic transducer to improve the generator's frequency lock capabilities. However, the tuning inductor must be matched with the specific static capacitance of an ultrasonic transducer at the operational resonance frequency. In other words, a different ultrasonic transducer having a different static capacitance requires a different tuning inductor.
- (5) Additionally, in some ultrasonic generator architectures, the generator's drive signal exhibits asymmetrical harmonic distortion that complicates impedance magnitude and phase measurements. For example, the accuracy of impedance phase measurements may be reduced due to harmonic distortion in the current and voltage signals.
- (6) Moreover, electromagnetic interference in noisy environments decreases the ability of the generator to maintain lock on the ultrasonic transducer's resonant frequency, increasing the likelihood of invalid control algorithm inputs.
- (7) Electrosurgical devices for applying electrical energy to tissue in order to treat and/or destroy the tissue are also finding increasingly widespread applications in surgical procedures. An electrosurgical device may comprise a handpiece and an instrument having a distally-mounted end effector (e.g., one or more electrodes). The end effector can be positioned against the tissue such that electrical current is introduced into the tissue. Electrosurgical devices can be configured for bipolar or monopolar operation. During bipolar operation, current is introduced into and returned from the tissue by active and return electrodes, respectively, of the end effector. During monopolar operation, current is introduced into the tissue by an active electrode of the end effector and returned through a return electrode (e.g., a grounding pad) separately located on a patient's body. Heat generated by the current flow through the tissue may form haemostatic seals within the tissue and/or between tissues and thus may be particularly useful for sealing blood vessels, for example. The end effector of an electrosurgical device may also comprise a cutting member that is movable relative to the tissue and the electrodes to transect the tissue.
- (8) Electrical energy applied by an electrosurgical device can be transmitted to the instrument by a generator in communication with the handpiece. The electrical energy may be in the form of radio frequency (“RF”) energy. RF energy is a form of electrical energy that may be in the frequency range of 300 KHz to 1 MHz. During its operation, an electrosurgical device can transmit low frequency RF energy through tissue, which causes ionic agitation, or friction, in effect resistive heating, thereby increasing the temperature of the tissue. Because a sharp boundary may be created between the affected tissue and the surrounding tissue, surgeons can operate with a high level of precision and control, without sacrificing un-targeted adjacent tissue. The low operating temperatures of RF energy may be useful for removing, shrinking, or sculpting soft tissue while simultaneously sealing blood vessels. RF energy may work particularly well on connective tissue, which is primarily comprised of collagen and shrinks when contacted by heat.
- (9) Due to their unique drive signal, sensing and feedback needs, ultrasonic and electrosurgical devices have generally required different generators. Additionally, in cases where the instrument is disposable or interchangeable with a handpiece, ultrasonic and electrosurgical generators are limited in their ability to recognize the particular instrument configuration being used and to optimize control and diagnostic processes accordingly. Moreover, capacitive coupling between the non-isolated and patient-isolated circuits of the generator, especially in cases where higher voltages and frequencies are used, may result in exposure of a patient to unacceptable levels of leakage

current.

(10) Furthermore, due to their unique drive signal, sensing and feedback needs, ultrasonic and electrosurgical devices have generally required different user interfaces for the different generators. In such conventional ultrasonic and electrosurgical devices, one user interface is configured for use with an ultrasonic instrument whereas a different user interface may be configured for use with an electrosurgical instrument. Such user interfaces include hand and/or foot activated user interfaces such as hand activated switches and/or foot activated switches. As various embodiments of combined generators for use with both ultrasonic and electrosurgical instruments are contemplated in the subsequent disclosure, additional user interfaces that are configured to operate with both ultrasonic and/or electrosurgical instrument generators also are contemplated.

(11) Additional user interfaces for providing feedback, whether to the user or other machine, are contemplated within the subsequent disclosure to provide feedback indicating an operating mode or status of either an ultrasonic and/or electrosurgical instrument. Providing user and/or machine feedback for operating a combination ultrasonic and/or electrosurgical instrument will require providing sensory feedback to a user and electrical/mechanical/electromechanical feedback to a machine. Feedback devices that incorporate visual feedback devices (e.g., an LCD display screen, LED indicators), audio feedback devices (e.g., a speaker, a buzzer) or tactile feedback devices (e.g., haptic actuators) for use in combined ultrasonic and/or electrosurgical instruments are contemplated in the subsequent disclosure.

SUMMARY

(12) Various embodiments of a generator to communicate a drive signal to a surgical device are disclosed. In accordance with various embodiments, the generator may comprise a power amplifier to receive a time-varying drive signal waveform. The drive signal waveform may be generated by a digital-to-analog conversion of at least a portion of a plurality of drive signal waveform samples. An output of the power amplifier may be for generating a drive signal. The drive signal may comprise one of: a first drive signal to be communicated to an ultrasonic surgical device, a second drive signal to be communicated to an electrosurgical device. The generator may also comprise a sampling circuit to generate samples of current and voltage of the drive signal when the drive signal is communicated to the surgical device. Generation of the samples may be synchronized with the digital-to-analog conversion of the drive signal waveform samples such that, for each digital-to-analog conversion of a drive signal waveform sample, the sampling circuit generates a corresponding set of current and voltage samples. The generator may also comprise at least one device programmed to, for each drive signal waveform sample and corresponding set of current and voltage samples, store the current and voltage samples in a memory of the at least one device to associate the stored samples with the drive signal waveform sample. The at least one device may also be programmed to, when the drive signal comprises the first drive signal: determine a motional branch current sample of the ultrasonic surgical device based on the stored current and voltage samples, compare the motional branch current sample to a target sample selected from a plurality of target samples that define a target waveform, the target sample selected based on the drive signal waveform sample, determine an amplitude error between the target sample and the motional branch current sample, and modify the drive signal waveform sample such that an amplitude error determined between the target sample and a subsequent motional branch current sample based on current and voltage samples associated with the modified drive signal waveform sample is reduced.

(13) In accordance with various embodiments, the generator may comprise a memory and a device coupled to the memory to receive for each of a plurality of drive signal waveform samples used to synthesize the drive signal, a corresponding set of current and voltage samples of the drive signal. For each drive signal waveform sample and corresponding set of current and voltage samples, the device may store the samples in a memory of the device to associate the stored samples with the drive signal waveform sample. Also, for each drive signal waveform sample and corresponding set of current and voltage samples, the device may, when the drive signal comprises a first drive signal

to be communicated to an ultrasonic surgical device, determine a motional branch current sample of the ultrasonic surgical device based on the stored samples, compare the motional branch current sample to a target sample selected from a plurality of target samples that define a target waveform, the target sample selected based on the drive signal waveform sample, determine an amplitude error between the target sample and the motional branch current sample, and modify the drive signal waveform sample such that an amplitude error determined between the target sample and a subsequent motional branch current sample based on current and voltage samples associated with the modified drive signal waveform sample is reduced.

(14) In accordance with various embodiments, methods for determining motional branch current in an ultrasonic transducer of an ultrasonic surgical device over multiple frequencies of a transducer drive signal are also disclosed. In one embodiment, the method may comprise, at each of a plurality of frequencies of the transducer drive signal, oversampling a current and voltage of the transducer drive signal, receiving, by a processor, the current and voltage samples, and determining, by the processor, the motional branch current based on the current and voltage samples, a static capacitance of the ultrasonic transducer and the frequency of the transducer drive signal.

(15) In accordance with various embodiments, methods for controlling a waveform shape of a motional branch current in an ultrasonic transducer of a surgical device are also disclosed. In one embodiment, the method may comprise generating a transducer drive signal by selectively recalling, using a direct digital synthesis (DDS) algorithm, drive signal waveform samples stored in a look-up table (LUT), generating samples of current and voltage of the transducer drive signal when the transducer drive signal is communicated to the surgical device, determining samples of the motional branch current based on the current and voltage samples, a static capacitance of the ultrasonic transducer and a frequency of the transducer drive signal, comparing each sample of the motional branch current to a respective target sample of a target waveform to determine an error amplitude, and modifying the drive signal waveform samples stored in the LUT such that an amplitude error between subsequent samples of the motional branch current and respective target samples is reduced.

(16) In accordance with various embodiments, a surgical generator for providing a drive signal to a surgical device may comprise a first transformer and a second transformer. The first transformer may comprise a first primary winding and a first secondary winding. The second transformer may comprise a second primary winding and a second secondary winding. The surgical generator may further comprise a generator circuit to generate the drive signal. The generator circuit may be electrically coupled to the first primary winding to provide the drive signal across the first primary winding. The surgical generator may also comprise a patient-side circuit electrically isolated from the generator circuit. The patient-side circuit may be electrically coupled to the first secondary winding. Further, the patient-side circuit may comprise first and second output lines to provide the drive signal to the surgical device. In addition, the surgical generator may comprise a capacitor. The capacitor and the second secondary winding may be electrically coupled in series between the first output line and ground.

(17) In accordance with various embodiments, a surgical generator for providing a drive signal to a surgical device may comprise a first transformer, a patient-side circuit, and a capacitor. The first transformer may comprise a primary winding, a first secondary winding, and a second secondary winding. A polarity of the first secondary winding relative to the primary winding may be opposite the polarity of the second secondary winding. The generator circuit may generate the drive signal and may be electrically coupled to the first primary winding to provide the drive signal across the first primary winding. The patient-side circuit may be electrically isolated from the generator circuit and may be electrically coupled to the first secondary winding. Also, the patient-side circuit may comprise first and second output lines to provide the drive signal to the surgical device. The capacitor and second secondary winding may be electrically coupled in series between the first output line and ground.

(18) In accordance with various embodiments, a surgical generator for providing a drive signal to a surgical device may comprise a first transformer, a generator circuit, a patient-side circuit and a capacitor. The first transformer may comprise a primary winding and a secondary winding. The generator circuit may generate the drive signal and may be electrically coupled to the first primary winding to provide the drive signal across the first primary winding. The patient-side circuit may be electrically isolated from the generator circuit and may be electrically coupled to the secondary winding. Further, the patient-side circuit may comprise first and second output lines to provide the drive signal to the surgical device. The capacitor may be electrically coupled to the primary winding and to the first output line.

(19) In accordance with various embodiments, a surgical generator for providing a drive signal to a surgical device may comprise a first transformer, a generator circuit, a patient-side circuit, as well as first, second and third capacitors. The first transformer may comprise a primary winding and a secondary winding. The generator circuit may generate the drive signal and may be electrically coupled to the first primary winding to provide the drive signal across the first primary winding. The patient-side circuit may be electrically isolated from the generator circuit and may be electrically coupled to the secondary winding. Further, the patient-side circuit may comprise first and second output lines to provide the drive signal to the surgical device. A first electrode of the first capacitor may be electrically coupled to the primary winding. A first electrode of the second capacitor may be electrically coupled to the first output line and a second electrode of the second capacitor may be electrically coupled to a second electrode of the first capacitor. A first electrode of the third capacitor may be electrically coupled to the second electrode of the first capacitor and the second electrode of the second capacitor. A second electrode of the third capacitor may be electrically coupled to ground.

(20) In accordance with various embodiments, control circuits for surgical devices are also disclosed. In one embodiment, the control circuit may comprise a first circuit portion comprising at least one first switch. The first circuit portion may communicate with a surgical generator over a conductor pair. The control circuit may also comprise a second circuit portion comprising a data circuit element. The data circuit element may be disposed in an instrument of the surgical device and transmit or receive data. The data circuit element may implement data communications with the surgical generator over at least one conductor of the conductor pair.

(21) In accordance with various embodiments, the control circuit may comprise a first circuit portion comprising at least one first switch. The first circuit portion may communicate with a surgical generator over a conductor pair. The control circuit may also comprise a second circuit portion comprising a data circuit element. The data circuit element may be disposed in an instrument of the surgical device and transmit or receive data. The data circuit element may implement data communications with the surgical generator over at least one conductor of the conductor pair. The first circuit portion may receive a first interrogation signal transmitted from the surgical generator in a first frequency band. The data circuit element may communicate with the surgical generator using an amplitude-modulated communication protocol transmitted in a second frequency band. The second frequency band may be higher than the first frequency band.

(22) In accordance with various embodiments, the control circuit may comprise a first circuit portion comprising at least one first switch. The first circuit portion may receive a first interrogation signal transmitted from a surgical generator over a conductor pair. The control circuit may also comprise a second circuit portion comprising at least one of a resistive element and an inductive element disposed in an instrument of the device. The second circuit portion may receive a second interrogation signal transmitted from the surgical generator over the conductor pair. The second circuit portion may be frequency-band separated from the first circuit portion. A characteristic of the first interrogation signal, when received through the first circuit portion, may be indicative of a state of the at least one first switch. A characteristic of the second interrogation signal, when received through the second circuit portion, may uniquely identify the instrument of

the device.

(23) In accordance with various embodiments, the control circuit may comprise a first circuit portion comprising a first switch network and a second switch network. The first switch network may comprise at least one first switch, and the second switch network may comprise at least one second switch. The first circuit portion may communicate with a surgical generator over a conductor pair. The control circuit may also comprise a second circuit portion comprising a data circuit element. The data circuit element may be disposed in an instrument of the surgical device and may transmit or receive data. The data circuit element may be in data communication with the surgical generator over at least one conductor of the conductor pair.

(24) In accordance with various embodiments, a surgical generator for providing a drive signal to a surgical device may comprise a surgical generator body having an aperture. The surgical generator may also comprise a receptacle assembly positioned in the aperture. The receptacle assembly may comprise a receptacle body and a flange having an inner wall and an outer wall. The inner wall may be comprised of at least one curved section and at least one linear section. The inner wall may define a cavity. A central protruding portion may be positioned in the cavity and may comprise a plurality of sockets and a magnet. An outer periphery of the central protruding portion may comprise at least one curved section and at least one linear section.

(25) In accordance with various embodiments, a surgical instrument may comprises an electrical connector assembly. The electrical connector assembly may comprise a flange defining a central cavity and a magnetically compatible pin extending into the central cavity. The electrical connector assembly may comprise a circuit board and a plurality of electrically conductive pins coupled to the circuit board. Each of the plurality of electrically conductive pins may extend into the central cavity. The electrical connector assembly may further comprise a strain relief member and a boot.

(26) In accordance with various embodiments, a surgical instrument system may comprise a surgical generator comprising a receptacle assembly. The receptacle assembly may comprise at least one curved section and at least one linear portion. The surgical instrument system may comprise a surgical instrument comprising a connector assembly and an adapter assembly operatively coupled to the receptacle assembly and the connector assembly. The adapter assembly may comprise a distal portion contacting the receptacle assembly. The distal portion may comprise a flange with the flange having at least one curved section and at least one linear portion. The adapter assembly may comprise a proximal portion contacting the connector assembly. The proximal portion may define a cavity dimensioned to receive at least a portion of the connector assembly. The adapter assembly may further comprise a circuit board.

(27) In accordance with various embodiments, methods may be utilized (e.g., in conjunction with surgical instruments) to accomplish various surgical objectives. For example, methods to control electrical power provided to tissue via first and second electrodes may comprise providing a drive signal to the tissue via the first and second electrodes and modulating a power provided to the tissue via the drive signal based on a sensed tissue impedance according to a first power curve. The first power curve may define, for each of a plurality of potential sensed tissue impedances, a first corresponding power. The methods may also comprise monitoring a total energy provided to the tissue via the first and second electrodes. When the total energy reaches a first energy threshold, the methods may comprise determining whether an impedance of the tissue has reached a first impedance threshold. The methods may further comprise, conditioned upon the impedance of the tissue failing to reach the first impedance threshold, modulating the power provided to the tissue via the drive signal based on the sensed tissue impedance according to a second power curve. The second power curve may define, for each of the plurality of potential sensed tissue impedances, a second corresponding power.

(28) In accordance with various embodiments, methods for controlling electrical power provided to tissue via first and second electrodes may comprise providing a drive signal to the tissue via the first and second electrodes and determining a power to be provided to the tissue. The determining

may comprise receiving an indication of a sensed tissue impedance; determining a first corresponding power for the sensed tissue impedance according to a power curve; and multiplying the corresponding power by a multiplier. The power curve may define a corresponding power for each of a plurality of potential sensed tissue impedances. The methods may further comprise modulating the drive signal to provide the determined power to the tissue and, conditioned upon the impedance of the tissue failing to reach a first impedance threshold, increasing the multiplier as a function of the total energy provided to the tissue.

(29) In accordance with various embodiments, methods for controlling electrical power provided to tissue via first and second electrodes may comprise providing a drive signal to the tissue via the first and second electrodes and determining a power to be provided to the tissue. The determining may comprise receiving an indication of a sensed tissue impedance; determining a first corresponding power for the sensed tissue impedance according to a power curve; and multiplying the corresponding power by a first multiplier to find a determined power. The power curve may define a corresponding power for each of a plurality of potential sensed tissue impedances. The methods may further comprise modulating the drive signal to provide the determined power to the tissue and monitoring a total energy provided to the tissue via the first and second electrodes. In addition, the methods may comprise, when the total energy reaches a first energy threshold, determining whether the impedance of the tissue has reached a first impedance threshold; and, conditioned upon the impedance of the tissue not reaching the first impedance threshold, increasing the first multiplier by a first amount.

(30) In accordance with various embodiments, methods for controlling electrical power provided to tissue via a surgical device may comprise providing a drive signal to a surgical device; receiving an indication of an impedance of the tissue; calculating a rate of increase of the impedance of the tissue; and modulating the drive signal to hold the rate of increase of the impedance greater than or equal to a predetermined constant.

(31) In accordance with various embodiments, methods for controlling electrical power provided to tissue via a surgical device may comprise providing a drive signal. A power of the drive signal may be proportional to a power provided to the tissue via the surgical device. The methods may also comprise periodically receiving indications of an impedance of the tissue and applying a first composite power curve to the tissue. Applying the first composite power curve to the tissue may comprise modulating a first predetermined number of first composite power curve pulses on the drive signal; and for each of the first composite power curve pulses, determining a pulse power and a pulse width according to a first function of the impedance of the tissue. The methods may also comprise applying a second composite power curve to the tissue. Applying the second composite power curve to the tissue may comprise modulating at least one second composite power curve pulse on the drive signal; and for each of the at least one second composite power curve pulses, determining a pulse power and a pulse width according to a second function of the impedance of the tissue.

(32) In accordance with various embodiments, a generator is provided to generate a drive signal to a surgical device. The generator includes an ultrasonic generator module to generate a first drive signal to drive an ultrasonic device, an electrosurgery/radio frequency (RF) generator module to generate a second drive signal to drive an electrosurgical device, and a foot switch coupled to each of the ultrasonic generator module and the electrosurgery/RF generator module. The foot switch is configured to operate in a first mode when the ultrasonic device is coupled to the ultrasonic generator module and the foot switch is configured to operate in a second mode when the electrosurgical device is coupled to the electrosurgery/RF generator module.

(33) In accordance with various embodiments, a generator is provided that includes a user interface to provide feedback in accordance with the operation of any one of the ultrasonic device and the electrosurgical device in accordance with a predetermined algorithm.

(34) In accordance with various embodiments, a control circuit of a surgical device is provided.

The control circuit comprises a first circuit portion coupled to at least one switch operable between an open state and a closed state. The first circuit portion communicates with a surgical generator over a conductor pair to receive a control signal to determine a state of the at least one switch.

(35) In accordance with various embodiments, a control circuit of a surgical device is provided. The control circuit comprises a first circuit portion coupled to at least one switch operable between an open state and a closed state. The first circuit portion communicates with a surgical generator over a conductor pair to receive a control signal from input terminals to determine a state of the at least one switch. The control signal having a positive phase and a negative phase. A first transistor is coupled between the input terminals, a first capacitor, and a first resistor is coupled in series with the first capacitor. During the positive phase of the control signal the first transistor is held in cutoff mode while the first capacitor charges to a predetermined voltage and during an initial portion of the negative phase of the control signal the first transistor transitions from cutoff mode to saturation mode and is held in saturation mode until the first capacitor discharges through the first resistor. During a final portion of the negative phase of the control signal the first transistor transitions from saturation mode to cutoff mode when the first capacitor voltage drops below a predetermined threshold.

(36) In accordance with various embodiments, a method is provided. The method comprises receiving a control signal at a control circuit of a surgical device and determining the state of the at least one switch based on the value of the resistor. The control circuit comprising a first circuit portion coupled to at least one switch operable between an open state and a closed state. The circuit portion to communicate with a surgical generator over a conductor pair to receive the control signal. The first circuit portion comprising at least one resistor coupled to the at least one switch.

Description

FIGURES

- (1) The novel features of the various embodiments are set forth with particularity in the appended claims. The described embodiments, however, both as to organization and methods of operation, may be best understood by reference to the following description, taken in conjunction with the accompanying drawings in which:
- (2) FIG. 1 illustrates one embodiment of a surgical system comprising a generator and various surgical instruments usable therewith;
- (3) FIG. 2 illustrates one embodiment of an example ultrasonic device that may be used for transection and/or sealing;
- (4) FIG. 3 illustrates one embodiment of the end effector of the example ultrasonic device of FIG. 2.
- (5) FIG. 4 illustrates one embodiment of an example electrosurgical device that may also be used for transection and sealing;
- (6) FIGS. 5, 6 and 7 illustrate one embodiment of the end effector shown in FIG. 4;
- (7) FIG. 8 is a diagram of the surgical system of FIG. 1;
- (8) FIG. 9 is a model illustrating motional branch current in one embodiment;
- (9) FIG. 10 is a structural view of a generator architecture in one embodiment;
- (10) FIGS. 11A-11C are functional views of a generator architecture in one embodiment;
- (11) FIG. 12 illustrates a controller for monitoring input devices and controlling output devices in one embodiment;
- (12) FIGS. 13A and 13B illustrate structural and functional aspects of one embodiment of the generator;
- (13) FIGS. 14-32 and 33A-33C illustrate embodiments of control circuits;
- (14) FIGS. 33D-33I illustrate embodiments of cabling and adaptor configurations for connecting

various generators and various surgical instruments;

(15) FIG. **34** illustrates one embodiment of a circuit **300** for active cancellation of leakage current.

(16) FIG. **35** illustrates one embodiment of a circuit that may be implemented by the generator of FIG. **1** to provide active cancellation of leakage current;

(17) FIG. **36** illustrates an alternative embodiment of a circuit that may be implemented by the generator of FIG. **1** to provide active cancellation of leakage current;

(18) FIG. **37** illustrates an alternative embodiment of a circuit that may be implemented by the generator of FIG. **1** to provide active cancellation of leakage current;

(19) FIG. **38** illustrates yet another embodiment of a circuit that may be implemented by the generator of FIG. **1** to provide active cancellation of leakage current;

(20) FIG. **39** illustrates an embodiment of a circuit that may be implemented by the generator of FIG. **1** to provide cancellation of leakage current;

(21) FIG. **40** illustrates another embodiment of a circuit that may be implemented by the generator of FIG. **1** to provide cancellation of leakage current;

(22) FIG. **41** illustrates a receptacle and connector interface in one embodiment;

(23) FIG. **42** is an exploded side view of the receptacle assembly in one embodiment;

(24) FIG. **43** is an exploded side view of the connector assembly in one embodiment;

(25) FIG. **44** is a perspective view of the receptacle assembly shown in FIG. **41**;

(26) FIG. **45** is an exploded perspective view of the receptacle assembly in one embodiment;

(27) FIG. **46** is a front elevation view of the receptacle assembly in one embodiment;

(28) FIG. **47** is a side elevation view of the receptacle assembly in one embodiment;

(29) FIG. **48** is an enlarged view of a socket in one embodiment;

(30) FIG. **49** is a perspective view of the connector assembly in one embodiment;

(31) FIG. **50** is an exploded perspective view of the connector assembly in one embodiment;

(32) FIG. **51** is a side elevation view of a connector body in one embodiment;

(33) FIG. **52** is perspective view of the distal end of a connector body in one embodiment;

(34) FIG. **53** is perspective view of the proximal end of a connector body in one embodiment;

(35) FIG. **54** illustrates a ferrous pin in one embodiment;

(36) FIG. **55** illustrates electrically conductive pins and a circuit board in one embodiment;

(37) FIG. **56** illustrates a strain relief member in one embodiment;

(38) FIG. **57** illustrates a boot in one embodiment;

(39) FIG. **58** illustrates two adaptor assemblies in accordance with various non-limiting embodiments;

(40) FIG. **59** illustrates a surgical generator in one embodiment;

(41) FIG. **60** illustrates a connector assembly connected to an adaptor assembly in one embodiment;

(42) FIG. **61** illustrates an adaptor assembly inserted into a receptacle assembly of a surgical generator in one embodiment;

(43) FIG. **62** illustrates a connector assembly connected to an adaptor assembly in one embodiment;

(44) FIG. **63** illustrates a perspective view of a back panel of a generator in one embodiment;

(45) FIG. **64** illustrates a back panel of a generator in one embodiment;

(46) FIGS. **65** and **66** illustrate different portions of a back panel of a generator in one embodiment;

(47) FIG. **67** illustrates a neural network for controlling a generator in one embodiment;

(48) FIG. **68** illustrates measured temperature versus estimated temperature output by a surgical instrument controlled by a generator in one embodiment;

(49) FIG. **69** illustrates one embodiment of a chart showing example power curves;

(50) FIG. **70** illustrates one embodiment of a process flow for applying one or more power curves to a tissue bite;

(51) FIG. **71** illustrates one embodiment of a chart showing example power curves that may be

used in conjunction with the process flow of FIG. 70;

(52) FIG. 72 illustrates one embodiment of a chart showing example common shape power curves that may be used in conjunction with the process flow of FIG. 70;

(53) FIG. 73A illustrates one embodiment of a routine that may be performed by a digital device of the generator of FIG. 1 to act upon a new tissue bite;

(54) FIG. 73B illustrates one embodiment of a routine that may be performed by a digital device of the generator of FIG. 1 to monitor tissue impedance;

(55) FIG. 73C illustrates one embodiment of a routine that may be performed by a digital device of the generator of FIG. 1 to provide one or more power curves to a tissue bite;

(56) FIG. 74 illustrates one embodiment of a process flow for applying one or more power curves to a tissue bite;

(57) FIG. 75 illustrates one embodiment of a block diagram describing the selection and application of composite load curves by the generator of FIG. 1;

(58) FIG. 76 illustrates shows a process flow illustrating one embodiment of the algorithm of FIG. 75, as implemented by the generator of FIG. 1;

(59) FIG. 77 illustrates one embodiment of a process flow for generating a first composite load curve pulse;

(60) FIG. 78 illustrates one embodiment of a pulse timing diagram illustrating an example application of the algorithm of FIG. 76 by the generator of FIG. 1;

(61) FIG. 79 illustrates a graphical representation of drive signal voltage, current and power according to an example composite load curve;

(62) FIGS. 80-85 illustrate a graphical representations of example composite load curves;

(63) FIG. 86 illustrates one embodiment of a block diagram describing the application of an algorithm for maintaining a constant tissue impedance rate of change;

(64) FIG. 87 illustrates one embodiment of a control circuit comprising parallel switched resistance circuit with high speed data communication support and at least one data element comprising at least one memory device;

(65) FIG. 88 is a graphical representation of one aspect of a control signal in the form of a constant current pulse waveform that may be generated by the signal conditioning circuit of the generator as shown in FIG. 10;

(66) FIG. 88A is a graphical representation of one aspect of a control signal in the form of the constant current pulse waveform of FIG. 88 showing numerical values of various features of the waveform according to one example embodiment;

(67) FIG. 89 is a graphical representation of various detection regions associated with the aspect of the control signal shown in FIG. 88;

(68) FIG. 90 is a graphical representation of one aspect of a control signal in the form of a current pulse waveform measured at the generator with SW1 closed and a zero ohm cable/connector impedance;

(69) FIG. 90A is a graphical representation of one aspect of a control signal in the form of the current pulse waveform of FIG. 90 showing numerical values of various features of the waveform according to one example embodiment;

(70) FIG. 91 is an oscilloscope trace of an actual measured analog-to-digital (ADC) input waveform using a generator and a control circuit;

(71) FIG. 92 illustrates another embodiment of a control circuit comprising parallel switched resistance circuit and at least one data element comprising at least one memory device memory device;

(72) FIG. 93 illustrates one embodiment of a control circuit comprising a serial switched resistance circuit and at least one data element comprising at least one memory device;

(73) FIG. 94 illustrates one embodiment of a control circuit comprising a serial switched resistance circuit with a precision voltage reference and at least one data element comprising at least one

memory device;

(74) FIG. **95** illustrates one embodiment of a control circuit comprising a variable frequency switched resistance circuit and at least one data element comprising at least one memory device;

(75) FIG. **96**, is a graphical representation of one embodiment of a detection method showing detection regions for the control circuit comprising a variable frequency switched resistance circuit and a memory device, as described in connection with FIG. **95**;

(76) FIG. **97** illustrates one embodiment of a control circuit comprising a parallel switched resistance circuit with a precision voltage reference and at least one data element comprising at least one memory device employing a variable slope waveform to determine switch states;

(77) FIG. **98** is a graphical representation of one embodiment of a detection method showing detection regions for the control circuit comprising a variable ramp/slope switched resistance circuit and a memory device, as described in connection with FIG. **97**; and

(78) FIG. **99** illustrates one embodiment of a control circuit comprising a one-wire multi-switch input device.

DESCRIPTION

(79) Before explaining various embodiments of surgical devices and generators in detail, it should be noted that the illustrative embodiments are not limited in application or use to the details of construction and arrangement of parts illustrated in the accompanying drawings and description. The illustrative embodiments may be implemented or incorporated in other embodiments, variations and modifications, and may be practiced or carried out in various ways. Further, unless otherwise indicated, the terms and expressions employed herein have been chosen for the purpose of describing the illustrative embodiments for the convenience of the reader and are not for the purpose of limitation thereof. Also, it will be appreciated that one or more of the following-described embodiments, expressions of embodiments and/or examples, can be combined with any one or more of the other following-described embodiments, expressions of embodiments and/or examples.

(80) Various embodiments are directed to improved ultrasonic surgical devices, electrosurgical devices and generators for use therewith. Embodiments of the ultrasonic surgical devices can be configured for transecting and/or coagulating tissue during surgical procedures, for example. Embodiments of the electrosurgical devices can be configured for transecting, coagulating, scaling, welding and/or desiccating tissue during surgical procedures, for example.

(81) Embodiments of the generator utilize high-speed analog-to-digital sampling (e.g., approximately 200× oversampling, depending on frequency) of the generator drive signal current and voltage, along with digital signal processing, to provide a number of advantages and benefits over known generator architectures. In one embodiment, for example, based on current and voltage feedback data, a value of the ultrasonic transducer static capacitance, and a value of the drive signal frequency, the generator may determine the motional branch current of an ultrasonic transducer. This provides the benefit of a virtually tuned system, and simulates the presence of a system that is tuned or resonant with any value of the static capacitance (e.g., C.sub.0 in FIG. **9**) at any frequency. Accordingly, control of the motional branch current may be realized by tuning out the effects of the static capacitance without the need for a tuning inductor. Additionally, the elimination of the tuning inductor may not degrade the generator's frequency lock capabilities, as frequency lock can be realized by suitably processing the current and voltage feedback data.

(82) High-speed analog-to-digital sampling of the generator drive signal current and voltage, along with digital signal processing, may also enable precise digital filtering of the samples. For example, embodiments of the generator may utilize a low-pass digital filter (e.g., a finite impulse response (FIR) filter) that rolls off between a fundamental drive signal frequency and a second-order harmonic to reduce the asymmetrical harmonic distortion and EMI-induced noise in current and voltage feedback samples. The filtered current and voltage feedback samples represent substantially the fundamental drive signal frequency, thus enabling a more accurate impedance

phase measurement with respect to the fundamental drive signal frequency and an improvement in the generator's ability to maintain resonant frequency lock. The accuracy of the impedance phase measurement may be further enhanced by averaging falling edge and rising edge phase measurements, and by regulating the measured impedance phase to 0°.

(83) Various embodiments of the generator may also utilize the high-speed analog-to-digital sampling of the generator drive signal current and voltage, along with digital signal processing, to determine real power consumption and other quantities with a high degree of precision. This may allow the generator to implement a number of useful algorithms, such as, for example, controlling the amount of power delivered to tissue as the impedance of the tissue changes and controlling the power delivery to maintain a constant rate of tissue impedance increase.

(84) Various embodiments of the generator may have a wide frequency range and increased output power necessary to drive both ultrasonic surgical devices and electrosurgical devices. The lower voltage, higher current demand of electrosurgical devices may be met by a dedicated tap on a wideband power transformer, thereby eliminating the need for a separate power amplifier and output transformer. Moreover, sensing and feedback circuits of the generator may support a large dynamic range that addresses the needs of both ultrasonic and electrosurgical applications with minimal distortion.

(85) Various embodiments may provide a simple, economical means for the generator to read from, and optionally write to, data circuit (e.g., a single-wire bus device, such as a one-wire protocol EEPROM known under the trade name “1-Wire”) disposed in an instrument attached to the handpiece using existing multi-conductor generator/handpiece cables. In this way, the generator is able to retrieve and process instrument-specific data from an instrument attached to the handpiece. This may enable the generator to provide better control and improved diagnostics and error detection. Additionally, the ability of the generator to write data to the instrument makes possible new functionality in terms of, for example, tracking instrument usage and capturing operational data. Moreover, the use of frequency band permits the backward compatibility of instruments containing a bus device with existing generators.

(86) Disclosed embodiments of the generator provide active cancellation of leakage current caused by unintended capacitive coupling between non-isolated and patient-isolated circuits of the generator. In addition to reducing patient risk, the reduction of leakage current may also lessen electromagnetic emissions.

(87) These and other benefits of embodiments of the present invention will be apparent from the description to follow.

(88) It will be appreciated that the terms “proximal” and “distal” are used herein with reference to a clinician gripping a handpiece. Thus, an end effector is distal with respect to the more proximal handpiece. It will be further appreciated that, for convenience and clarity, spatial terms such as “top” and “bottom” may also be used herein with respect to the clinician gripping the handpiece. However, surgical devices are used in many orientations and positions, and these terms are not intended to be limiting and absolute.

(89) FIG. 1 illustrates one embodiment of a surgical system **100** comprising a generator **102** configurable for use with surgical devices. According to various embodiments, the generator **102** may be configurable for use with surgical devices of different types, including, for example, the ultrasonic surgical device **104** and electrosurgical or RF surgical device **106**. Although in the embodiment of FIG. 1 the generator **102** is shown separate from the surgical devices **104**, **106**, in certain embodiments the generator **102** may be formed integrally with either of the surgical devices **104**, **106** to form a unitary surgical system.

(90) FIG. 2 illustrates one embodiment of an example ultrasonic device **104** that may be used for transection and/or sealing. The device **104** may comprise a hand piece **116** which may, in turn, comprise an ultrasonic transducer **114**. The transducer **114** may be in electrical communication with the generator **102**, for example, via a cable **122** (e.g., a multi-conductor cable). The transducer **114**

may comprise piezoceramic elements, or other elements or components suitable for converting the electrical energy of a drive signal into mechanical vibrations. When activated by the generator **102**, the ultrasonic transducer **114** may cause longitudinal vibration. The vibration may be transmitted through an instrument portion **124** of the device **104** (e.g., via a waveguide embedded in an outer sheath) to an end effector **126** of the instrument portion **124**.

(91) FIG. 3 illustrates one embodiment of the end effector **126** of the example ultrasonic device **104**. The end effector **126** may comprise a blade **151** that may be coupled to the ultrasonic transducer **114** via the wave guide (not shown). When driven by the transducer **114**, the blade **151** may vibrate and, when brought into contact with tissue, may cut and/or coagulate the tissue, as described herein. According to various embodiments, and as illustrated in FIG. 3, the end effector **126** may also comprise a clamp arm **155** that may be configured for cooperative action with the blade **151** of the end effector **126**. With the blade **151**, the clamp arm **155** may comprise a set of jaws **140**. The clamp arm **155** may be pivotally connected at a distal end of a shaft **153** of the instrument portion **124**. The clamp arm **155** may include a clamp arm tissue pad **163**, which may be formed from TEFLON® or other suitable low-friction material. The pad **163** may be mounted for cooperation with the blade **151**, with pivotal movement of the clamp arm **155** positioning the clamp pad **163** in substantially parallel relationship to, and in contact with, the blade **151**. By this construction, a tissue bite to be clamped may be grasped between the tissue pad **163** and the blade **151**. The tissue pad **163** may be provided with a sawtooth-like configuration including a plurality of axially spaced, proximally extending gripping teeth **161** to enhance the gripping of tissue in cooperation with the blade **151**. The clamp arm **155** may transition from the open position shown in FIG. 3 to a closed position (with the clamp arm **155** in contact with or proximity to the blade **151**) in any suitable manner. For example, the hand piece **116** may comprise a jaw closure trigger **138**. When actuated by a clinician, the jaw closure trigger **138** may pivot the clamp arm **155** in any suitable manner.

(92) The generator **102** may be activated to provide the drive signal to the transducer **114** in any suitable manner. For example, the generator **102** may comprise a foot switch **120** coupled to the generator **102** via a footswitch cable **122** (FIG. 8). A clinician may activate the transducer **114**, and thereby the transducer **114** and blade **151**, by depressing the foot switch **120**. In addition, or instead of the foot switch **120** some embodiments of the device **104** may utilize one or more switches positioned on the hand piece **116** that, when activated, may cause the generator **102** to activate the transducer **114**. In one embodiment, for example, the one or more switches may comprise a pair of toggle buttons **136a**, **136b**, for example, to determine an operating mode of the device **104**. When the toggle button **136a** is depressed, for example, the ultrasonic generator **102** may provide a maximum drive signal to the transducer **114**, causing it to produce maximum ultrasonic energy output. Depressing toggle button **136b** may cause the ultrasonic generator **102** to provide a user-selectable drive signal to the transducer **114**, causing it to produce less than the maximum ultrasonic energy output. The device **104** additionally or alternatively may comprise a second switch to, for example, indicate a position of a jaw closure trigger **138** for operating jaws **140** of the end effector **126**. Also, in some embodiments, the ultrasonic generator **102** may be activated based on the position of the jaw closure trigger **138**, (e.g., as the clinician depresses the jaw closure trigger **138** to close the jaws **140**, ultrasonic energy may be applied.

(93) Additionally or alternatively, the one or more switches may comprises a toggle button **136c** that, when depressed, causes the generator **102** to provide a pulsed output. The pulses may be provided at any suitable frequency and grouping, for example. In certain embodiments, the power level of the pulses may be the power levels associated with toggle buttons **136a**, **b** (maximum, less than maximum), for example.

(94) It will be appreciated that a device **104** may comprise any combination of the toggle buttons **136a**, **b**, **c**. For example, the device **104** could be configured to have only two toggle buttons: a toggle button **136a** for producing maximum ultrasonic energy output and a toggle button **136c** for

producing a pulsed output at either the maximum or less than maximum power level per. In this way, the drive signal output configuration of the generator **102** could be 5 continuous signals and 5 or 4 or 3 or 2 or 1 pulsed signals. In certain embodiments, the specific drive signal configuration may be controlled based upon, for example, EEPROM settings in the generator **102** and/or user power level selection(s).

(95) In certain embodiments, a two-position switch may be provided as an alternative to a toggle button **136c**. For example, a device **104** may include a toggle button **136a** for producing a continuous output at a maximum power level and a two-position toggle button **136b**. In a first detented position, toggle button **136b** may produce a continuous output at a less than maximum power level, and in a second detented position the toggle button **136b** may produce a pulsed output (e.g., at either a maximum or less than maximum power level, depending upon the EEPROM settings).

(96) In some embodiments, the end effector **126** may also comprise a pair of electrodes **159**, **157**. The electrodes **159**, **157** may be in communication with the generator **102**, for example, via the cable **122**. The electrodes **159**, **157** may be used, for example, to measure an impedance of a tissue bite present between the clamp arm **155** and the blade **151**. The generator **102** may provide a signal (e.g., a non-therapeutic signal) to the electrodes **159**, **157**. The impedance of the tissue bite may be found, for example, by monitoring the current, voltage, etc. of the signal.

(97) FIG. 4 illustrates one embodiment of an example electrosurgical device **106** that may also be used for transection and sealing. According to various embodiments, the transection and sealing device **106** may comprise a hand piece assembly **130**, a shaft **165** and an end effector **132**. The shaft **165** may be rigid (e.g., for laparoscopic and/or open surgical application) or flexible, as shown, (e.g., for endoscopic application). In various embodiments, the shaft **165** may comprise one or more articulation points. The end effector **132** may comprise jaws **144** having a first jaw member **167** and a second jaw member **169**. The first jaw member **167** and second jaw member **169** may be connected to a clevis **171**, which, in turn, may be coupled to the shaft **165**. A translating member **173** may extend within the shaft **165** from the end effector **132** to the hand piece **130**. At the hand piece **130**, the shaft **165** may be directly or indirectly coupled to a jaw closure trigger **142** (FIG. 4).

(98) The jaw members **167**, **169** of the end effector **132** may comprise respective electrodes **177**, **179**. The electrodes **177**, **179** may be connected to the generator **102** via electrical leads **187a**, **187b** (FIG. 5) extending from the end effector **132** through the shaft **165** and hand piece **130** and ultimately to the generator **102** (e.g., by a multiconductor cable **128**). The generator **102** may provide a drive signal to the electrodes **177**, **179** to bring about a therapeutic effect to tissue present within the jaw members **167**, **169**. The electrodes **177**, **179** may comprise an active electrode and a return electrode, wherein the active electrode and the return electrode can be positioned against, or adjacent to, the tissue to be treated such that current can flow from the active electrode to the return electrode through the tissue. As illustrated in FIG. 4, the end effector **132** is shown with the jaw members **167**, **169** in an open position. A reciprocating blade **175** is illustrated between the jaw members **167**, **169**.

(99) FIGS. 5, 6 and 7 illustrate one embodiment of the end effector **132** shown in FIG. 4. To close the jaws **144** of the end effector **132**, a clinician may cause the jaw closure trigger **142** to pivot along arrow **183** from a first position to a second position. This may cause the jaws **144** to open and close according to any suitable method. For example, motion of the jaw closure trigger **142** may, in turn, cause the translating member **173** to translate within a bore **185** of the shaft **165**. A distal portion of the translating member **173** may be coupled to a reciprocating member **197** such that distal and proximal motion of the translating member **173** causes corresponding distal and proximal motion of the reciprocating member. The reciprocating member **197** may have shoulder portions **191a**, **191b**, while the jaw members **167**, **169** may have corresponding cam surfaces **189a**, **189b**. As the reciprocating member **197** is translated distally from the position shown in FIG. 6 to the position shown in FIG. 7, the shoulder portions **191a**, **191b** may contact the cam surfaces **189a**,

189b, causing the jaw members **167**, **169** to transition to the closed position. Also, in various embodiments, the blade **175** may be positioned at a distal end of the reciprocating member **197**. As the reciprocating member extends to the fully distal position shown in FIG. 7, the blade **175** may be pushed through any tissue present between the jaw members **167**, **169**, in the process, severing it. (100) In use, a clinician may place the end effector **132** and close the jaws **144** around a tissue bite to be acted upon, for example, by pivoting the jaw closure trigger **142** along arrow **183** as described. Once the tissue bite is secure between the jaws **144**, the clinician may initiate the provision of RF or other electro-surgical energy by the generator **102** and through the electrodes **177**, **179**. The provision of RF energy may be accomplished in any suitable way. For example, the clinician may activate the foot switch **120** (FIG. 8) of the generator **102** to initiate the provision of RF energy. Also, for example, the hand piece **130** may comprise one or more switches **181** that may be actuated by the clinician to cause the generator **102** to begin providing RF energy. Additionally, in some embodiments, RF energy may be provided based on the position of the jaw closure trigger **142**. For example, when the trigger **142** is fully depressed (indicating that the jaws **144** are closed), RF energy may be provided. Also, according to various embodiments, the blade **175** may be advanced during closure of the jaws **144** or may be separately advanced by the clinician after closure of the jaws **144** (e.g., after a RF energy has been applied to the tissue).

(101) FIG. 8 is a diagram of the surgical system **100** of FIG. 1. In various embodiments, the generator **102** may comprise several separate functional elements, such as modules and/or blocks. Different functional elements or modules may be configured for driving the different kinds of surgical devices **104**, **106**. For example an ultrasonic generator module **108** may drive an ultrasonic device, such as the ultrasonic device **104**. An electrosurgery/RF generator module **110** may drive the electrosurgical device **106**. For example, the respective modules **108**, **110** may generate respective drive signals for driving the surgical devices **104**, **106**. In various embodiments, the ultrasonic generator module **108** and/or the electrosurgery/RF generator module **110** each may be formed integrally with the generator **102**. Alternatively, one or more of the modules **108**, **110** may be provided as a separate circuit module electrically coupled to the generator **102**. (The modules **108** and **110** are shown in phantom to illustrate this option.) Also, in some embodiments, the electrosurgery/RF generator module **110** may be formed integrally with the ultrasonic generator module **108**, or vice versa.

(102) In accordance with the described embodiments, the ultrasonic generator module **108** may produce a drive signal or signals of particular voltages, currents, and frequencies, e.g. 55,500 cycles per second (Hz). The drive signal or signals may be provided to the ultrasonic device **104**, and specifically to the transducer **114**, which may operate, for example, as described above. In one embodiment, the generator **102** may be configured to produce a drive signal of a particular voltage, current, and/or frequency output signal that can be stepped with high resolution, accuracy, and repeatability.

(103) In accordance with the described embodiments, the electrosurgery/RF generator module **110** may generate a drive signal or signals with output power sufficient to perform bipolar electrosurgery using radio frequency (RF) energy. In bipolar electrosurgery applications. The drive signal may be provided, for example, to the electrodes **177**, **179** of the electrosurgical device **106**, for example, as described above. Accordingly, the generator **102** may be configured for therapeutic purposes by applying electrical energy to the tissue sufficient for treating the tissue (e.g., coagulation, cauterization, tissue welding, etc.).

(104) The generator **102** may comprise an input device **145** (FIG. 1) located, for example, on a front panel of the generator **102** console. The input device **145** may comprise any suitable device that generates signals suitable for programming the operation of the generator **102**. In operation, the user can program or otherwise control operation of the generator **102** using the input device **145**. The input device **145** may comprise any suitable device that generates signals that can be used by the generator (e.g., by one or more processors contained in the generator) to control the operation

of the generator **102** (e.g., operation of the ultrasonic generator module **108** and/or electrosurgery/RF generator module **110**). In various embodiments, the input device **145** includes one or more of buttons, switches, thumbwheels, keyboard, keypad, touch screen monitor, pointing device, remote connection to a general purpose or dedicated computer. In other embodiments, the input device **145** may comprise a suitable user interface, such as one or more user interface screens displayed on a touch screen monitor, for example. Accordingly, by way of the input device **145**, the user can set or program various operating parameters of the generator, such as, for example, current (I), voltage (V), frequency (f), and/or period (T) of a drive signal or signals generated by the ultrasonic generator module **108** and/or electrosurgery/RF generator module **110**.

(105) The generator **102** may also comprise an output device **147** (FIG. 1) located, for example, on a front panel of the generator **102** console. The output device **147** includes one or more devices for providing a sensory feedback to a user. Such devices may comprise, for example, visual feedback devices (e.g., an LCD display screen, LED indicators), audio feedback devices (e.g., a speaker, a buzzer) or tactile feedback devices (e.g., haptic actuators).

(106) Although certain modules and/or blocks of the generator **102** may be described by way of example, it can be appreciated that a greater or lesser number of modules and/or blocks may be used and still fall within the scope of the embodiments. Further, although various embodiments may be described in terms of modules and/or blocks to facilitate description, such modules and/or blocks may be implemented by one or more hardware components, e.g., processors, Digital Signal Processors (DSPs), Programmable Logic Devices (PLDs), Application Specific Integrated Circuits (ASICs), circuits, registers and/or software components, e.g., programs, subroutines, logic and/or combinations of hardware and software components.

(107) In one embodiment, the ultrasonic generator drive module **108** and electrosurgery/RF drive module **110** may comprise one or more embedded applications implemented as firmware, software, hardware, or any combination thereof. The modules **108**, **110** may comprise various executable modules such as software, programs, data, drivers, application program interfaces (APIs), and so forth. The firmware may be stored in nonvolatile memory (NVM), such as in bit-masked read-only memory (ROM) or flash memory. In various implementations, storing the firmware in ROM may preserve flash memory. The NVM may comprise other types of memory including, for example, programmable ROM (PROM), erasable programmable ROM (EPROM), electrically erasable programmable ROM (EEPROM), or battery backed random-access memory (RAM) such as dynamic RAM (DRAM), Double-Data-Rate DRAM (DDRAM), and/or synchronous DRAM (SDRAM).

(108) In one embodiment, the modules **108**, **110** comprise a hardware component implemented as a processor for executing program instructions for monitoring various measurable characteristics of the devices **104**, **106** and generating a corresponding output drive signal or signals for operating the devices **104**, **106**. In embodiments in which the generator **102** is used in conjunction with the device **104**, the drive signal may drive the ultrasonic transducer **114** in cutting and/or coagulation operating modes. Electrical characteristics of the device **104** and/or tissue may be measured and used to control operational aspects of the generator **102** and/or provided as feedback to the user. In embodiments in which the generator **102** is used in conjunction with the device **106**, the drive signal may supply electrical energy (e.g., RF energy) to the end effector **132** in cutting, coagulation and/or desiccation modes. Electrical characteristics of the device **106** and/or tissue may be measured and used to control operational aspects of the generator **102** and/or provided as feedback to the user. In various embodiments, as previously discussed, the hardware components may be implemented as DSP, PLD, ASIC, circuits, and/or registers. In one embodiment, the processor may be configured to store and execute computer software program instructions to generate the step function output signals for driving various components of the devices **104**, **106**, such as the ultrasonic transducer **114** and the end effectors **126**, **132**.

(109) FIG. 9 illustrates an equivalent circuit **150** of an ultrasonic transducer, such as the ultrasonic

transducer **114**, according to one embodiment. The circuit **150** comprises a first “motional” branch having a serially connected inductance $L_{sub.s}$, resistance $R_{sub.s}$ and capacitance $C_{sub.s}$ that define the electromechanical properties of the resonator, and a second capacitive branch having a static capacitance $C_{sub.0}$. Drive current $I_{sub.g}$ may be received from a generator at a drive voltage $V_{sub.g}$, with motional current $I_{sub.m}$ flowing through the first branch and current $I_{sub.g} - I_{sub.m}$ flowing through the capacitive branch. Control of the electromechanical properties of the ultrasonic transducer may be achieved by suitably controlling $I_{sub.g}$ and $V_{sub.g}$. As explained above, known generator architectures may include a tuning inductor $L_{sub.t}$ (shown in phantom in FIG. **9**) for tuning out in a parallel resonance circuit the static capacitance $C_{sub.0}$ at a resonant frequency so that substantially all of generator's current output $I_{sub.g}$ flows through the motional branch. In this way, control of the motional branch current $I_{sub.m}$ is achieved by controlling the generator current output $I_{sub.g}$. The tuning inductor $L_{sub.t}$ is specific to the static capacitance $C_{sub.0}$ of an ultrasonic transducer, however, and a different ultrasonic transducer having a different static capacitance requires a different tuning inductor $L_{sub.t}$. Moreover, because the tuning inductor L is matched to the nominal value of the static capacitance $C_{sub.0}$ at a single resonant frequency, accurate control of the motional branch current $I_{sub.m}$ is assured only at that frequency, and as frequency shifts down with transducer temperature, accurate control of the motional branch current is compromised.

(110) Various embodiments of the generator **102** may not rely on a tuning inductor $L_{sub.t}$ to monitor the motional branch current $I_{sub.m}$. Instead, the generator **102** may use the measured value of the static capacitance $C_{sub.0}$ in between applications of power for a specific ultrasonic surgical device **104** (along with drive signal voltage and current feedback data) to determine values of the motional branch current $I_{sub.m}$ on a dynamic and ongoing basis (e.g., in real-time). Such embodiments of the generator **102** are therefore able to provide virtual tuning to simulate a system that is tuned or resonant with any value of static capacitance $C_{sub.0}$ at any frequency, and not just at a single resonant frequency dictated by a nominal value of the static capacitance $C_{sub.0}$.

(111) FIG. **10** is a simplified block diagram of one embodiment of the generator **102** for providing inductorless tuning as described above, among other benefits. FIGS. **11A-11C** illustrate an architecture of the generator **102** of FIG. **10** according to one embodiment. With reference to FIG. **10**, the generator **102** may comprise a patient isolated stage **152** in communication with a non-isolated stage **154** via a power transformer **156**. A secondary winding **158** of the power transformer **156** is contained in the isolated stage **152** and may comprise a tapped configuration (e.g., a center-tapped or non-center tapped configuration) to define drive signal outputs **160a**, **160b**, **160c** for outputting drive signals to different surgical devices, such as, for example, an ultrasonic surgical device **104** and an electrosurgical device **106**. In particular, drive signal outputs **160a**, **160c** may output a drive signal (e.g., a 420V RMS drive signal) to an ultrasonic surgical device **104**, and drive signal outputs **160b**, **160c** may output a drive signal (e.g., a 100V RMS drive signal) to an electrosurgical device **106**, with output **160b** corresponding to the center tap of the power transformer **156**. The non-isolated stage **154** may comprise a power amplifier **162** having an output connected to a primary winding **164** of the power transformer **156**. In certain embodiments the power amplifier **162** may comprise a push-pull amplifier, for example. The non-isolated stage **154** may further comprise a programmable logic device **166** for supplying a digital output to a digital-to-analog converter (DAC) **168**, which in turn supplies a corresponding analog signal to an input of the power amplifier **162**. In certain embodiments the programmable logic device **166** may comprise a field-programmable gate array (FPGA), for example. The programmable logic device **166**, by virtue of controlling the power amplifier's **162** input via the DAC **168**, may therefore control any of a number of parameters (e.g., frequency, waveform shape, waveform amplitude) of drive signals appearing at the drive signal outputs **160a**, **160b**, **160c**. In certain embodiments and as discussed below, the programmable logic device **166**, in conjunction with a processor (e.g., processor **174** discussed below), may implement a number of digital signal processing (DSP)-based and/or other

control algorithms to control parameters of the drive signals output by the generator **102**.

(112) Power may be supplied to a power rail of the power amplifier **162** by a switch-mode regulator **170**. In certain embodiments the switch-mode regulator **170** may comprise an adjustable buck regulator, for example. The non-isolated stage **154** may further comprise a processor **174**, which in one embodiment may comprise a DSP processor such as an Analog Devices ADSP-21469 SHARC DSP, available from Analog Devices, Norwood, MA, for example. In certain embodiments the processor **174** may control operation of the switch-mode power converter **170** responsive to voltage feedback data received from the power amplifier **162** by the processor **174** via an analog-to-digital converter (ADC) **176**. In one embodiment, for example, the processor **174** may receive as input, via the ADC **176**, the waveform envelope of a signal (e.g., an RF signal) being amplified by the power amplifier **162**. The processor **174** may then control the switch-mode regulator **170** (e.g., via a pulse-width modulated (PWM) output) such that the rail voltage supplied to the power amplifier **162** tracks the waveform envelope of the amplified signal. By dynamically modulating the rail voltage of the power amplifier **162** based on the waveform envelope, the efficiency of the power amplifier **162** may be significantly improved relative to a fixed rail voltage amplifier schemes.

(113) In certain embodiments and as discussed in further detail in connection with FIGS. **13A** and **13B**, the programmable logic device **166**, in conjunction with the processor **174**, may implement a direct digital synthesizer (DDS) control scheme to control the waveform shape, frequency and/or amplitude of drive signals output by the generator **102**. In one embodiment, for example, the programmable logic device **166** may implement a DDS control algorithm **268** by recalling waveform samples stored in a dynamically-updated look-up table (LUT), such as a RAM LUT which may be embedded in an FPGA. This control algorithm is particularly useful for ultrasonic applications in which an ultrasonic transducer, such as the ultrasonic transducer **114**, may be driven by a clean sinusoidal current at its resonant frequency. Because other frequencies may excite parasitic resonances, minimizing or reducing the total distortion of the motional branch current may correspondingly minimize or reduce undesirable resonance effects. Because the waveform shape of a drive signal output by the generator **102** is impacted by various sources of distortion present in the output drive circuit (e.g., the power transformer **156**, the power amplifier **162**), voltage and current feedback data based on the drive signal may be input into an algorithm, such as an error control algorithm implemented by the processor **174**, which compensates for distortion by suitably pre-distorting or modifying the waveform samples stored in the LUT on a dynamic, ongoing basis (e.g., in real-time). In one embodiment, the amount or degree of pre-distortion applied to the LUT samples may be based on the error between a computed motional branch current and a desired current waveform shape, with the error being determined on a sample-by sample basis. In this way, the pre-distorted LUT samples, when processed through the drive circuit, may result in a motional branch drive signal having the desired waveform shape (e.g., sinusoidal) for optimally driving the ultrasonic transducer. In such embodiments, the LUT waveform samples will therefore not represent the desired waveform shape of the drive signal, but rather the waveform shape that is required to ultimately produce the desired waveform shape of the motional branch drive signal when distortion effects are taken into account.

(114) The non-isolated stage **154** may further comprise an ADC **178** and an ADC **180** coupled to the output of the power transformer **156** via respective isolation transformers **182**, **184** for respectively sampling the voltage and current of drive signals output by the generator **102**. In certain embodiments, the ADCs **178**, **180** may be configured to sample at high speeds (e.g., 80 Msps) to enable oversampling of the drive signals. In one embodiment, for example, the sampling speed of the ADCs **178**, **180** may enable approximately 200× (depending on drive frequency) oversampling of the drive signals. In certain embodiments, the sampling operations of the ADCs **178**, **180** may be performed by a single ADC receiving input voltage and current signals via a two-way multiplexer. The use of high-speed sampling in embodiments of the generator **102** may enable,

among other things, calculation of the complex current flowing through the motional branch (which may be used in certain embodiments to implement DDS-based waveform shape control described above), accurate digital filtering of the sampled signals, and calculation of real power consumption with a high degree of precision. Voltage and current feedback data output by the ADCs **178**, **180** may be received and processed (e.g., FIFO buffering, multiplexing) by the programmable logic device **166** and stored in data memory for subsequent retrieval by, for example, the processor **174**. As noted above, voltage and current feedback data may be used as input to an algorithm for pre-distorting or modifying LUT waveform samples on a dynamic and ongoing basis. In certain embodiments, this may require each stored voltage and current feedback data pair to be indexed based on, or otherwise associated with, a corresponding LUT sample that was output by the programmable logic device **166** when the voltage and current feedback data pair was acquired. Synchronization of the LUT samples and the voltage and current feedback data in this manner contributes to the correct timing and stability of the pre-distortion algorithm.

(115) In certain embodiments, the voltage and current feedback data may be used to control the frequency and/or amplitude (e.g., current amplitude) of the drive signals. In one embodiment, for example, voltage and current feedback data may be used to determine impedance phase. The frequency of the drive signal may then be controlled to minimize or reduce the difference between the determined impedance phase and an impedance phase setpoint (e.g., 0°), thereby minimizing or reducing the effects of harmonic distortion and correspondingly enhancing impedance phase measurement accuracy. The determination of phase impedance and a frequency control signal may be implemented in the processor **174**, for example, with the frequency control signal being supplied as input to a DDS control algorithm implemented by the programmable logic device **166**.

(116) In another embodiment, for example, the current feedback data may be monitored in order to maintain the current amplitude of the drive signal at a current amplitude setpoint. The current amplitude setpoint may be specified directly or determined indirectly based on specified voltage amplitude and power setpoints. In certain embodiments, control of the current amplitude may be implemented by control algorithm, such as, for example, a proportional-integral-derivative (PID) control algorithm, in the processor **174**. Variables controlled by the control algorithm to suitably control the current amplitude of the drive signal may include, for example, the scaling of the LUT waveform samples stored in the programmable logic device **166** and/or the full-scale output voltage of the DAC **168** (which supplies the input to the power amplifier **162**) via a DAC **186**.

(117) The non-isolated stage **154** may further comprise a processor **190** for providing, among other things user interface (UI) functionality. In one embodiment, the processor **190** may comprise an Atmel AT91SAM9263 processor having an ARM 926EJ-S core, available from Atmel Corporation, San Jose, California, for example. Examples of UI functionality supported by the processor **190** may include audible and visual user feedback, communication with peripheral devices (e.g., via a Universal Serial Bus (USB) interface), communication with the footswitch **120**, communication with an input device **112** (e.g., a touch screen display) and communication with an output device **147** (e.g., a speaker). The processor **190** may communicate with the processor **174** and the programmable logic device (e.g., via serial peripheral interface (SPI) buses). Although the processor **190** may primarily support UI functionality, it may also coordinate with the processor **174** to implement hazard mitigation in certain embodiments. For example, the processor **190** may be programmed to monitor various aspects of user input and/or other inputs (e.g., touch screen inputs, footswitch **120** inputs, temperature sensor inputs) and may disable the drive output of the generator **102** when an erroneous condition is detected.

(118) In certain embodiments, both the processor **174** and the processor **190** may determine and monitor the operating state of the generator **102**. For the processor **174**, the operating state of the generator **102** may dictate, for example, which control and/or diagnostic processes are implemented by the processor **174**. For the processor **190**, the operating state of the generator **102** may dictate, for example, which elements of a user interface (e.g., display screens, sounds) are presented to a

user. The processors **174**, **190** may independently maintain the current operating state of the generator **102** and recognize and evaluate possible transitions out of the current operating state. The processor **174** may function as the master in this relationship and determine when transitions between operating states are to occur. The processor **190** may be aware of valid transitions between operating states and may confirm if a particular transition is appropriate. For example, when the processor **174** instructs the processor **190** to transition to a specific state, the processor **190** may verify that requested transition is valid. In the event that a requested transition between states is determined to be invalid by the processor **190**, the processor **190** may cause the generator **102** to enter a failure mode.

(119) The non-isolated stage **154** may further comprise a controller **196** for monitoring input devices **145** (e.g., a capacitive touch sensor used for turning the generator **102** on and off, a capacitive touch screen). In certain embodiments, the controller **196** may comprise at least one processor and/or other controller device in communication with the processor **190**. In one embodiment, for example, the controller **196** may comprise a processor (e.g., a Mega168 8-bit controller available from Atmel) configured to monitor user input provided via one or more capacitive touch sensors. In one embodiment, the controller **196** may comprise a touch screen controller (e.g., a QT5480 touch screen controller available from Atmel) to control and manage the acquisition of touch data from a capacitive touch screen.

(120) In certain embodiments, when the generator **102** is in a “power off” state, the controller **196** may continue to receive operating power (e.g., via a line from a power supply of the generator **102**, such as the power supply **211** discussed below). In this way, the controller **196** may continue to monitor an input device **145** (e.g., a capacitive touch sensor located on a front panel of the generator **102**) for turning the generator **102** on and off. When the generator **102** is in the power off state, the controller **196** may wake the power supply (e.g., enable operation of one or more DC/DC voltage converters **213** of the power supply **211**) if activation of the “on/off” input device **145** by a user is detected. The controller **196** may therefore initiate a sequence for transitioning the generator **102** to a “power on” state. Conversely, the controller **196** may initiate a sequence for transitioning the generator **102** to the power off state if activation of the “on/off” input device **145** is detected when the generator **102** is in the power on state. In certain embodiments, for example, the controller **196** may report activation of the “on/off” input device **145** to the processor **190**, which in turn implements the necessary process sequence for transitioning the generator **102** to the power off state. In such embodiments, the controller **196** may have no independent ability for causing the removal of power from the generator **102** after its power on state has been established.

(121) In certain embodiments, the controller **196** may cause the generator **102** to provide audible or other sensory feedback for alerting the user that a power on or power off sequence has been initiated. Such an alert may be provided at the beginning of a power on or power off sequence and prior to the commencement of other processes associated with the sequence.

(122) In certain embodiments, the isolated stage **152** may comprise an instrument interface circuit **198** to, for example, provide a communication interface between a control circuit of a surgical device (e.g., a control circuit comprising handpiece switches) and components of the non-isolated stage **154**, such as, for example, the programmable logic device **166**, the processor **174** and/or the processor **190**. The instrument interface circuit **198** may exchange information with components of the non-isolated stage **154** via a communication link that maintains a suitable degree of electrical isolation between the stages **152**, **154**, such as, for example, an infrared (IR)-based communication link. Power may be supplied to the instrument interface circuit **198** using, for example, a low-dropout voltage regulator powered by an isolation transformer driven from the non-isolated stage **154**.

(123) In one embodiment, the instrument interface circuit **198** may comprise a programmable logic device **200** (e.g., an FPGA) in communication with a signal conditioning circuit **202**. The signal conditioning circuit **202** may be configured to receive a periodic signal from the programmable

logic device **200** (e.g., a 2 kHz square wave) to generate a bipolar interrogation signal having an identical frequency. The interrogation signal may be generated, for example, using a bipolar current source fed by a differential amplifier. The interrogation signal may be communicated to a surgical device control circuit (e.g., by using a conductive pair in a cable that connects the generator **102** to the surgical device) and monitored to determine a state or configuration of the control circuit. As discussed below in connection with FIGS. **16-32**, for example, the control circuit may comprise a number of switches, resistors and/or diodes to modify one or more characteristics (e.g., amplitude, rectification) of the interrogation signal such that a state or configuration of the control circuit is uniquely discernable based on the one or more characteristics. In one embodiment, for example, the signal conditioning circuit **202** may comprise an ADC for generating samples of a voltage signal appearing across inputs of the control circuit resulting from passage of interrogation signal therethrough. The programmable logic device **200** (or a component of the non-isolated stage **154**) may then determine the state or configuration of the control circuit based on the ADC samples. (124) In one embodiment, the instrument interface circuit **198** may comprise a first data circuit interface **204** to enable information exchange between the programmable logic device **200** (or other element of the instrument interface circuit **198**) and a first data circuit disposed in or otherwise associated with a surgical device. In certain embodiments and with reference to FIGS. **33E-33G**, for example, a first data circuit **206** may be disposed in a cable integrally attached to a surgical device handpiece, or in an adaptor for interfacing a specific surgical device type or model with the generator **102**. In certain embodiments, the first data circuit may comprise a non-volatile storage device, such as an electrically erasable programmable read-only memory (EEPROM) device. In certain embodiments and referring again to FIG. **10**, the first data circuit interface **204** may be implemented separately from the programmable logic device **200** and comprise suitable circuitry (e.g., discrete logic devices, a processor) to enable communication between the programmable logic device **200** and the first data circuit. In other embodiments, the first data circuit interface **204** may be integral with the programmable logic device **200**.

(125) In certain embodiments, the first data circuit **206** may store information pertaining to the particular surgical device with which it is associated. Such information may include, for example, a model number, a serial number, a number of operations in which the surgical device has been used, and/or any other type of information. This information may be read by the instrument interface circuit **198** (e.g., by the programmable logic device **200**), transferred to a component of the non-isolated stage **154** (e.g., to programmable logic device **166**, processor **174** and/or processor **190**) for presentation to a user via an output device **147** and/or for controlling a function or operation of the generator **102**. Additionally, any type of information may be communicated to first data circuit **206** for storage therein via the first data circuit interface **204** (e.g., using the programmable logic device **200**). Such information may comprise, for example, an updated number of operations in which the surgical device has been used and/or dates and/or times of its usage.

(126) As discussed previously, a surgical instrument may be detachable from a handpiece (e.g., instrument **124** may be detachable from handpiece **116**) to promote instrument interchangeability and/or disposability. In such cases, known generators may be limited in their ability to recognize particular instrument configurations being used and to optimize control and diagnostic processes accordingly. The addition of readable data circuits to surgical device instruments to address this issue is problematic from a compatibility standpoint, however. For example, designing a surgical device to remain backwardly compatible with generators that lack the requisite data reading functionality may be impractical due to, for example, differing signal schemes, design complexity and cost. Embodiments of instruments discussed below in connection with FIGS. **16-32** address these concerns by using data circuits that may be implemented in existing surgical instruments economically and with minimal design changes to preserve compatibility of the surgical devices with current generator platforms.

(127) Additionally, embodiments of the generator **102** may enable communication with instrument-

based data circuits, such as those described below in connection with FIGS. 16-32 and FIGS. 33A-33C. For example, the generator **102** may be configured to communicate with a second data circuit (e.g., data circuit **284** of FIG. 16) contained in an instrument (e.g., instrument **124** or **134**) of a surgical device. The instrument interface circuit **198** may comprise a second data circuit interface **210** to enable this communication. In one embodiment, the second data circuit interface **210** may comprise a tri-state digital interface, although other interfaces may also be used. In certain embodiments, the second data circuit may generally be any circuit for transmitting and/or receiving data. In one embodiment, for example, the second data circuit may store information pertaining to the particular surgical instrument with which it is associated. Such information may include, for example, a model number, a serial number, a number of operations in which the surgical instrument has been used, and/or any other type of information. Additionally or alternatively, any type of information may be communicated to second data circuit for storage therein via the second data circuit interface **210** (e.g., using the programmable logic device **200**). Such information may comprise, for example, an updated number of operations in which the instrument has been used and/or dates and/or times of its usage. In certain embodiments, the second data circuit may transmit data acquired by one or more sensors (e.g., an instrument-based temperature sensor). In certain embodiments, the second data circuit may receive data from the generator **102** and provide an indication to a user (e.g., an LED indication or other visible indication) based on the received data. (128) In certain embodiments, the second data circuit and the second data circuit interface **210** may be configured such that communication between the programmable logic device **200** and the second data circuit can be effected without the need to provide additional conductors for this purpose (e.g., dedicated conductors of a cable connecting a handpiece to the generator **102**). In one embodiment, for example, information may be communicated to and from the second data circuit using a one-wire bus communication scheme implemented on existing cabling, such as one of the conductors used transmit interrogation signals from the signal conditioning circuit **202** to a control circuit in a handpiece. In this way, design changes or modifications to the surgical device that might otherwise be necessary are minimized or reduced. Moreover, as discussed in further detail below in connection with FIGS. 16-32 and FIGS. 33A-33C, because different types of communications can be implemented over a common physical channel (either with or without frequency-band separation), the presence of a second data circuit may be “invisible” to generators that do not have the requisite data reading functionality, thus enabling backward compatibility of the surgical device instrument.

(129) In certain embodiments, the isolated stage **152** may comprise at least one blocking capacitor **296-1** connected to the drive signal output **160b** to prevent passage of DC current to a patient. A single blocking capacitor may be required to comply with medical regulations or standards, for example. While failure in single-capacitor designs is relatively uncommon, such failure may nonetheless have negative consequences. In one embodiment, a second blocking capacitor **296-2** may be provided in series with the blocking capacitor **296-1**, with current leakage from a point between the blocking capacitors **296-1**, **296-2** being monitored by, for example, an ADC **298** for sampling a voltage induced by leakage current. The samples may be received by the programmable logic device **200**, for example. Based on changes in the leakage current (as indicated by the voltage samples in the embodiment of FIG. 10), the generator **102** may determine when at least one of the blocking capacitors **296-1**, **296-2** has failed. Accordingly, the embodiment of FIG. 10 may provide a benefit over single-capacitor designs having a single point of failure.

(130) In certain embodiments, the non-isolated stage **154** may comprise a power supply **211** for outputting DC power at a suitable voltage and current. The power supply may comprise, for example, a 400 W power supply for outputting a 48 VDC system voltage. The power supply **211** may further comprise one or more DC/DC voltage converters **213** for receiving the output of the power supply to generate DC outputs at the voltages and currents required by the various components of the generator **102**. As discussed above in connection with the controller **196**, one or

more of the DC/DC voltage converters **213** may receive an input from the controller **196** when activation of the “on/off” input device **145** by a user is detected by the controller **196** to enable operation of, or wake, the DC/DC voltage converters **213**.

(131) FIGS. **13A** and **13B** illustrate certain functional and structural aspects of one embodiment of the generator **102**. Feedback indicating current and voltage output from the secondary winding **158** of the power transformer **156** is received by the ADCs **178**, **180**, respectively. As shown, the ADCs **178**, **180** may be implemented as a 2-channel ADC and may sample the feedback signals at a high speed (e.g., 80 Msps) to enable oversampling (e.g., approximately 200× oversampling) of the drive signals. The current and voltage feedback signals may be suitably conditioned in the analog domain (e.g., amplified, filtered) prior to processing by the ADCs **178**, **180**. Current and voltage feedback samples from the ADCs **178**, **180** may be individually buffered and subsequently multiplexed or interleaved into a single data stream within block **212** of the programmable logic device **166**. In the embodiment of FIGS. **13A** and **13B**, the programmable logic device **166** comprises an FPGA.

(132) The multiplexed current and voltage feedback samples may be received by a parallel data acquisition port (PDAP) implemented within block **214** of the processor **174**. The PDAP may comprise a packing unit for implementing any of a number of methodologies for correlating the multiplexed feedback samples with a memory address. In one embodiment, for example, feedback samples corresponding to a particular LUT sample output by the programmable logic device **166** may be stored at one or more memory addresses that are correlated or indexed with the LUT address of the LUT sample. In another embodiment, feedback samples corresponding to a particular LUT sample output by the programmable logic device **166** may be stored, along with the LUT address of the LUT sample, at a common memory location. In any event, the feedback samples may be stored such that the address of an LUT sample from which a particular set of feedback samples originated may be subsequently ascertained. As discussed above, synchronization of the LUT sample addresses and the feedback samples in this way contributes to the correct timing and stability of the pre-distortion algorithm. A direct memory access (DMA) controller implemented at block **216** of the processor **174** may store the feedback samples (and any LUT sample address data, where applicable) at a designated memory location **218** of the processor **174** (e.g., internal RAM).

(133) Block **220** of the processor **174** may implement a pre-distortion algorithm for pre-distorting or modifying the LUT samples stored in the programmable logic device **166** on a dynamic, ongoing basis. As discussed above, pre-distortion of the LUT samples may compensate for various sources of distortion present in the output drive circuit of the generator **102**. The pre-distorted LUT samples, when processed through the drive circuit, will therefore result in a drive signal having the desired waveform shape (e.g., sinusoidal) for optimally driving the ultrasonic transducer.

(134) At block **222** of the pre-distortion algorithm, the current through the motional branch of the ultrasonic transducer is determined. The motional branch current may be determined using Kirchoff's Current Law based on, for example, the current and voltage feedback samples stored at memory location **218** (which, when suitably scaled, may be representative of $I_{\text{sub.g}}$ and $V_{\text{sub.g}}$ in the model of FIG. **9** discussed above), a value of the ultrasonic transducer static capacitance $C_{\text{sub.0}}$ (measured or known a priori) and a known value of the drive frequency. A motional branch current sample for each set of stored current and voltage feedback samples associated with a LUT sample may be determined.

(135) At block **224** of the pre-distortion algorithm, each motional branch current sample determined at block **222** is compared to a sample of a desired current waveform shape to determine a difference, or sample amplitude error, between the compared samples. For this determination, the sample of the desired current waveform shape may be supplied, for example, from a waveform shape LUT **226** containing amplitude samples for one cycle of a desired current waveform shape. The particular sample of the desired current waveform shape from the LUT **226** used for the comparison may be dictated by the LUT sample address associated with the motional branch

current sample used in the comparison. Accordingly, the input of the motional branch current to block **224** may be synchronized with the input of its associated LUT sample address to block **224**. The LUT samples stored in the programmable logic device **166** and the LUT samples stored in the waveform shape LUT **226** may therefore be equal in number. In certain embodiments, the desired current waveform shape represented by the LUT samples stored in the waveform shape LUT **226** may be a fundamental sine wave. Other waveform shapes may be desirable. For example, it is contemplated that a fundamental sine wave for driving main longitudinal motion of an ultrasonic transducer superimposed with one or more other drive signals at other frequencies, such as a third order harmonic for driving at least two mechanical resonances for beneficial vibrations of transverse or other modes, could be used.

(136) Each value of the sample amplitude error determined at block **224** may be transmitted to the LUT of the programmable logic device **166** (shown at block **228** in FIG. **13A**) along with an indication of its associated LUT address. Based on the value of the sample amplitude error and its associated address (and, optionally, values of sample amplitude error for the same LUT address previously received), the LUT **228** (or other control block of the programmable logic device **166**) may pre-distort or modify the value of the LUT sample stored at the LUT address such that the sample amplitude error is reduced or minimized. It will be appreciated that such pre-distortion or modification of each LUT sample in an iterative manner across the entire range of LUT addresses will cause the waveform shape of the generator's output current to match or conform to the desired current waveform shape represented by the samples of the waveform shape LUT **226**.

(137) Current and voltage amplitude measurements, power measurements and impedance measurements may be determined at block **230** of the processor **174** based on the current and voltage feedback samples stored at memory location **218**. Prior to the determination of these quantities, the feedback samples may be suitably scaled and, in certain embodiments, processed through a suitable filter **232** to remove noise resulting from, for example, the data acquisition process and induced harmonic components. The filtered voltage and current samples may therefore substantially represent the fundamental frequency of the generator's drive output signal. In certain embodiments, the filter **232** may be a finite impulse response (FIR) filter applied in the frequency domain. Such embodiments may use the fast Fourier transform (FFT) of the output drive signal current and voltage signals. In certain embodiments, the resulting frequency spectrum may be used to provide additional generator functionality. In one embodiment, for example, the ratio of the second and/or third order harmonic component relative to the fundamental frequency component may be used as a diagnostic indicator.

(138) At block **234**, a root mean square (RMS) calculation may be applied to a sample size of the current feedback samples representing an integral number of cycles of the drive signal to generate a measurement $I_{sub.rms}$ representing the drive signal output current.

(139) At block **236**, a root mean square (RMS) calculation may be applied to a sample size of the voltage feedback samples representing an integral number of cycles of the drive signal to determine a measurement $V_{sub.rms}$ representing the drive signal output voltage.

(140) At block **238**, the current and voltage feedback samples may be multiplied point by point, and a mean calculation is applied to samples representing an integral number of cycles of the drive signal to determine a measurement $P_{sub.r}$ of the generator's real output power.

(141) At block **240**, measurement $P_{sub.a}$ of the generator's apparent output power may be determined as the product $V_{sub.rms} \cdot I_{sub.rms}$.

(142) At block **242**, measurement $Z_{sub.m}$ of the load impedance magnitude may be determined as the quotient $V_{sub.rms}/I_{sub.rms}$.

(143) In certain embodiments, the quantities $I_{sub.rms}$, $V_{sub.rms}$, $P_{sub.r}$, $P_{sub.a}$ and $Z_{sub.m}$ determined at blocks **234**, **236**, **238**, **240** and **242** may be used by the generator **102** to implement any of number of control and/or diagnostic processes. In certain embodiments, any of these quantities may be communicated to a user via, for example, an output device **147** integral with the

generator **102** or an output device **147** connected to the generator **102** through a suitable communication interface (e.g., a USB interface). Various diagnostic processes may include, without limitation, handpiece integrity, instrument integrity, instrument attachment integrity, instrument overload, approaching instrument overload, frequency lock failure, over-voltage, over-current, over-power, voltage sense failure, current sense failure, audio indication failure, visual indication failure, short circuit, power delivery failure, blocking capacitor failure, for example.

(144) Block **244** of the processor **174** may implement a phase control algorithm for determining and controlling the impedance phase of an electrical load (e.g., the ultrasonic transducer) driven by the generator **102**. As discussed above, by controlling the frequency of the drive signal to minimize or reduce the difference between the determined impedance phase and an impedance phase setpoint (e.g., 0°), the effects of harmonic distortion may be minimized or reduced, and the accuracy of the phase measurement increased.

(145) The phase control algorithm receives as input the current and voltage feedback samples stored in the memory location **218**. Prior to their use in the phase control algorithm, the feedback samples may be suitably scaled and, in certain embodiments, processed through a suitable filter **246** (which may be identical to filter **232**) to remove noise resulting from the data acquisition process and induced harmonic components, for example. The filtered voltage and current samples may therefore substantially represent the fundamental frequency of the generator's drive output signal.

(146) At block **248** of the phase control algorithm, the current through the motional branch of the ultrasonic transducer is determined. This determination may be identical to that described above in connection with block **222** of the pre-distortion algorithm. The output of block **248** may thus be, for each set of stored current and voltage feedback samples associated with a LUT sample, a motional branch current sample.

(147) At block **250** of the phase control algorithm, impedance phase is determined based on the synchronized input of motional branch current samples determined at block **248** and corresponding voltage feedback samples. In certain embodiments, the impedance phase is determined as the average of the impedance phase measured at the rising edge of the waveforms and the impedance phase measured at the falling edge of the waveforms.

(148) At block **252** of the of the phase control algorithm, the value of the impedance phase determined at block **222** is compared to phase setpoint **254** to determine a difference, or phase error, between the compared values.

(149) At block **256** of the phase control algorithm, based on a value of phase error determined at block **252** and the impedance magnitude determined at block **242**, a frequency output for controlling the frequency of the drive signal is determined. The value of the frequency output may be continuously adjusted by the block **256** and transferred to a DDS control block **268** (discussed below) in order to maintain the impedance phase determined at block **250** at the phase setpoint (e.g., zero phase error). In certain embodiments, the impedance phase may be regulated to a 0° phase setpoint. In this way, any harmonic distortion will be centered about the crest of the voltage waveform, enhancing the accuracy of phase impedance determination.

(150) Block **258** of the processor **174** may implement an algorithm for modulating the current amplitude of the drive signal in order to control the drive signal current, voltage and power in accordance with user specified setpoints, or in accordance with requirements specified by other processes or algorithms implemented by the generator **102**. Control of these quantities may be realized, for example, by scaling the LUT samples in the LUT **228** and/or by adjusting the full-scale output voltage of the DAC **168** (which supplies the input to the power amplifier **162**) via a DAC **186**. Block **260** (which may be implemented as a PID controller in certain embodiments) may receive as input current feedback samples (which may be suitably scaled and filtered) from the memory location **218**. The current feedback samples may be compared to a "current demand" I.sub.d value dictated by the controlled variable (e.g., current, voltage or power) to determine if the drive signal is supplying the necessary current. In embodiments in which drive signal current is the

control variable, the current demand $I_{sub.d}$ may be specified directly by a current setpoint **262A** ($I_{sub.sp}$). For example, an RMS value of the current feedback data (determined as in block **234**) may be compared to user-specified RMS current setpoint $I_{sub.sp}$ to determine the appropriate controller action. If, for example, the current feedback data indicates an RMS value less than the current setpoint $I_{sub.sp}$, LUT scaling and/or the full-scale output voltage of the DAC **168** may be adjusted by the block **260** such that the drive signal current is increased. Conversely, block **260** may adjust LUT scaling and/or the full-scale output voltage of the DAC **168** to decrease the drive signal current when the current feedback data indicates an RMS value greater than the current setpoint $I_{sub.sp}$.

(151) In embodiments in which the drive signal voltage is the control variable, the current demand $I_{sub.d}$ may be specified indirectly, for example, based on the current required maintain a desired voltage setpoint **262B** ($V_{sub.sp}$) given the load impedance magnitude $Z_{sub.m}$ measured at block **242** (e.g. $I_{sub.d}=V_{sub.sp}/Z_{sub.m}$). Similarly, in embodiments in which drive signal power is the control variable, the current demand $I_{sub.d}$ may be specified indirectly, for example, based on the current required to maintain a desired power setpoint **262C** ($P_{sub.sp}$) given the voltage $V_{sub.rms}$ measured at blocks **236** (e.g. $I_{sub.d}=P_{sub.sp}/V_{sub.rms}$).

(152) Block **268** may implement a DDS control algorithm for controlling the drive signal by recalling LUT samples stored in the LUT **228**. In certain embodiments, the DDS control algorithm be a numerically-controlled oscillator (NCO) algorithm for generating samples of a waveform at a fixed clock rate using a point (memory location)-skipping technique. The NCO algorithm may implement a phase accumulator, or frequency-to-phase converter, that functions as an address pointer for recalling LUT samples from the LUT **228**. In one embodiment, the phase accumulator may be a D step size, modulo N phase accumulator, where D is a positive integer representing a frequency control value, and N is the number of LUT samples in the LUT **228**. A frequency control value of $D=1$, for example, may cause the phase accumulator to sequentially point to every address of the LUT **228**, resulting in a waveform output replicating the waveform stored in the LUT **228**. When $D>1$, the phase accumulator may skip addresses in the LUT **228**, resulting in a waveform output having a higher frequency. Accordingly, the frequency of the waveform generated by the DDS control algorithm may therefore be controlled by suitably varying the frequency control value. In certain embodiments, the frequency control value may be determined based on the output of the phase control algorithm implemented at block **244**. The output of block **268** may supply the input of DAC **168**, which in turn supplies a corresponding analog signal to an input of the power amplifier **162**.

(153) Block **270** of the processor **174** may implement a switch-mode converter control algorithm for dynamically modulating the rail voltage of the power amplifier **162** based on the waveform envelope of the signal being amplified, thereby improving the efficiency of the power amplifier **162**. In certain embodiments, characteristics of the waveform envelope may be determined by monitoring one or more signals contained in the power amplifier **162**. In one embodiment, for example, characteristics of the waveform envelope may be determined by monitoring the minima of a drain voltage (e.g., a MOSFET drain voltage) that is modulated in accordance with the envelope of the amplified signal. A minima voltage signal may be generated, for example, by a voltage minima detector coupled to the drain voltage. The minima voltage signal may be sampled by ADC **176**, with the output minima voltage samples being received at block **272** of the switch-mode converter control algorithm. Based on the values of the minima voltage samples, block **274** may control a PWM signal output by a PWM generator **276**, which, in turn, controls the rail voltage supplied to the power amplifier **162** by the switch-mode regulator **170**. In certain embodiments, as long as the values of the minima voltage samples are less than a minima target **278** input into block **262**, the rail voltage may be modulated in accordance with the waveform envelope as characterized by the minima voltage samples. When the minima voltage samples indicate low envelope power levels, for example, block **274** may cause a low rail voltage to be

supplied to the power amplifier **162**, with the full rail voltage being supplied only when the minima voltage samples indicate maximum envelope power levels. When the minima voltage samples fall below the minima target **278**, block **274** may cause the rail voltage to be maintained at a minimum value suitable for ensuring proper operation of the power amplifier **162**.

(154) FIGS. **33A-33C** illustrate control circuits of surgical devices according to various embodiments. As discussed above in connection with FIG. **10**, a control circuit may modify characteristics of an interrogation signal transmitted by the generator **102**. The characteristics of the interrogation signal, which may uniquely indicate a state or configuration of the control circuit, can be discerned by the generator **102** and used to control aspects of its operation. The control circuits may be contained in an ultrasonic surgical device (e.g., in the handpiece **116** of the ultrasonic surgical device **104**), or in an electrosurgical device (e.g., in the handpiece **130** of the electrosurgical device **106**).

(155) Referring to the embodiment of FIG. **33A**, a control circuit **300-1** may be connected to the generator **102** to receive an interrogation signal (e.g., a bipolar interrogation signal at 2 kHz) from the signal conditioning circuit **202** (e.g., from generator terminals HS and SR (FIG. **10**) via a conductive pair of cable **112** or cable **128**). The control circuit **300-1** may comprise a first branch that includes series-connected diodes **D1** and **D2** and a switch **SW1** connected in parallel with **D2**. The control circuit **300-1** may also comprise a second branch that includes series-connected diodes **D3**, **D4** and **D5**, a switch **SW2** connected in parallel with **D4**, and a resistor **R1** connected in parallel with **D5**. In certain embodiments and as shown, **D5** may be a Zener diode. The control circuit **300-1** may additionally comprise a data storage element **302** that, together with one or more components of the second branch (e.g., **D5**, **R1**), define a data circuit **304**. In certain embodiments, the data storage element **302**, and possibly other components of the data circuit **304**, may be contained in the instrument (e.g., instrument **124**, instrument **134**) of the surgical device, with other components of the control circuit **300-1** (e.g., **SW1**, **SW2**, **D1**, **D2**, **D3**, **D4**) being contained in the handpiece (e.g., handpiece **116**, handpiece **130**). In certain embodiments, the data storage element **302** may be a single-wire bus device (e.g., a single-wire protocol EEPROM), or other single-wire protocol or local interconnect network (LIN) protocol device. In one embodiment, for example, the data storage element **302** may comprise a Maxim DS28EC20 EEPROM, available from Maxim Integrated Products, Inc., Sunnyvale, CA, known under the trade name “1-Wire.” The data storage element **302** is one example of a circuit element that may be contained in the data circuit **304**. The data circuit **304** may additionally or alternatively comprise one or more other circuit elements or components capable of transmitting or receiving data. Such circuit elements or components may be configured to, for example, transmit data acquired by one or more sensors (e.g., an instrument-based temperature sensor) and/or receive data from the generator **102** and provide an indication to a user (e.g., an LED indication or other visible indication) based on the received data.

(156) During operation, an interrogation signal (e.g., a bipolar interrogation signal at 2 kHz) from the signal conditioning circuit **202** may be applied across both branches of the control circuit **300-1**. In this way, the voltage appearing across the branches may be uniquely determined by the states of **SW1** and **SW2**. For example, when **SW1** is open, the voltage drop across the control circuit **300-1** for negative values of the interrogation signal will be sum of the forward voltage drops across **D1** and **D2**. When **SW1** is closed, the voltage drop for negative values of the interrogation signal will be determined by the forward voltage drop of **D1** only. Thus, for example, with a forward voltage drop of 0.7 volts for each of **D1** and **D2**, open and closed states of **SW1** may correspond to voltage drops of 1.4 volts and 0.7 volts, respectively. In the same way, the voltage drop across the control circuit **300-1** for positive values of the interrogation signal may be uniquely determined by the state of **SW2**. For example, when **SW2** is open, the voltage drop across the control circuit **300-1** will be the sum of the forward voltage drops across **D3** and **D4** (e.g., 1.4 volts) and the breakdown voltage of **D5** (e.g., 3.3 volts). When **SW2** is closed, the voltage drop across the control circuit **300-1** will be the sum of the forward voltage drop across **D3** and the breakdown voltage of **D5**. Accordingly,

the state or configuration of SW1 and SW2 may be discerned by the generator 102 based on the interrogation signal voltage appearing across the inputs of the control circuit 300-1 (e.g., as measured by an ADC of the signal conditioning circuit 202).

(157) In certain embodiments, the generator 102 may be configured to communicate with the data circuit 304, and, in particular, with the data storage element 302, via the second data circuit interface 210 (FIG. 10) and the conductive pair of cable 112 or cable 128. The frequency band of the communication protocol used to communicate with the data circuit 304 may be higher than the frequency band of the interrogation signal. In certain embodiments, for example, the frequency of the communication protocol for the data storage element 302 may be, for example, 200 kHz or a significantly higher frequency, whereas the frequency of the interrogation signal used to determine the different states of SW1 and SW2 may be, for example, 2 kHz. Diode D5 may limit the voltage supplied to the data storage element 302 to a suitable operating range (e.g., 3.3-5V).

(158) As explained above in connection with FIG. 10, the data circuit 304, and, in particular, the data storage element 302, may store information pertaining to the particular surgical instrument with which it is associated. Such information may be retrieved by the generator 102 and include, for example, a model number, a serial number, a number of operations in which the surgical instrument has been used, and/or any other type of information. Additionally, any type of information may be communicated from the generator 102 to the data circuit 304 for storage in the data storage element 302. Such information may comprise, for example, an updated number of operations in which the instrument has been used and/or dates and/or times of its usage.

(159) As noted above, the data circuit 304 may additionally or alternatively comprise components or elements other than the data storage element 302 for transmitting or receiving data. Such components or elements may be configured to, for example, transmit data acquired by one or more sensors (e.g., an instrument-based temperature sensor) and/or receive data from the generator 102 and provide an indication to a user (e.g., an LED indication or other visible indication) based on the received data.

(160) Embodiments of the control circuit may comprise additional switches. With reference to the embodiment of FIG. 33B, for example, control circuit 300-2 may comprise a first branch having a first switch SW1 and a second switch SW2 (for a total of three switches), with each combination of SW1 and SW2 states corresponding to a unique voltage drop across the control circuit 300-2 for negative values of the interrogation signal. For example, the open and closed states of SW1 add or remove, respectively, the forward voltage drops of D2 and D3, and the open and closed states of SW2 add or remove, respectively, the forward voltage drop of D4. In the embodiment of FIG. 33C, the first branch of control circuit 300-3 comprises three switches (for a total of four switches), with the breakdown voltage of Zener diode D2 being used to distinguish changes in the voltage drop resulting from the operation of SW1 from voltage changes resulting from the operation of SW2 and SW3.

(161) FIGS. 14 and 15 illustrate control circuits of surgical devices according to various embodiments. As discussed above in connection with FIG. 10, a control circuit may modify characteristics of an interrogation signal transmitted by the generator 102. The characteristics of the interrogation signal, which may uniquely indicate the state or configuration of the control circuit, can be discerned by the generator 102 and used to control aspects of its operation. The control circuit 280 of FIG. 14 may be contained in an ultrasonic surgical device (e.g., in handpiece 116 of ultrasonic surgical device 104), and the control circuit 282 of FIG. 15 may be contained in an electrosurgical device (e.g., in handpiece 130 of electrosurgical device 106).

(162) Referring to FIG. 14, control circuit 280 may be connected to the generator 102 to receive an interrogation signal (e.g., a bipolar interrogation signal at 2 kHz) from the signal conditioning circuit 202 (e.g., from generator terminals HS and SR (FIG. 10) via a conductive pair of cable 112). The control circuit 280 may comprise a first switch SW1 in series with a first diode D1 to define a first branch, and a second switch SW2 in series with a second diode D2 to define a second branch.

The first and second branches may be connected in parallel such that the forward conduction direction of D2 is opposite that of D1. The interrogation signal may be applied across both branches. When both SW1 and SW2 are open, the control circuit 280 may define an open circuit. When SW1 is closed and SW2 is open, the interrogation signal may undergo half-wave rectification in a first direction (e.g., positive half of interrogation signal blocked). When SW1 is open and SW2 is closed, the interrogation signal may undergo half-wave rectification in a second direction (e.g., negative half of interrogation signal blocked). When both SW1 and SW2 are closed, no rectification may occur. Accordingly, based on the different characteristics of the interrogation signal corresponding to the different states of SW1 and SW2, the state or configuration of the control circuit 280 may be discerned by the generator 102 based on a voltage signal appearing across the inputs of the control circuit 280 (e.g., as measured by an ADC of the signal conditioning circuit 202).

(163) In certain embodiments and as shown in FIG. 14, the cable 112 may comprise a data circuit 206. The data circuit 206 may comprise, for example, a non-volatile storage device, such as an EEPROM device. The generator 102 may exchange information with the data circuit 206 via the first data circuit interface 204 as discussed above in connection with FIG. 10. Such information may be specific to a surgical device integral with, or configured for use with, the cable 112 and may comprise, for example, a model number, a serial number, a number of operations in which the surgical device has been used, and/or any other type of information. Information may also be communicated from the generator 102 to the data circuit 206 for storage therein, as discussed above in connection with FIG. 10. In certain embodiments and with reference to FIGS. 33E-33G, the data circuit 206 may be disposed in an adaptor for interfacing a specific surgical device type or model with the generator 102.

(164) Referring to FIG. 15, control circuit 282 may be connected to the generator 102 to receive an interrogation signal (e.g., a bipolar interrogation signal at 2 kHz) from the signal conditioning circuit 202 (e.g., from generator terminals HS and SR (FIG. 10) via a conductive pair of cable 128). The control circuit 282 may comprise series-connected resistors R2, R3 and R4, with switches SW1 and SW2 connected across R2 and R4, respectively. The interrogation signal may be applied across at least one of the series-connected resistors to generate a voltage drop across the control circuit 282. For example, when both SW1 and SW2 are open, the voltage drop may be determined by R2, R3 and R4. When SW1 is closed and SW2 is open, the voltage drop may be determined by R3 and R4. When SW1 is open and SW2 is closed, the voltage drop may be determined by R2 and R3. When both SW1 and SW2 are closed, the voltage drop may be determined by R3. Accordingly, based on the voltage drop across the control circuit 282 (e.g., as measured by an ADC of the signal conditioning circuit 202), the state or configuration of the control circuit 282 may be discerned by the generator 102.

(165) FIG. 16 illustrates one embodiment of a control circuit 280-1 of an ultrasonic surgical device, such as the ultrasonic surgical device 104. The control circuit 280-1, in addition to comprising components of the control circuit 280 of FIG. 14, may comprise a data circuit 284 having a data storage element 286. In certain embodiments, the data storage element 286, and possibly other components of the data circuit 284, may be contained in the instrument (e.g., instrument 124) of the ultrasonic surgical device, with other components of the control circuit 280-1 (e.g., SW1, SW2, D1, D2, D3, D4, C1) being contained in the handpiece (e.g., handpiece 116). In certain embodiments, the data storage element 286 may be a single-wire bus device (e.g., a single-wire protocol EEPROM), or other single-wire protocol or local interconnect network (LIN) protocol device. In one embodiment, for example, the data storage element 286 may comprise a Maxim DS28EC20 one-wire EEPROM, available from Maxim Integrated Products, Inc., Sunnyvale, CA, known under the trade name "1-Wire."

(166) In certain embodiments, the generator 102 may be configured to communicate with the data circuit 284, and, in particular, with the data storage element 286, via the second data circuit

interface **210** (FIG. **10**) and the conductive pair of the cable **112**. In particular, the frequency band of the communication protocol used to communicate with the data circuit **284** may be higher than the frequency band of the interrogation signal. In certain embodiments, for example, the frequency of the communication protocol for the data storage element **286** may be, for example, 200 kHz or a significantly higher frequency, whereas the frequency of the interrogation signal used to determine the different states of SW**1** and SW**2** may be, for example, 2 kHz. Accordingly, the value of capacitor C**1** of the data circuit **284** may be selected such that the data storage element **286** is “hidden” from the relatively low frequency of the interrogation signal while allowing the generator **102** to communicate with the data storage element **286** at the higher frequency of the communication protocol. A series diode D**3** may protect the data storage element **286** from negative cycles of the interrogation signal, and a parallel Zener diode D**4** may limit the voltage supplied to the data storage element **286** to a suitable operating range (e.g., 3.3-5V). When in the forward conduction mode, D**4** may also clamp negative cycles of the interrogation signal to ground.

(167) As explained above in connection with FIG. **10**, the data circuit **284**, and, in particular, the data storage element **286**, may store information pertaining to the particular surgical instrument with which it is associated. Such information may be retrieved by the generator **102** and include, for example, a model number, a serial number, a number of operations in which the surgical instrument has been used, and/or any other type of information. Additionally, any type of information may be communicated from the generator **102** to the data circuit **284** for storage in the data storage element **286**. Such information may comprise, for example, an updated number of operations in which the instrument has been used and/or dates and/or times of its usage. Moreover, because the different types of communications between the generator **102** and the surgical device may be frequency-band separated, the presence of the data storage element **286** may be “invisible” to generators that do not have the requisite data reading functionality, thus enabling backward compatibility of the surgical device.

(168) In certain embodiments and as shown in FIG. **17**, the data circuit **284-1** may comprise an inductor L**1** to provide isolation of the data storage element **286** from the states of SW**1** and SW**2**. The addition of L**1** may additionally enable use of the data circuit **284-1** in electrosurgical devices. FIG. **18**, for example, illustrates one embodiment of a control circuit **282-1** that combines the control circuit **282** of FIG. **15** with the data circuit **284-1** of FIG. **17**.

(169) In certain embodiments, a data circuit may comprise one or more switches to modify one or more characteristics (e.g., amplitude, rectification) of an interrogation signal received by the data circuit such that a state or configuration of the one or more switches is uniquely discernable based on the one or more characteristics. FIG. **19**, for example, illustrates one embodiment of a control circuit **282-2** in which the data circuit **284-2** comprises a switch SW**3** connected in parallel with D**4**. An interrogation signal may be communicated from the generator **102** (e.g., from the signal conditioning circuit **202** of FIG. **10**) at a frequency sufficient for the interrogation signal to be received by the data circuit **284-2** via C**1** but blocked from other portions of the control circuit **282-2** by L**1**. In this way, one or more characteristics of a first interrogation signal (e.g., a bipolar interrogation signal at 25 kHz) may be used to discern the state of SW**3**, and one or more characteristics of a second interrogation signal at a lower frequency (e.g., a bipolar interrogation signal at 2 kHz) may be used to discern the states of SW**1** and SW**2**. Although the addition of SW**3** is illustrated in connection with the control circuit **282-2** in an electrosurgical device, it will be appreciated that SW**3** may be added to a control circuit of an ultrasonic surgical device, such as, for example, the control circuit **280-2** of FIG. **17**.

(170) Additionally, it will be appreciated that switches in addition to SW**3** may be added to a data circuit. As shown in FIGS. **20** and **21**, for example, embodiments of the data circuit **284-3** and **284-4**, respectively, may comprise a second switch SW**4**. In FIG. **20**, voltage values of Zener diodes D**5** and D**6** may be selected such that their voltage values sufficiently differ to allow reliable discrimination of the interrogation signal in the presence of noise. The sum of the voltage values

of D5 and D6 may be equal to or less than the voltage value of D4. In certain embodiments, depending upon the voltages values of D5 and D6, it may be possible to eliminate D4 from the embodiment of the data circuit **284-3** illustrated in FIG. **20**.

(171) In certain cases, the switches (e.g., SW1-SW4) may impede the ability of the generator **102** to communicate with the data storage element **286**. In one embodiment, this issue may be addressed by declaring an error if the states of the switches are such that they will interfere with communication between the generator **102** and the data storage element **286**. In another embodiment, the generator **102** may only permit communication with the data storage element **286** when determined by the generator **102** that the states of the switches will not interfere with the communication. Because the states of the switches may be unpredictable to an extent, the generator **102** may make this determination on a recurring basis. The addition of L1 in certain embodiments may prevent interference caused by switches external to the data circuit (e.g., SW1 and SW2). For switches contained within the data circuit (e.g., SW3 and SW4), isolation of the switches by frequency band separation may be realized by the addition of a capacitor C2 having a capacitance value significantly smaller than C1 (e.g., $C2 \ll C1$). Embodiments of data circuits **284-5**, **284-6**, **284-7** comprising C2 are shown in FIGS. **22-24**, respectively.

(172) In any of the embodiments of FIGS. **16-24**, depending on the frequency response characteristics of D4, it may be desirable or necessary to add a fast diode in parallel with D4 and pointing in the same direction.

(173) FIG. **25** illustrates one embodiment of a control circuit **280-5** in which communication between the generator **102** and a data storage element is implemented using an amplitude-modulated communication protocol (e.g., amplitude-modulated one-wire protocol [known under the trade name “1-Wire”] amplitude-modulated LIN protocol). Amplitude modulation of the communication protocol on a high-frequency carrier (e.g., 8 MHz or higher) substantially increases frequency band separation between low frequency interrogation signals (e.g., interrogation signals at 2 kHz) and the native “baseband” frequency of the communication protocol used in the embodiments of FIGS. **16-24**. The control circuit **280-5** may be similar to the control circuit **280-1** of FIG. **16**, with the data circuit **288** comprising an additional capacitor C3 and resistor R5, which, in conjunction with D3, demodulate the amplitude-modulated communication protocol for receipt by the data storage element **286**. As in the embodiment of FIG. **16**, D3 may protect the data storage element **286** from negative cycles of the interrogation signal, and D4 may limit the voltage supplied to the data storage element **286** to a suitable operating range (e.g., 3.3-5V) and clamp negative cycles of the interrogation signal to ground when in the forward conduction mode. The increased frequency separation may allow C1 to be somewhat small relative to the embodiments of FIGS. **16-24**. Additionally, the higher frequency of the carrier signal may also improve noise immunity of communications with the data storage element because it is further removed from the frequency range of electrical noise that may be generated by other surgical devices used in the same operating room environment. In certain embodiments, the relatively high frequency of the carrier in combination with the frequency response characteristics of D4 may make it desirable or necessary to add a fast diode in parallel with D4 and pointing in the same direction.

(174) With the addition of an inductor L1 to prevent interference with data storage element **286** communications caused by switches external to the data circuit **288** (e.g., SW1 and SW2), the data circuit **288** may be used in control circuits of electrosurgical instruments, as shown in the embodiment of the data circuit **288-1** of FIG. **26**.

(175) With the exception of C2 and R3, and the more likely need for D7, the embodiments of FIGS. **25** and **26** are similar to the “baseband” embodiments of FIGS. **16-24**. For example, the manner in which switches may be added to the data circuits of FIGS. **19-21** is directly applicable to the embodiments of FIGS. **25** and **26** (including the possibility of eliminating D4 from the modulated-carrier equivalent of the FIG. **20**). Modulated-carrier equivalents of the data circuits embodied in FIGS. **22-24** may simply require the addition of an appropriately-sized inductor L2 in

series with C2 in order to isolate the interrogation frequency for the additional switches (e.g., SW3, SW4) to an intermediate frequency band between the carrier frequency and the lower interrogation frequency for switches external to the data circuit. An embodiment of one such data circuit **282-7** is shown in FIG. 27.

(176) In the embodiment of FIG. 27, any interference with the generator's ability to communicate with the data storage element **286** caused by states of SW1 and SW2 may be addressed as described above in connection with the embodiments of FIGS. 19-24. For example, the generator **102** may declare an error if switch states will prevent communication, or the generator **102** may only permit communication when determined by the generator **102** that the switch states will not cause interference.

(177) In certain embodiments, the data circuit may not comprise a data storage element **286** (e.g., an EEPROM device) to store information. FIGS. 28-32 illustrate embodiments of control circuits that utilize resistive and/or inductive elements to modify one or more characteristics of an interrogation signal (e.g., amplitude, phase) such that a state or configuration of the control circuit may be uniquely discerned based on the one or more characteristics.

(178) In FIG. 28, for example, the data circuit **290** may comprise an identification resistor R1, with the value of C1 selected such that R1 is "hidden" from a first low frequency interrogation signal (e.g., an interrogation signal at 2 kHz) for determining the states of SW1 and SW2. By measuring the voltage and/or current (e.g., amplitude, phase) at the inputs of the control circuit **280-6** resulting from a second interrogation signal within a substantially higher frequency band, the generator **102** may measure the value of R1 through C1 in order to determine which of a plurality of identification resistors is contained in the instrument. Such information may be used by the generator **102** to identify the instrument, or a particular characteristic of the instrument, so that control and diagnostic processes may be optimized. Any interference with the generator's ability to measure R1 caused by states of SW1 and SW2 may be addressed by declaring an error if switch states will prevent measurement, or by maintaining the voltage of the second higher-frequency interrogation signal below the turn-on voltages of D1 and D2. Such interference may also be addressed by adding an inductor in series with the switch circuitry (L1 in FIG. 29) to block the second higher-frequency interrogation signal while passing the first, lower-frequency interrogation signal. The addition of an inductor in this manner may also enable the use of the data circuit **290** in control circuits of electrosurgical instruments, as shown in the embodiment of the data circuit **290-2** of FIG. 30.

(179) In certain embodiments, multiple capacitors C1 for allowing interrogation at multiple frequencies could be used to differentiate between a larger number of distinct R1 values for a given signal-to-noise ratio, or for a given set of component tolerances. In one such embodiment, inductors may be placed in series with all but the lowest value of C1 to create specific pass bands for different interrogation frequencies, as shown in the embodiment of the data circuit **290-3** in FIG. 31.

(180) In embodiments of control circuits based on the control circuit **280** of FIG. 14, identification resistors may be measured without the need for frequency band separation. FIG. 32 illustrates one such embodiment, with R1 selected to have a relatively high value.

(181) FIGS. 33D-33 illustrate embodiments of multi-conductor cables and adaptors that may be used to establish electrical communication between the generator **102** and a handpiece of a surgical device. In particular, the cables may transmit the generator drive signal to surgical device and enable control-based communications between the generator **102** and a control circuit of the surgical device. In certain embodiments, the cables may be integrally formed with the surgical device or configured for removable engagement by a suitable connector of the surgical device. Cables **112-1**, **112-2** and **112-3** (FIGS. 33E-33G, respectively) may be configured for use with an ultrasonic surgical device (e.g., ultrasonic surgical device **104**), and cable **128-1** (FIG. 33D) may be configured for use with an electrosurgical device (e.g., electrosurgical device **106**). One or more of

the cables may be configured to connect directly with the generator **102**, such as cable **112-1**, for example. In such embodiments, the cable may comprise a data circuit (e.g., data circuit **206**) for storing information pertaining to the particular surgical device with which it is associated (e.g., a model number, a serial number, a number of operations in which the surgical device has been used, and/or any other type of information). In certain embodiments, one or more of the cables may connect to the generator **102** via an adaptor. For example, cables **112-2** and **112-3** may connect to the generator **102** via a first adaptor **292** (FIG. 33I), and cable **128-1** may connect to the generator **102** via a second adaptor **294** (FIG. 33H). In such embodiments, a data circuit (e.g., data circuit **206**) may be disposed in the cable (e.g., cables **112-2** and **112-3**) or in the adaptor (e.g., second adaptor **294**).

(182) In various embodiments, the generator **102** may be electrically isolated from the surgical devices **104**, **106** in order to prevent undesired and potentially harmful currents in the patient. For example, if the generator **102** and the surgical devices **104**, **106** were not electrically isolated, voltage provided to the devices **104**, **106** via the drive signal could potentially change the electrical potential of patent tissue being acted upon by the device or devices **104**, **106** and, thereby, result in undesired currents in the patient. It will be appreciated that such concerns may be more acute when using an ultrasonic surgical device **104** that is not intended to pass any current through tissue. Accordingly, the remainder of the description of active cancellation of leakage current is described in terms of an ultrasonic surgical device **104**. It will be appreciated, however, that the systems and methods described herein may be applicable to electrosurgical devices **106** as well.

(183) According to various embodiments, an isolation transformer, such as the isolation transformer **156**, may be used to provide electrical isolation between the generator **102** and the surgical device **104**. For example, the transformer **156** may provide isolation between the non-isolated stage **154** and the isolated stage **152** described above. The isolated stage **154** may be in communication with the surgical device **104**. The drive signal may be provided by the generator **102** (e.g., the generator module **108**) to the primary winding **164** of the isolation transformer **156** and provided to the surgical device **104** from the secondary winding **158** of the isolation transformer. Considering the non-idealities of real transformers, however, this arrangement may not provide complete electrical isolation. For example, a real transformer may have stray capacitance between the primary and secondary windings. The stray capacitance may prevent complete electrical isolation and allow electrical potential present on the primary winding to affect the potential of the secondary winding. This may result in leakage currents within the patient.

(184) Contemporary industry standards, such as the International Electrotechnical Commission (IEC) 60601-1 standard limit allowable patient leakage current to 10 μ A or less. Leakage current may be passively reduced by providing a leakage capacitor between the secondary winding of the isolation transformer and ground (e.g., earth ground). The leakage capacitor may operate to smooth changes in patient-side potential coupled from the non-isolated side via the stray capacitance of the isolation transformer and thereby reduce leakage current. As the voltage, current, power and/or frequency of the drive signal provided by the generator **102** increase, however, the leakage current may also increase. In various embodiments, induced leakage current may increase beyond the capability of a passive leakage capacitor to keep it below 10 μ A and/or other leakage current standards.

(185) Accordingly, various embodiments are directed to systems and methods for actively cancelling leakage current. FIG. 34 illustrates one embodiment of a circuit **800** for active cancellation of leakage current. The circuit **800** may be implemented as a part of or in conjunction with the generator **102**. The circuit may comprise an isolation transformer **802** having a primary winding **804** and a secondary winding **806**. The drive signal **816** may be provided across the primary winding **804**, generating an isolated drive signal across the secondary winding **806**. In addition to the isolated drive signal, stray capacitance **808** of the isolation transformer **802** may couple some component of the potential of the drive signal relative to ground **818** to the secondary

winding **806** on the patient side.

(186) A leakage capacitor **810** and active cancellation circuit **812** may be provided, as shown, connected between the secondary winding **806** and ground **818**. The active cancellation circuit **812** may generate an inverse drive signal **814** that may be about 180° out of phase with the drive signal **816**. The active cancellation circuit **812** may be electrically coupled to the leakage capacitor **810** to drive the leakage capacitor to a potential that, relative to ground **818**, is about 180° out of phase with the drive signal **816**. Accordingly, electrical charge on the patient-side secondary winding **806** may reach ground **818** via the leakage capacitor **810** instead of through the patient, reducing leakage current. According to various embodiments, the leakage capacitor **810** may be designed to meet adequate, industry, government and/or design standards for robustness. For example, the leakage capacitor **810** may be a Y-type capacitor complying with the IEC 60384-14 standard and/or may comprise multiple physical capacitors in series.

(187) FIG. **35** illustrates one embodiment of a circuit **820** that may be implemented by the generator **102** to provide active cancellation of leakage current. The circuit **820** may comprise a generator circuit **824** and a patient-side circuit **822**. The generator circuit **824** may generate and/or modulate the drive signal, as described herein. For example, in some embodiments, the generator circuit **824** may operate similar to the non-isolated stage **154** described above. Also, for example, the patient-side circuit **822** may operate similar to the isolated state **152** described above.

(188) Electrical isolation between the generator circuit **824** and the patient-side circuit **822** may be provided by an isolation transformer **826**. The primary winding **828** of the isolation transformer **826** may be coupled to the generator circuit **824**. For example, the generator circuit **824** may generate the drive signal across the primary winding **828**. The drive signal may be generated across the primary winding **828** according to any suitable method. For example, according to various embodiments, the primary winding **828** may comprise a center tap **829** that may be held to a DC voltage (e.g., 48 volts). The generator circuit **824** may comprise output stages **825**, **827** that are, respectively, coupled to the other ends of the primary winding **828**. Output stages **825**, **827** may cause currents corresponding to the drive signal to flow in the primary winding **828**. For example, positive portions of the drive signal may be realized when the output stage **827** pulls its output voltage lower than the center tap voltage, causing the output stage **827** to sink current from across the primary winding **828**. A corresponding current may be induced in the secondary winding **830**. Likewise, negative portions of the drive signal may be implemented when the output stage **827** pulls its output voltage lower than the center tap voltage, causing the output stage **825** to sink an opposite current across the primary winding **828**. This may induce a corresponding, opposite current in the secondary winding **830**. The patient-side circuit **822** may perform various signal conditioning and/or other processing to the isolated drive signal, which may be provided to a device **104** via output lines **821**, **823**.

(189) An active cancellation transformer **832** may have a primary winding **834** and a secondary winding **836**. The primary winding **834** may be electrically coupled to the primary winding **828** of the isolation transformer **826** such that the drive signal is provided across the winding **834**. For example, the primary winding **834** may comprise two windings **843**, **845**. A first end **835** of the first winding **845** and a first end **839** of the second winding **843** may be electrically coupled to the center tap **829** of the winding **828**. A second end **841** of the first winding **845** may be electrically coupled to the output stage **827**, while a second end **837** of the second winding **843** may be electrically coupled to the output stage **825**. The secondary winding **836** of the cancellation transformer **832** may be coupled to ground **818** and to a first electrode of a cancellation capacitor **840**. The other electrode of the cancellation capacitor **840** may be coupled to the output line **823**. An optional load resistor **838** may also be electrically coupled in parallel across the secondary winding **836**.

(190) According to various embodiments, the secondary winding **836** of the active cancellation transformer may be wound and/or wired to the other components **840**, **838**, **818**, such that its

polarity is opposite the polarity of the primary winding **834**. For example, an inverse drive signal may be induced across the secondary winding **836**. Relative to ground **818**, the inverse drive signal may be 180° out of phase with the drive signal provided across the primary winding **834** of the active cancellation transform **832**. In conjunction with the load resistor **838**, the secondary winding **836** may provide the inverse drive signal at the cancellation capacitor **840**. Accordingly, charge causing leakage potential appearing at the patient-side circuit **822** due to the drive signal may be drawn to the cancellation capacitor **840**. In this way, the capacitor **840**, secondary winding **836** and load resistor **838** may sink potential leakage current to ground **818**, minimizing patient leakage current.

(191) According to various embodiments, the parameters of the components **832**, **838**, **840** may be selected to maximize leakage current cancellation and, in various embodiments, to lessen electromagnetic emissions. For example, the active cancellation transformer **832** may be made from materials and according to a construction that allows it to match the frequency, temperature, humidity and other characteristics of the isolation transformer **826**. Other parameters of the active transformer **832** (e.g., number of turns, turn ratios, etc.) may be selected to achieve a balance between minimizing output-induced current, electromagnetic (EM) emissions and leakage current due to applied external voltage. For example, the circuit **820** may be configured to meet the IEC 60601 or other suitable industry or government standards. The value of the load resistor **838** may be similarly chosen. In addition, the parameters of the cancellation capacitor **840** (e.g., capacitance, etc.) may be selected to match, as well as possible, the characteristics of the stray capacitances responsible for the inducing leakage current.

(192) FIG. **36** illustrates an alternate embodiment of a circuit **842** that may be implemented by the generator **102** to provide active cancellation of leakage current. The circuit **842** may be similar to the circuit **820**, however, the secondary winding **836** of the active cancellation transformation **832** may be electrically coupled to the output line **823**. The cancellation capacitor **823** may be connected in series between the secondary winding **836** and ground **818**. The circuit **842** may operate in a manner similar to that of the circuit **820**. According to various embodiments, (e.g., when the active cancellation transformer **832** is a step-up transformer), the total working voltage, for example, as defined in IEC 60601-1, may be minimized.

(193) FIG. **37** illustrates an alternate embodiment of a circuit **844** that may be implemented by the generator **102** to provide active cancellation of leakage current. The circuit **844** may omit the active cancellation transformer **832** and replace it with a second secondary winding **846** of the isolation transformer **826**. The second secondary winding **846** may be connected to the output line **823**. The cancellation capacitor **840** may be connected in series between the second secondary winding **846** and ground. The second secondary winding may be wound and or wired with a polarity opposite that of the primary winding **828** and the secondary winding **830**. Accordingly, when the drive signal is present across the primary winding **828**, the inverse drive signal, as described above, may be present across the secondary winding **846**. Accordingly, the circuit **844** may cancel leakage current in a manner similar to that described above with respect to the circuits **820** and **842**. Omitting the active cancellation transformer **832**, as shown in circuit **844**, may reduce part count, cost and complexity.

(194) FIG. **38** illustrates yet another embodiment of a circuit **848** that may be implemented by the generator **102** to provide active cancellation of leakage current. The circuit **848** may be configured to cancel extraneous currents in the patient side circuit **822** due to capacitive coupling, as described above, as well as other external effects such as, for example, frequency-specific effects (e.g., 60 Hz or other frequency noise from power supplies), path effects, load effects, etc. Instead of being electrically coupled to ground **818**, the cancellation capacitor **840**, as shown in the circuit **848**, may be coupled to an correction control circuit **851**. The circuit **851** may comprise a digital signal processor (DSP) **850** or other processor. The DSP **850** may receive inputs **858** (e.g., via an analog-to-digital converter). The inputs **858** may be values tending to indicate external effects that may

cause additional leakage current. Examples of such inputs may be, for example, power supply parameters, load data such as impedance, impedance or other values describing the path from the circuit **848** to the device **104**, etc. Based on the inputs **858**, the DSP **85** may derive a cancellation potential that, when provided to the cancellation capacitor **840**, may cancel patient-side currents due to the external effects. The cancellation potential may be provided, digitally, to digital-to-analog converter **852**, which may provide an analog version of the cancellation potential to the cancellation capacitor **840**. Accordingly, the voltage drop across the cancellation capacitor **840** may be a function of the inverse drive signal, present across the second secondary winding **846** and the cancellation potential found by the circuit **851**.

(195) The circuit **848** is shown with the active cancellation transformer **832** omitted and the capacitor **840** and second secondary winding **846** in the configuration of the circuit **844**. It will be appreciated, however, that the correction control circuit **851** may be utilized in any of the configurations described herein (e.g., **820**, **842**, **844**, etc.). For example, the correction control circuit **851** may be substituted for ground **818** in any of the circuits **820**, **842**, **844**.

(196) FIG. **39** illustrated an embodiment of a circuit **860** that may be implemented by the generator **102** to provide cancellation of leakage current. According to the circuit **860**, the cancellation capacitor **840** may be connected between the primary winding **828** of the isolation transformer **826** and the output line **823** (e.g., the common output line). In this way, the inverse of the drive signal may appear across the cancellation capacitor **840**, bringing about a similar leakage current cancellation effect to those described above.

(197) FIG. **40** illustrates another embodiment of a circuit **862** that may be implemented by the generator **102** to provide cancellation of leakage current. The circuit **862** may be similar to the circuit **860** with the exception that the cancellation capacitor may be connected between the output line **823** (e.g., the common output line) and two additional capacitors **864**, **866**. Capacitor **864** may be connected between the cancellation capacitor **840** and the primary winding **828** of the isolation transformer **826**. Capacitor **866** may be connected between the cancellation capacitor **840** and ground **818**. The combination of the capacitors **864**, **866** may provide a radio frequency (RF) path to ground that may enhance the RF performance of the generator **102** (e.g., by decreasing electromagnetic emissions).

(198) A surgical generator, such as the generator **102** schematically illustrated in FIG. **10**, for example, may be electrically coupled to a variety of surgical instruments. The surgical instruments may include, for example, both RF-based instruments and ultrasonic-based devices. FIG. **41** illustrates a receptacle and connector interface **900** in accordance with one non-limiting embodiment. In one embodiment, the interface **900** comprises a receptacle assembly **902** and a connector assembly **920**. The connector assembly **920** may be electrically coupled to the distal end of a cable **921** that is ultimately connected to a handheld surgical instrument, for example. FIG. **59** illustrates a surgical generator **1050** in accordance with one non-limiting embodiment. The surgical generator **1050** may comprise a surgical generator body **1052** that generally includes the outer shell of the generator. The surgical body **1052** may define an aperture **1054** for receiving a receptacle assembly, such as the receptacle assembly **1058** illustrated in FIG. **59**. Referring now to FIGS. **41** and **59**, the receptacle assembly **902** may comprise a seal **906** to generally prevent fluid ingress into the surgical generator **1050** by way of the aperture **1054**. In one embodiment, the seal **906** is an epoxy seal.

(199) FIG. **42** is an exploded side view of the receptacle assembly **902** in accordance with one non-limiting embodiment. The receptacle assembly **902** may include a variety of components, such as a magnet **212**, for example. The receptacle assembly **902** may also comprise a plurality of sockets **908** that may be arranged in a generally circular formation, or any other suitable formation. FIG. **48** is an enlarged view of a socket **908** in accordance with one non-limiting embodiment. In one embodiment, the socket **908** is bifurcated and the receptacle assembly **902** includes nine bifurcated sockets **908**, while greater or few sockets may be utilized in other embodiments. Each of the

sockets **908** may define an inner cavity **910** for receiving electrically conductive pins, as discussed in more detail below. In some embodiments, various sockets **908** will be mounted within the receptacle assembly **902** at different elevations such that certain sockets are contacted prior to other sockets when a connector assembly is inserted into the receptacle assembly.

(200) FIG. **43** is an exploded side view of the connector assembly **920** in accordance with one non-limiting embodiment. The connector assembly **920** may comprise, for example, a connector body **922** that includes an insertion portion **924** that is sized to be received by the receptacle assembly **902**, as described in more detail below. The connector assembly **920** may comprise a variety of other components, such as a ferrous pin **926**, a circuit board **928**, and a plurality of electrically conductive pins **930**. As shown in FIG. **54**, the ferrous pin **926** may be cylindrical. In other embodiments, the ferrous pin **926** may be other shapes, such as rectangular, for example. The ferrous pin **926** may be steel, iron, or any other magnetically compatible material that is attracted to magnetic fields or that may be magnetizable. The ferrous pin **926** may also have a shoulder **927**, or other type of laterally extending feature. Referring now to FIG. **55**, the electrical conductive pins **930** may be affixed to and extend from the circuit board **928**. The circuit board **928** may also include device identification circuitry, such as the circuits illustrated in FIGS. **33E-33G**, for example. Thus, in various embodiments, the circuit board **928** may carry EEPROM, resistors, or any other electrical components. In some embodiments, portions of the circuit board **928** may be potted, or otherwise encapsulated, to improve the sterility of the surgical device and assist in water resistance.

(201) Referring again to FIG. **43**, the connector assembly **920** may also include a strain relief member **932**. As shown in FIG. **56**, the strain relief member **932** generally accepts cable loading to prevent that loading from being applied to the circuit board **928** and/or the sockets **908**. In some embodiments, the strain relief member **932** may include an alignment notch **934** to aid in assembly. Referring again to FIG. **43**, the connector assembly **920** may also include a boot **936** that is coupled to the connector body **922**. FIG. **57** illustrates the boot **936** in accordance with one non-limiting embodiment. The boot **936** may generally serve as bend relief for an associated cable and assist in sealing the connector assembly **920**. In some embodiments, the boot **936** may snap onto the connector body **922**. For autoclave applications, the boot **936** may be an overmolded component. In other embodiments, other attachment techniques may be used, such as adhesives or spin welding, for example.

(202) FIG. **44** is a perspective view of the receptacle assembly **902** shown in FIG. **41**. FIG. **45** is an exploded perspective view of the receptacle assembly **902**. FIG. **46** is a front elevation view of the receptacle assembly **902**. FIG. **47** is a side elevation view of the receptacle assembly **902**. Referring to FIGS. **44-47**, the receptacle assembly **902** may comprise a flange **950**. The flange **950** may have an inner wall **952** and an outer wall **954**. Spanning the inner wall **952** and the outer wall **954** is a flange surface **956**. The inner wall **952** may include at least one curved portion and at least one linear portion. The inner wall **952** of the flange **950** defines a cavity **960** having a unique geometry. In one embodiment, the cavity **960** is defined by about 270 degrees of a circle and two linear segments that are tangential to the circle and intersect to form an angle θ . In one embodiment, angle θ is about 90 degrees. In one embodiment, a central protruding portion **962** having an outer periphery **964** is positioned in the cavity **960**. The central protruding portion **962** may have a central surface **966** that defines a recess **968**. The magnet **912** (FIG. **42**) may be positioned proximate the recess **968**. As illustrated, the sockets **908** may be positioned through apertures **972** defined by the central surface **966** of the central protruding portion **962**. In embodiments utilizing a circular arrangement of sockets **908**, the magnet **912** may be positioned internal to the circle defined by the sockets. The receptacle body **904** may also define a rear recess **976** (FIG. **47**). The rear recess **976** may be sized to receive the seal **906**. The flange face **966** may be slanted at an angle β (FIG. **47**). As illustrated in FIG. **61**, a face of the body **1052** of the surgical generator **1050** also may be slanted at the angle β as well.

(203) FIG. 49 is a perspective view of the connector assembly 920 and FIG. 50 is an exploded perspective view of the connector assembly 920. FIG. 51 is a side elevation view of the connector body 922 with FIGS. 52 and 53 illustrating perspective views of the distal and proximal ends, respectively, of the connector body 922. Referring now to FIGS. 49-53, connector body 922 may have a flange 980. The flange 980 may comprise at least one curved portion and at least one linear portion.

(204) The adapter assemblies 1002 and 1004 may comprise substantially the similar components that are contained by the connector body 922 (FIG. 50). For example, the adapter assemblies 1002 and 1004 may each house a circuit board with device identification circuitry. The adapter assemblies 1002 and 1004 may also each house one of a ferrous pin and a magnet to aid in the connection with the surgical generator. An outer wall 982 of the flange 980 may generally be shaped similarly to the inner wall 952 of the receptacle assembly 902 (FIG. 46). An inner wall 984 of the flange 980 may be shaped similarly to the outer periphery 964 of the central protruding portion 962. The connector body 922 may also have a wall 988 that includes a plurality of apertures 990. The apertures 990 may be sized to receive the electrically conductive pins 930 and the ferrous pin 926. In one embodiment, the shoulder 927 of the ferrous pin 926 is sized so that it can not pass through the aperture 990. In some embodiments, the ferrous pin 926 may be able to translate with respect to the wall 988. When assembled, the shoulder 927 of the ferrous pin 926 may be positioned intermediate the wall 988 and the circuit board 928. The ferrous pin 926 may be positioned such that it encounters the magnetic field of the magnet 912 when the connector assembly 920 is inserted into the receptacle assembly 902. In some embodiments, a proper connection will be denoted by an audible click when the ferrous pin 926 translates to the wall 988 and strikes the magnet 912. As is to be appreciated, various components may be positioned intermediate the ferrous pin 926 and the magnet 912, such as a washer, for example, to reduce incidental wear to the interfacing components. Additionally, in some embodiments the magnet 912 may be coupled to the connector assembly 920 and the ferrous pin 926 may be coupled to the receptacle assembly 902.

(205) FIG. 58 illustrates two adaptor assemblies 1002 and 1004 in accordance with various non-limiting embodiments. The adaptor assemblies 1002 and 1004 allow of connector assemblies having various geometries to be electrically coupled to a receptacle assembly of a surgical generator. Adaptor assembly 1002 is configured to accommodate a surgical instrument having connector assembly 1006 and adaptor assembly 1004 is configured to accommodate a surgical instrument having a connector assembly 1008. In one embodiment, the connector assembly 1006 is associated with an RF-based surgical device via a cable 1060 and the connector assembly 1008 is associated with an ultrasonic-based device via a cable 1062. As is to be appreciated, other embodiments of adaptor assemblies may accommodate surgical instruments have connector assemblies different than those illustrated in FIG. 58. FIG. 59 illustrates the adaptor assembly 1002 after being inserting into the receptacle assembly 1058 of a surgical generator 1050 in accordance with one non-limiting embodiment. FIG. 60 illustrates the connector assembly 1006 after being inserted into the adaptor assembly 1002 and therefore electrically coupled to the surgical generator 1050. Similarly, FIG. 61 illustrates the adaptor assembly 1004 after being inserted into the receptacle assembly 1058 of a surgical generator 1050 in accordance with one non-limiting embodiment. FIG. 62 illustrates the connector assembly 1008 after being inserted into the adaptor assembly 1004. Accordingly, while connector assemblies 1006 and 1008 each having different geometries, both may be used with the surgical generator 1050.

(206) Referring to FIGS. 58-62, in one embodiment, the adaptor assembly 1002 has a distal portion 1010 that comprises a flange 1012. The flange 1012 is configured to be inserted into the receptacle assembly 1058 of the surgical instrument 1050 and may be similar to the flange 980 illustrated in FIG. 52, for example. Any number of electrically conductive pins, or other connection components, may be positioned in the distal portion to engage the receptacle assembly 1058. In one

embodiment, the adaptor assembly **1002** also has a proximal portion **1014** that defines a cavity **1016**. The cavity **1016** may be configured to accept a particular connector assembly, such as connector assembly **1006**. As is to be appreciated, the proximal portion **1014** may be configured appropriately based on the type of connector assembly with which it will be used. In one embodiment, the adaptor assembly **1006** has a distal portion **1020** that comprises a flange **1022**. The flange **1022** is configured to be inserted into the receptacle assembly **1058** of the surgical instrument **1050** and may be similar to the flange **980** illustrated in FIG. **52**, for example. The adaptor assembly **1004** also has a proximal portion **1024** that defines a cavity **1026**. In the illustrated embodiment, the central portion **1028** is positioned in the cavity **1026** and is configured to accept the connector assembly **1008**.

(207) FIG. **63** illustrates a perspective view of a back panel **1100** of a generator **1102** in accordance with one non-limiting embodiment. The generator **1102** may be similar to generator **102** illustrated in FIG. **10**, for example. The back panel **1100** may comprise various input and/or output ports **1104**. The back panel **1110** may also comprise an electronic paper display device **1106**. The electronic paper display device **1106** may be based on electrophoresis in which an electromagnetic field is applied to a conductive material such that the conductive material has mobility. Micro particles having conductivity are distributed between thin-type flexible substrates, and positions of the micro particles (or toner particles) are changed due to the change of the polarities of an electromagnetic field, whereby data is displayed. The technical approach to realize the electronic paper may be accomplished using any suitable technique, such as liquid crystals, organic electro luminescence (EL), reflective film reflection-type display, electrophoresis, twist balls, or mechanical reflection-type display, for example. Generally, electrophoresis is a phenomenon in which, when particles are suspended in a medium (i.e., a dispersion medium), the particles are electrically charged, and, when an electric field is applied to the charged particles, the particles move to an electrode having opposite charge through the dispersion medium. Further discussion regarding electronic paper display devices may be found in U.S. Pat. No. 7,751,115 entitled ELECTRONIC PAPER DISPLAY DEVICE, MANUFACTURING METHOD AND DRIVING METHOD THEREOF, the entirety of which is incorporated by reference.

(208) FIG. **64** illustrates the back panel **1110** illustrated in FIG. **63**. FIGS. **65** and **66** provide enlarged views of the back panel **1110**. Referring to FIGS. **64-66**, the electronic paper display device **1106** may display a variety of information, such a serial number, a part number, patent numbers, warning labels, port identifiers, instructions, vendor information, service information, manufacturer information, operational information, or any other type of information. In one embodiment, the information displayed on the electronic paper display device **1106** may be changed or updated through connecting a computing device to a communication port (e.g., a USB port) of the generator **1102**.

(209) As shown in FIG. **66**, in some embodiments, the back panel **1100** may comprise an interactive portion **1108**. In one embodiment, the interactive portion **1108** allows a user to input information to the generator **1102** using input devices, such as buttons **1110**. The interactive portion **1108** may also display information that is simultaneously displayed on a front panel (not shown) of the generator **1102**.

(210) In a surgical procedure utilizing an ultrasonic surgical device, such as the ultrasonic surgical device **104**, the end effector **126** transmits ultrasonic energy to tissue brought into contact with the end effector **126** to realize cutting and sealing action. The application of ultrasonic energy in this manner may cause localized heating of the tissue. Monitoring and controlling such heating may be desirable to minimize unintended tissue damage and/or to optimize the effectiveness of the cutting and sealing action. Direct measurement of ultrasonic heating requires temperature sensing devices in or near the end effector **126**. Although sensor-based measurements of ultrasonic heating is technically feasible, design complexity and other considerations may make direct measurement impractical. Various embodiments of the generator **102** may address this problem by generating an

estimate of temperature or heating resulting from an application of ultrasonic energy.

(211) In particular, one embodiment of the generator **102** may implement an artificial neural network to estimate ultrasonic heating based on a number of input variables **1218**. Artificial neural networks are mathematical models that learn complex, nonlinear relationships between inputs and outputs based on exposure to known input and output patterns, a process commonly referred to as “training.” An artificial neural network may comprise a network of simple processing units, or nodes, connected together to perform data processing tasks. The structure of an artificial neural network may be somewhat analogous to the structure of biological neural networks in the brain. When an artificial neural network is presented with an input data pattern, it produces an output pattern. An artificial neural network may be trained for a specific processing task by presentation of large amounts of training data. In this way, the artificial neural network may modify its structure by changing the “strength” of communication between nodes to improve its performance on the training data.

(212) FIG. **67** illustrates one embodiment of an artificial neural network **1200** for generating an estimated temperature $T_{sub.est}$ resulting from an application of ultrasonic energy using an ultrasonic surgical device, such as the ultrasonic surgical device **104**. In certain embodiments, the neural network may be implemented in the processor **174** and/or the programmable logic device **166** of the generator **102**. The neural network **1200** may comprise an input layer **1202**, one or more nodes **1204** defining a hidden layer **1206**, and one or more nodes **1208** defining an output layer **1210**. For the sake of clarity, only one hidden layer **1206** is shown. In certain embodiments, the neural network **1200** may comprise one or more additional hidden layers in a cascaded arrangement, with each additional hidden layer having a number of nodes **1204** that may be equal to or different from the number of nodes **1204** in the hidden layer **1206**.

(213) Each node **1204**, **1208** in the layers **1202**, **1210** may include one or more weight values w **1212**, a bias value b **1214**, and a transform function f **1216**. In FIG. **67**, the use of different subscripts for these values and functions is intended to illustrate that each of these values and functions may be different from the other values and functions. The input layer **1202** comprises one or more input variables p **1218**, with each node **1204** of the hidden layer **1206** receiving as input at least one of the input variables p **1218**. As shown in FIG. **67**, for example, each node **1204** may receive all of the input variables p **1218**. In other embodiments, less than all of the input variables p **1218** may be received by a node **1204**. Each input variable p **1218** received by a particular node **1204** is weighted by a corresponding weight value w **1212**, then added to any other similarly weighted input variables p **1218**, and to the bias value b **1214**. The transform function f **1216** of the node **1204** is then applied to the resulting sum to generate the node's output. In FIG. **67**, for example, the output of node **1204-1** may be given as $f_{sub.1}(n_{sub.1})$, where $n_{sub.1} = (W_{sub.1,1} \cdot Math.p_{sub.1} + W_{sub.1,2} \cdot Math.p_{sub.2} + \dots + W_{sub.1,j'} \cdot Math.p_{sub.j}) + b_{sub.1}$.

(214) A particular node **1208** of the output layer **1210** may receive an output from one or more of the nodes **1204** of the hidden layer **1206** (e.g., each node **1208** receives outputs $f_{sub.1}(Math.)$, $f_{sub.2}(Math.)$, $\dots$, $f_{sub.i}(Math.)$ from respective nodes **1204-1**, **1204-2**, $\dots$, **1204-i** in FIG. **67**), with each received output being weighted by a corresponding weight value w **1212** and subsequently added to any other similarly weighted received outputs, and to a bias value b **1214**. The transform function f **1216** of the node **1208** is then applied to the resulting sum to generate the node's output, which corresponds to an output of the neural network **1200** (e.g., the estimated temperature $T_{sub.est}$ in the embodiment of FIG. **67**). Although the embodiment of the neural network **1200** in FIG. **67** comprises only one node **1208** in the output layer **1210**, in other embodiments the neural network **1200** may comprise more than one output, in which case the output layer **1210** may comprise multiple nodes **1208**.

(215) In certain embodiments, the transform function f **1216** of a node **1204**, **1208** may be a nonlinear transfer function. In one embodiment, for example, one or more of the transform functions f **1216** may be a sigmoid function. In other embodiments, the transform functions f **1216**

may include a tangent sigmoid, a hyperbolic tangent sigmoid, a logarithmic sigmoid, a linear transfer function, a saturated linear transfer function, a radial basis transfer function, or some other type of transfer function. The transform function f **1216** of a particular node **1204**, **1208** may be the same as, or different from, a transform function f **1216** in another node **1204**, **1208**.

(216) In certain embodiments, the input variables p **1218** received by the nodes **1204** of the hidden layer **1206** may represent, for example, signals and/or other quantities or conditions known or believed to have an effect on the temperature or heating resulting from an application of ultrasonic energy. Such variables may comprise, for example, one or more of: drive voltage output by the generator **102**, drive current output by the generator **102**, drive frequency of the generator output **102**, drive power output by the generator **102**, drive energy output by the generator **102**, impedance of the ultrasonic transducer **114**, and time duration over which ultrasonic energy is applied.

Additionally, one or more of the input variables p **1218** may be unrelated to outputs of the generator **102** and may comprise, for example, characteristics of the end effector **126** (e.g., blade tip size, geometry, and/or material) and a particular type of tissue targeted by the ultrasonic energy.

(217) The neural network **1200** may be trained (e.g., by changing or varying the weight values w **1212**, the bias values b **1214**, and the transform functions f **1216**) such that its output (e.g., estimated temperature $T_{sub.est}$ in the embodiment of FIG. **67**) suitably approximates a measured dependency of the output for known values of the input variables p **1218**. Training may be performed, for example, by supplying known sets of input variables p **1218**, comparing output of the neural network **1200** to measured outputs corresponding to the known sets of input variables p **1218**, and modifying the weight values w **1212**, the bias values b **1214**, and/or the transform functions f **1216** until the error between the outputs of the neural network **1200** and the corresponding measured outputs is below a predetermined error level. For example, the neural network **1200** may be trained until the mean square error is below a predetermined error threshold. In certain embodiments, aspects of the training process may be implemented by the neural network **1200** (e.g., by propagating errors back through the network **1200** to adaptively adjust the weight values w **1212** and/or the bias values b **1214**).

(218) FIG. **68** illustrates a comparison between estimated temperature values $T_{sub.est}$ and measured temperature values $T_{sub.m}$ for an implementation of one embodiment of the neural network **1200**. The neural network **1200** used to generate $T_{sub.est}$ in FIG. **68** comprised six input variables p **1218**: drive voltage, drive current, drive frequency, drive power, impedance of the ultrasonic transducer, and time duration over which ultrasonic energy was applied. The hidden layer **1206** comprised 25 nodes, and the output layer **1210** comprised a single node **1208**. Training data was generated based on **13** applications of ultrasonic energy to carotid vessels. Actual temperature ($T_{sub.m}$) was determined based on IR measurements over a 250-sample range for varying values of the input variables p **1218**, with estimated temperatures $T_{sub.est}$ being generated by the neural network **1200** based on corresponding values of the input variables p **1218**. The data shown in FIG. **68** was generated on a run that was excluded from the training data. The estimated temperatures $T_{sub.est}$ demonstrate a reasonably accurate approximation of the measured temperatures $T_{sub.m}$ in the region of 110-190° F. It is believed that inconsistencies in estimated temperatures $T_{sub.est}$ appearing in certain regions, such as the region following 110° F., may be minimized or reduced by implementing additional neural networks specific to those regions. Additionally, inconsistencies in the data that may skew the trained output of the neural network **1200** may be identified and programmed in as special cases to further improve performance.

(219) In certain embodiments, when the estimated temperature exceeds a user-defined temperature threshold $T_{sub.th}$, the generator **102** may be configured to control the application of ultrasonic energy such that the estimated temperature $T_{sub.est}$ is maintained at or below the temperature threshold $T_{sub.th}$. For example, in embodiments in which the drive current is an input variable p **1218** to the neural network **1200**, the drive current may be treated as a control variable and modulated to minimize or reduce the difference between $T_{sub.est}$ and $T_{sub.th}$. Such embodiments

may be implemented using a feedback control algorithm (e.g., a PID control algorithm), with T.sub.th being input to the control algorithm as a setpoint, T.sub.est being input to the algorithm as process variable feedback, and drive current corresponding to the controlled output of the algorithm. In cases where the drive current serves as the control variable, suitable variations in drive current value should be represented in the sets of input variables p **1218** used to train the neural network **1200**. In particular, the effectiveness of drive current as a control variable may be reduced if the training data reflects constant drive current values, as the neural network **1200** may reduce the weight values w **1212** associated with drive current due to its apparent lack of effect on temperature. It will be appreciated that input variables p **1218** other than drive current (e.g., drive voltage) may be used to minimize or reduce the difference between T.sub.est and T.sub.th.

(220) According to various embodiments, the generator **102** may provide power to a tissue bite according to one or more power curves. A power curve may define a relationship between power delivered to the tissue and the impedance of the tissue. For example as the impedance of the tissue changes (e.g., increases) during coagulation, the power provided by the generator **102** may also change (e.g., decrease) according to the applied power curve.

(221) Different power curves may be particularly suited, or ill-suited, to different types and/or sizes of tissue bites. Aggressive power curves (e.g., power curves calling for high power levels) may be suited for large tissue bites. When applied to smaller tissue bites, such as small vessels, more aggressive power curves may lead to exterior searing. Exterior searing may reduce the coagulation/weld quality at the exterior and can also prevent complete coagulation of interior portions of the tissue. Similarly, less aggressive power curves may fail to achieve hemostasis when applied to larger tissue bites (e.g., larger bundles).

(222) FIG. **69** illustrates one embodiment of a chart **1300** showing example power curves **1306**, **1308**, **1310**. The chart **1300** comprises an impedance axis **1302** illustrating increasing potential tissue impedances from left to right. A power axis **1304** illustrates increasing power from down to up. Each of the power curves **1306**, **1308**, **1310** may define a set of power levels, on the power axis **1304**, corresponding to a plurality of potential sensed tissue impedances, in the impedance axis **1302**. In general, power curves may take different shapes, and this is illustrated in FIG. **69**. Power curve **1306** is shown with a step-wise shape, while power curves **1308**, **1310** are shown with curved shapes. It will be appreciated that power curves utilized by various embodiments may take any usable continuous or non-continuous shape. The rate of power delivery or aggressiveness of a power curve may be indicated by its position on the chart **1300**. For example, power curves that deliver higher power for a given tissue impedance may be considered more aggressive.

Accordingly, between two power curves, the curve positioned highest on the power axis **1304** may be the more aggressive. It will be appreciated that some power curves may overlap.

(223) The aggressiveness of two power curves may be compared according to any suitable method. For example, a first power curve may be considered more aggressive than a second power curve over a given range of potential tissue impedances if the first power curve has a higher delivered power corresponding to at least half of the range of potential tissue impedances. Also, for example, a first power curve may be considered more aggressive than a second power curve over a given range of potential tissue impedances if the area under the first curve over the range is larger than the area under the second curve over the range. Equivalently, when power curves are expressed discretely, a first power curve may be considered more aggressive than a second power curve over a given set of potential tissue impedances if the sum of the power values for the first power curve over the set of potential tissue impedances is greater than the sum of the power values for the second power curve over the set of potential tissue impedances.

(224) According to various embodiments, the power curve shifting algorithms described herein may be used with any kind of surgical device (e.g., ultrasonic device **104**, electrosurgical device **106**). In embodiments utilizing a ultrasonic device **104**, tissue impedance readings may be taken utilizing electrodes **157**, **159**. With an electrosurgical device, such as **106**, tissue impedance

readings may be taken utilizing first and second electrodes **177**, **179**.

(225) In some embodiments, an electrosurgical device **104** may comprise a positive temperature coefficient (PTC) material positioned between one or both of the electrodes **177**, **179** and the tissue bite. The PTC material may have an impedance profile that remains relatively low and relatively constant until it reaches a threshold or trigger temperature, at which point the impedance of the PTC material may increase. In use, the PTC material may be placed in contact with the tissue while power is applied. The trigger temperature of the PTC material may be selected such that it corresponds to a tissue temperature indicating the completion of welding or coagulation.

Accordingly, as a welding or coagulation process is completed, the impedance of the PTC material may increase, bringing about a corresponding decrease in power actually provided to the tissue.

(226) It will be appreciated that during the coagulation or welding process, tissue impedance may generally increase. In some embodiments, tissue impedance may display a sudden impedance increase indicating successful coagulation. The increase may be due to physiological changes in the tissue, a PTC material reaching its trigger threshold, etc., and may occur at any point in the coagulation process. The amount of energy that may be required to bring about the sudden impedance increase may be related to the thermal mass of the tissue being acted upon. The thermal mass of any given tissue bite, in turn, may be related to the type and amount of tissue in the bite.

(227) Various embodiments may utilize this sudden increase in tissue impedance to select an appropriate power curve for a given tissue bite. For example, the generator **102** may select and apply successively more aggressive power curves until the tissue impedance reaches an impedance threshold indicating that the sudden increase has occurred. For example, reaching the impedance threshold may indicate that coagulation is progressing appropriately with the currently applied power curve. The impedance threshold may be a tissue impedance value, a rate of change of tissue impedance, and/or a combination of impedance and rate of change. For example, the impedance threshold may be met when a certain impedance value and/or rate of change are observed.

According to various embodiments, different power curves may have different impedance thresholds, as described herein.

(228) FIG. **70** illustrates one embodiment of a process flow **1330** for applying one or more power curves to a tissue bite. Any suitable number of power curves may be used. The power curves may be successively applied in order of aggressiveness until one of the power curves drives the tissue to the impedance threshold. At **1332**, the generator **102** may apply a first power curve. According to various embodiments, the first power curve may be selected to deliver power at a relatively low rate. For example, the first power curve may be selected to avoid tissue searing with the smallest and most vulnerable expected tissue bites.

(229) The first power curve may be applied to the tissue in any suitable manner. For example, the generator **102** may generate a drive signal implementing the first power curve. The power curve may be implemented by modulating the power of the drive signal. The power of the drive signal may be modulated in any suitable manner. For example, the voltage and/or current of the signal may be modulated. Also, in various embodiments, the drive signal may be pulsed. For example, the generator **102** may modulate the average power by changing the pulse width, duty cycle, etc. of the drive signal. The drive signal may be provided to the first and second electrodes **177**, **179** of the electrosurgical device **106**. Also, in some embodiments the drive signal implementing the first power curve may be provided to an ultrasonic generator **114** of the ultrasonic device **104** described above.

(230) While applying the first power curve, the generator **102** may monitor the total energy provided to the tissue. The impedance of the tissue may be compared to the impedance threshold at one or more energy thresholds. There may be any suitable number of energy thresholds, which may be selected according to any suitable methodology. For example, the energy thresholds may be selected to correspond to known points where different tissue types achieve the impedance threshold. At **1334**, the generator **102** may determine whether the total energy delivered to the

tissue has met or exceeded a first energy threshold. If the total energy has not yet reached the first energy threshold, the generator **102** may continue to apply the first power curve at **1332**.

(231) If the total energy has reached the first energy threshold, the generator **102** may determine whether the impedance threshold has been reached (**1336**). As described above, the impedance threshold may be a predetermined rate of impedance change (e.g., increase) a predetermined impedance, or combination of the two. If the impedance threshold is reached, the generator **102** may continue to apply the first power curve at **1332**. For example, reaching the impedance threshold in the first power curve may indicate that the aggressiveness of the first power curve is sufficient to bring about suitable coagulation or welding.

(232) In the event that the impedance threshold is not reached at **1336**, the generator **102** may increment to the next most aggressive power curve at **1338** and apply the power curve as the current power curve at **1332**. When the next energy threshold is reached at **1334**, the generator **102** again may determine whether the impedance threshold is reached at **1336**. If it is not reached, the generator **102** may again increment to the next most aggressive power curve at **1338** and deliver that power curve at **1332**.

(233) The process flow **1330** may continue until terminated. For example, the process flow **1330** may be terminated when the impedance threshold is reached at **1336**. Upon reaching the impedance threshold, the generator **102** may apply the then-current power curve until coagulation or welding is complete. Also, for example, the process flow **1330** may terminate upon the exhaustion of all available power curves. Any suitable number of power curves may be used. If the most aggressive power curve fails to drive the tissue to the impedance threshold, the generator **102** may continue to apply the most aggressive power curve until the process is otherwise terminated (e.g., by a clinician or upon reaching a final energy threshold).

(234) According to various embodiments, the process flow **1330** may continue until the occurrence of a termination threshold. The termination threshold may indicate that coagulation and/or welding is complete. For example, the termination threshold may be based on one or more of tissue impedance, tissue temperature, tissue capacitance, tissue inductance, elapsed time, etc. These may be a single termination threshold or, in various embodiments, different power curves may have different termination thresholds. According to various embodiments, different power curves may utilize different impedance thresholds. For example, the process flow **1330** may transition from a first to a second power curve if the first power curve has failed to drive the tissue to a first tissue impedance threshold and may, subsequently, shift from the second to a third power curve if the second power curve has failed to drive the tissue to a second impedance threshold.

(235) FIG. **71** illustrates one embodiment of a chart **1380** showing example power curves **1382**, **1384**, **1386**, **1388** that may be used in conjunction with the process flow **1330**. Although four power curves **1382**, **1384**, **1386**, **1388** are shown, it will be appreciated that any suitable number of power curves may be utilized. Power curve **1382** may represent the least aggressive power curve and may be applied first. If the impedance threshold is not reached at the first energy threshold, then the generator **102** may provide the second power curve **1384**. The other power curves **1386**, **1388** may be utilized, as needed, for example in the manner described above.

(236) As illustrated in FIG. **71**, the power curves **1382**, **1384**, **1386**, **1388** are of different shapes. It will be appreciated, however, that some or all of a set of power curves implemented by the process flow **1330** may be of the same shape. FIG. **72** illustrates one embodiment of a chart showing example common shape power curves **1392**, **1394**, **1396**, **1398** that may be used in conjunction with the process flow of FIG. **70**. According to various embodiments, common shape power curves, such as **1392**, **1394**, **1396**, **1398** may be constant multiples of one another. Accordingly, the generator **102** may implement the common shape power curves **1392**, **1394**, **1396**, **1398** by applying different multiples to a single power curve. For example, the curve **1394** may be implemented by multiplying the curve **1392** by a first constant multiplier. The curve **1396** may be generated by multiplying the curve **1392** by a second constant multiplier. Likewise, the curve **1398**

may be generated by multiplying the curve **1392** by a third constant multiplier. Accordingly, in various embodiments, the generator **102** may increment to a next most aggressive power curve at **1338** by changing the constant multiplier.

(237) According to various embodiments, the process flow **1330** may be implemented by a digital device (e.g., a processor, digital signal processor, field programmable gate array (FPGA), etc.) of the generator **102**. Examples of such digital devices include, for example, processor **174**, programmable logic device **166**, processor **190**, etc.). FIGS. **73A-73C** illustrate process flows describing routines that may be executed by a digital device of the generator **102** to generally implement the process flow **1330** described above. FIG. **73A** illustrates one embodiment of a routine **1340** for preparing the generator **102** to act upon a new tissue bite. The activation or start of the new tissue bite may be initiated at **1342**. At **1344**, the digital device may point to a first power curve. The first power curve, as described above, may be the least aggressive power curve to be implemented as a part of the process flow **1330**. Pointing to the first power curve may comprise pointing to a deterministic formula indicating the first power curve, pointing to a look-up table representing the first power curve, pointing to a first power curve multiplier, etc.

(238) At **1346**, the digital device may reset an impedance threshold flag. As described below, setting the impedance threshold flag may indicate that the impedance threshold has been met. Accordingly, resetting the flag may indicate that the impedance threshold has not been met, as may be appropriate at the outset of the process flow **1330**. At **1348**, the digital device may continue to the next routine **1350**.

(239) FIG. **73B** illustrates one embodiment of a routine **1350** that may be performed by the digital device to monitor tissue impedance. At **1352**, load or tissue impedance may be measured. Tissue impedance may be measured according to any suitable method and utilizing any suitable hardware. For example, according to various embodiments, tissue impedance may be calculated according to Ohm's law utilizing the current and voltage provided to the tissue. At **1354**, the digital device may calculate a rate of change of the impedance. The impedance rate of change may likewise be calculated according to any suitable manner. For example, the digital device may maintain prior values of tissue impedance and calculate a rate of change by comparing a current tissue impedance value or values with the prior values. Also, it will be appreciated that the routine **1350** assumes that the impedance threshold is a rate of change. In embodiments where the impedance threshold is a value, **1354** may be omitted. If the tissue impedance rate of change (or impedance itself) is greater than the threshold (**1356**), then the impedance threshold flag may be set. The digital device may continue to the next routing at **1360**.

(240) FIG. **73C** illustrates one embodiment of a routine **1362** that may be performed by the digital device to provide one or more power curves to a tissue bite. At **1364**, power may be delivered to the tissue, for example, as described above with respect to **1334** of FIG. **70**. The digital device may direct the delivery of the power curve, for example, by applying the power curve to find a corresponding power for each sensed tissue impedance, modulating the corresponding power onto a drive signal provided to the first and second electrodes **A20**, **A22**, the transducer **114**, etc.

(241) At **1366**, the digital device may calculate the total accumulated energy delivered to the tissue. For example, the digital device may monitor the total time of power curve delivery and the power delivered at each time. Total energy may be calculated from these values. At **1368**, the digital device may determine whether the total energy is greater than or equal to a next energy threshold, for example, similar to the manner described above with respect to **1334** of FIG. **70**. If the next energy threshold is not met, the current power curve may continue to be applied at **1378** and **1364**.

(242) If the next energy threshold is met at **1368**, then at **1370**, the digital device may determine whether the impedance threshold flag is set. The state of the impedance threshold flag may indicate whether the impedance threshold has been met. For example, the impedance threshold flag may have been set by the routine **1350** if the impedance threshold has been met. If the impedance flag is not set (e.g., the impedance threshold is not met), then the digital device may determine, at **1372**,

whether any more aggressive power curves remain to be implemented. If so, the digital device may point the routine **1362** to the next, more aggressive power curve at **1374**. The routine **1362** may continue (**1378**) to deliver power according to the new power curve at **1364**. If all available power curves have been applied, then the digital device may disable calculating and checking of accumulated energy for the remainder of the tissue operation at **1376**.

(243) If the impedance flag is set at **1370** (e.g., the impedance threshold has been met), then the digital device may disable calculating and checking of accumulated energy for the remainder of the tissue operation at **1376**. It will be appreciated that, in some embodiments, accumulated energy calculation may be continued, while **1370**, **1372**, **1374**, and **1376** may be discontinued. For example, the generator **102** and/or digital device may implement an automated shut-off when accumulated energy reaches a predetermined value.

(244) FIG. **74** illustrates one embodiment of a process flow **1400** for applying one or more power curves to a tissue bite. For example, the process flow **1400** may be implemented by the generator **102** (e.g., the digital device of the generator **102**). At **1402**, the generator **102** may deliver a power curve to the tissue. The power curve may be derived by applying a multiplier to a first power curve. At **1404**, the generator **102** may determine if the impedance threshold has been met. If the impedance threshold has not been met, the generator **102** may increase the multiplier as a function of the total applied energy. This may have the effect of increasing the aggressiveness of the applied power curve. It will be appreciated that the multiplier may be increased periodically or continuously. For example, the generator **102** may check the impedance threshold (**1404**) and increase the multiplier (**1406**) at a predetermined periodic interval. In various embodiments, the generator **102** may continuously check the impedance threshold (**1404**) and increase the multiplier (**1406**). Increasing the multiplier as a function of total applied energy may be accomplished in any suitable manner. For example, the generator **102** may apply a deterministic equation that receives total received energy as input and provides a corresponding multiplier value as output. Also, for example, the generator **102** may store a look-up table that comprises a list of potential values for total applied energy and corresponding multiplier values. According to various embodiments, the generator **102** may provide a pulsed drive signal to tissue (e.g., via one of the surgical devices **104**, **106**). According to various embodiments, when the impedance threshold is met, the multiplier may be held constant. The generator **102** may continue to apply power, for example, until a termination threshold is reached. The termination threshold may be constant, or may depend on the final value of the multiplier.

(245) In some embodiments utilizing a pulsed drive signal, the generator **102** may apply one or more composite load curves to the drive signal, and ultimately to the tissue. Composite load curves, like other power curves described herein, may define a level of power to be delivered to the tissue as a function of a measured tissue property or properties (e.g., impedance). Composite load curves may, additionally, define pulse characteristics, such as pulse width, in terms of the measured tissue properties.

(246) FIG. **75** illustrates one embodiment of a block diagram **1450** describing the selection and application of composite load curves by the generator **102**. It will be appreciated that the block diagram **1450** may be implemented with any suitable type of generator or surgical device. According to various embodiments, the block diagram **1450** may be implemented utilizing an electrosurgical device, such as the device **106** described above with respect to FIGS. **4-7**. Also, in various embodiments, the block diagram **1450** may be implemented with a ultrasonic surgical device, such as the surgical device **104** described above with respect to FIGS. **2-3**. In some embodiments, the block diagram **1450** may be utilized with a surgical device having cutting as well as coagulating capabilities. For example, an RF surgical device, such as the device **106**, may comprise a cutting edge, such as the blade **175** for severing tissue either before or during coagulation.

(247) Referring back to FIG. **75**, an algorithm **1452** may be executed, for example by a digital

device of the generator **102** to select and apply composite load curves **1456, 1458, 1460, 1462**. The algorithm **1452** may receive a time input from a clock **1454** and may also receive loop input **1472** from sensors **1468**. The loop input **1472** may represent properties or characteristics of the tissue that may be utilized in the algorithm **1452** to select and/or apply a composite load curve. Examples of such characteristics may comprise, for example, current, voltage, temperature, reflectivity, force applied to the tissue, resonant frequency, rate of change of resonant frequency, etc. The sensors **1468** may be dedicated sensors (e.g., thermometers, pressure sensors, etc.) or may be software implemented sensors for deriving tissue characteristics based on other system values (e.g., for observing and/or calculating voltage, current, tissue temperature, etc., based on the drive signal). The algorithm **1452** may select one of the composite load curves **1456, 1458, 1460, 1462** to apply, for example based on the loop input **1472** and/or the time input from the clock **1454**. Although four composite load curves are shown, it will be appreciated that any suitable number of composite load curves may be used.

(248) The algorithm **1452** may apply a selected composite load curve in any suitable manner. For example, the algorithm **1452** may use the selected composite load curve to calculate a power level and one or more pulse characteristics based on tissue impedance (e.g., currently measured tissue impedance may be a part of, or may be derived from, the loop input) or resonant frequency characteristics of a ultrasonic device **104**. Examples of pulse characteristics that may be determined based on tissue impedance according to a composite load curve may include pulse width, ramp time, and off time.

(249) At set point **1464**, the derived power and pulse characteristics may be applied to the drive signal. In various embodiments, a feedback loop **1474** may be implemented to allow for more accurate modulation of the drive signal. At the output of the set point **1464**, the drive signal may be provided to an amplifier **1466**, which may provide suitable amplification. The amplified drive signal may be provided to a load **1470** (e.g., via sensors **1468**). The load **1470** may comprise the tissue, the surgical device **104, 106**, and/or any cable electrically coupling the generator **102** with the surgical device **104, 106** (e.g., cables **112, 128**).

(250) FIG. **76** illustrates shows a process flow illustrating one embodiment of the algorithm **1452**, as implemented by the generator **102** (e.g., by a digital device of the generator **102**). The algorithm **1452** may be activated at **1476**. It will be appreciated that the algorithm **1452** may be activated in any suitable manner. For example, the algorithm **1452** may be activated by a clinician upon actuation of the surgical device **104, 106** (e.g., by pulling or otherwise actuating a jaw closure trigger **138, 142**, switch, handle, etc.).

(251) According to various embodiments, the algorithm **1452** may comprise a plurality of regions **1478, 1480, 1482, 1484**. Each region may represent a different stage of the cutting and coagulation of a tissue bite. For example, in the first region **1478**, the generator **102** may perform an analysis of initial tissue conditions (e.g., impedance). In the second region **1480**, the generator **102** may apply energy to the tissue in order to prepare the tissue for cutting. In the third or cut region **1482**, the generator **102** may continue to apply energy while the surgical device **104, 106** cuts the tissue (e.g., with the electrosurgical device **106**, cutting may be performed by advancing the blade **A18**). In the fourth or completion region **1484**, the generator **102** may apply energy post-cut to complete coagulation.

(252) Referring now to the first region **1478**, the generator **102** may measure any suitable tissue condition or conditions including, for example, current, voltage, temperature, reflectivity, force applied to the tissue, etc. In various embodiments, an initial impedance of the tissue may be measured according to any suitable manner. For example, the generator **102** may modulate the drive signal to provide a known voltage or current to the tissue. Impedance may be derived from the known voltage and the measured current or vice versa. It will be appreciated that tissue impedance may alternately or additionally be measured in any other suitable manner. According to the algorithm **1452**, the generator **102** may proceed from the first region **1478** to the second region

1480. In various embodiments, the clinician may end the algorithm **1452** in the first region **1478**, for example, by deactivating the generator **102** and/or the surgical device **104**, **106**. If the clinician terminates the algorithm **1542**, RF (and/or ultrasonic) delivery may also be terminated at **1486**. (253) In the second region **1480**, the generator **102** may begin to apply energy to the tissue via the drive signal to prepare the tissue for cutting. Energy may be applied according to the composite load curves **1456**, **1458**, **1460**, **1462**, as described below. Applying energy according to the second region **1480** may comprise modulating pulses onto the drive signal according to some or all of the composite load curves **1456**, **1458**, **1460**, **1462**. In various embodiments, the composite load curves **1456**, **1458**, **1460**, **1462** may be successively applied in order of aggressiveness (e.g., to accommodate various types of tissue-volume clamped in the instrument jaws).

(254) The first composite load curve **1456** may be applied first. The generator **102** may apply the first composite load curve **1456** by modulating one or more first composite load curve pulses onto the drive signal. Each first composite load curve pulse may have a power and pulse characteristics determined according to the first composite load curve and considering measured tissue impedance. Measured tissue impedance for the first pulse may be the impedance measured at the first region **1478**. In various embodiments, the generator **102** may utilize all or a portion of the first composite load curve pulses to take additional measurements of tissue impedance or resonant frequency. The additional measurements may be used to determine the power and other pulse characteristics of a subsequent pulse or pulses.

(255) FIG. 77 illustrates one embodiment of a process flow **1488** for generating a first composite load curve pulse. The process flow **1488** may be executed by the generator **102** (e.g., by a digital device of the generator **102**), for example, as a part of the algorithm **1452**. At **1490**, the generator **102** may calculate a pulse width (T.sub.pw). The pulse width may be determined considering the most recent measured tissue impedance (Z) and according to the first composite load curve **1456**.

(256) At **1492**, the generator **102** may ramp the power of the drive signal up to a pulse power (P.sub.Limit) over a ramp time (tramp), thereby applying the pulse to the tissue. The pulse power may be determined, again, considering the most recent measured tissue impedance (Z) and according to the first composite load curve **1456**. The ramp time may be determined according to the composite load curve considering tissue impedance or may be constant (e.g., constant for all first composite load curve pulses, constant for all pulses, etc.). The generator **102** may apply the pulse power to the drive signal in any suitable manner including, for example, modulating a current and/or voltage provided by the drive signal. According to various embodiments, the drive signal may be an alternating current (A/C) signal, and therefore the pulse itself may comprise multiple cycles of the drive signal.

(257) The drive signal may be held at the pulse power for the pulse width at **1494**. At the conclusion of the pulse, the drive signal may be ramped down, at **1496**, over a fall time (T.sub.fall). The fall time may be determined according to the first composite load curve considering tissue impedance, or may be constant (e.g., constant for all first composite load curve pulses, constant for all pulses, etc.). It will be appreciated that, depending on the embodiment, the ramp time and fall time may or may not be considered part of the pulse width. At **1498**, the generator **102** may pause for an off time (T.sub.off). Like the ramp time and fall time, the off time may be determined according to the first composite load curve considering tissue impedance, or may be constant (e.g., constant for all first composite load curve pulses, constant for all pulses, etc.).

(258) At the completion of the off time, the generator **102** may repeat the process flow **1488** as long as the first composite load curve **1456** is applied. According to various embodiments, the generator **102** may apply the first composite load curve **1456** for a predetermined amount of time.

Accordingly, the process flow **1488** may be repeated until the predetermined amount of time has elapsed (e.g., as determined based on the time input received from the clock **1454**). Also, in various embodiments, the first composite load curve may be applied for a predetermined number of pulses. Because the applied pulse width varies according to measured tissue impedance, the total time that

the first composite load curve is applied may also vary with measured tissue impedance. According to various embodiments, the first composite load curve **1456** (as well as the other composite load curves **1458**, **1460**, **1462**) may specify decreasing pulse widths as tissue impedance increases. Therefore, a higher initial tissue impedance may lead to less time spent in the first composite load curve.

(259) Upon completion of the first composite load curve **1456**, the generator **102** may successively apply the remaining consolidated load curves **1458**, **1460**, **1462** throughout the application of the second region **1480**. Each load curve **1458**, **1460**, **1462** may be applied in a manner similar to that of the load curve **1456** described above. For example, pulses according to a current load curve may be generated until the completion of that load curve (e.g., the expiration of a predetermined amount of time or a predetermined number of pulses). The predetermined number of pulses may be the same for each composite load curve **1456**, **1458**, **1460**, **1462** or may be different. According to various embodiments, pulses according to the load curves **1458**, **1460**, **1462** may be generated in a manner similar to process flow **1488**, except that pulse power, pulse width and, in some embodiments, ramp time, fall time, and off time, may be derived according to the current composite load curve.

(260) The second region **1480** may be terminated upon the occurrence of various events. For example, if the total RF application time has exceeded a timeout time, then the generator **102** may end the tissue operation by terminating RF (and/or ultrasonic) delivery at **1486**. Also, various events may cause the generator **102** to transition from the second region **1480** to the third region **1482**. For example, the generator **102** may transition to the third region **1482** when the tissue impedance (Z) exceeds a threshold tissue impedance ($Z_{sub.term}$) and RF energy has been delivered for at least more than a minimum time ($T_{sub.start}$). The threshold tissue impedance may be an impedance and/or an impedance rate of change indicating that the tissue bite is adequately prepared for cutting by the blade **175**.

(261) According to various embodiments, if the final load curve **1462** is completed in the second region **1480** before completion of the second region **1480**, then the final power curve **1462** may be continuously applied, for example, until the tissue impedance threshold is met, the maximum second region time is reached and/or the timeout time is reached. Also, it will be appreciated that, with some tissue cuts, the second region **1480** may be completed before all available consolidated load curves **1456**, **1458**, **1460**, **1462** are executed.

(262) At the third region **1482**, the generator **102** may continue to modulate pulses onto the drive signal. Generally, third region pulses may be modulated onto the drive signal according to any suitable manner including, for example, that described above with reference to the process flow **1488**. The power and pulse characteristics of the third region pulses may be determined according to any suitable method and, in various embodiments, may be determined based on the composite load curve that was being executed at the completion of the second region **1480** (the current load curve). According to various embodiments, the current load curve may be utilized to determine the pulse power of third region pulses, while the pulse characteristics (e.g., pulse width, ramp time, fall time, off time, etc.) may be constant regardless of composite load curve. In some embodiments, the third region **1482** may utilize a third-region-specific composite load curve that may be one of the load curves **1456**, **1458**, **1460**, **1462** utilized in the second region **1480**, or may be a different composite load curve (not shown).

(263) The generator **102** may continue to execute the third region **1482** until receiving an indication that the tissue cut is complete. In embodiments utilizing surgical implements having a blade, such as **175**, the indication may be received when the blade **175** reaches its distal-most position, as shown in FIG. **6**. This may trip a knife limit sensor (not shown) indicating that the blade **175** has reached the end of its throw. Upon receiving the indication that the tissue cut is complete, the generator **102** may continue to the fourth region **1484**. It will also be appreciated that, in some embodiments, the generator **102** may transition from the third region **1482** directly to RF (and/or

ultrasonic) termination at **1486**, for example, if the timeout time has been reached.

(264) In the fourth region **1484**, the generator **102** may provide an energy profile designed to complete coagulation of the now-cut tissue. For example, according to various embodiments, the generator **102** may provide a predetermined number of pulses. The pulses may be provided in a manner similar to that described above with respect to the process flow **1488**. The power and pulse characteristics of the pulses may be determined according to any suitable manner. For example, power and pulse characteristics of the fourth region pulses may be determined based on the current composite load curve, the third-region-specific load curve, or a fourth-region-specific composite load curve. In some embodiments, power may be determined based on the current composite load curve, while pulse characteristics may be fourth region-specific. Also, according to various embodiments, the power and pulse characteristics of fourth region pulses may be determined independent of the current composite load curve.

(265) FIG. **78** illustrates one embodiment of a pulse timing diagram **1474** illustrating an example application of the algorithm **1452** by the generator **102** (e.g., by a digital device of the generator **102**). A first region pulse **1502** is shown in the first region **1478**. The first region pulse **1502** may be utilized, as described, to measure an initial tissue impedance. At the completion of the first region pulse (**1509**), second region **1480** may begin with second region pulses **1504** applied. The second region pulses **1504** may be applied according to the various composite load curves **1456**, **1458**, **1460**, **1462**, for example, as described herein. In the example diagram **1474**, the second region **1480** concludes at **1510** when the tissue reaches the threshold impedance ($Z_{sub.term}$). The third region **1482** is then implemented, with third region pulses **1506**, as described above, applied until a knife limit signal is received at **1512**. At that point, the fourth region **1484** may commence, with fourth region pulses **1508**, as described above, applied until cycle completion at **1514**.

(266) According to various embodiments, the generator **102** may implement a user interface in conjunction with the algorithm **1452**. For example, the user interface may indicate the current region of the algorithm. The user interface may be implemented visually and/or audibly. For example, the generator **102** may comprise a speaker for generating audible tones or other audible indication. At least one audible indication may correspond to the second region **1480**. The third and fourth regions **1482**, **1484** may also have region-specific audible indications. According to various embodiments, the first region **1478** may have a region-specific audible indication as well. According to various embodiments, the audible indications may comprise pulsed tones generated by the generator **102**. The frequency of the tones and/or the pitch of the tones themselves may indicate the current region. In addition to, or instead of, the audible indications, the generator **102** may also provide a visual indication of the current region (e.g., on output device **147**). It will be appreciated that the clinician may utilize the described user interface to properly use the generator **102** and associated surgical devices **104**, **106**. For example, the indication of the second region **1480** may let the clinician know that tissue treatment has begun. The indication of the third region **1482** may let the clinician know that the tissue is ready for the cutting operation. The indication of the fourth region **1484** may let the clinician know that the cutting operation is complete. The cessation of the indication and/or a final indication may indicate that the total cutting/coagulation operation is complete.

(267) FIG. **79** illustrates a graphical representation of drive signal voltage, current and power according to an example load curve **1520**. In the chart **1520**, drive signal voltage is represented by line **1522**, drive signal current is represented by line **1524** and drive signal power is represented by line **1526**. Pulse width is not indicated in FIG. **79**. In various embodiments, the values for voltage **1522**, current **1524** and power **1526** indicated by the graph **1520** may represent possible values within a single pulse. Accordingly, the load curve **1520** may be expressed as a composite load curve by adding a curve (not shown) indicating a pulse width as a function of tissue impedance or another tissue condition. As shown for the load curve **1520**, the maximum voltage **1522** is 100 Volts Root Mean Square (RMS), the maximum current is 3 Amps RMS and the maximum power is

135 Watts RMS.

(268) FIGS. **79-84** illustrate graphical representations of various example composite load curves **1530, 1532, 1534, 1536, 1538, 1540**. Each of the composite load curves **1530, 1532, 1534, 1536, 1538, 1540** may indicate both pulse power and pulse width in terms of measured tissue impedance. The composite load curves **1530, 1532, 1534, 1536** may be implemented either in isolation or as part of a pattern of successively more aggressive composite load curves, as described above with respect to the algorithm **1452**.

(269) FIG. **80** illustrates a graphical representation of a first example composite load curve **1530**. The composite load curve **1530** may have a maximum pulse power of 45 Watts RMS and a maximum pulse width of 0.35 seconds. In FIG. **80**, the power as a function of tissue impedance is indicated by **1542**, while the pulse width as a function of tissue impedance is indicated by **1544**. Table 1 below illustrates values for the composite load curve **1530** for tissue impedances from 0Ω to 475Ω.

(270) TABLE-US-00001 TABLE 1 V I P Load, Lim, Lim, Lim, PW, Ohms RMS RMS W Sec 0-24 85 1.4 45 0.35 25-49 85 1.4 45 0.35 50-74 85 1.4 45 0.3 75-99 85 1.4 45 0.3 100-124 85 1.4 45 0.25 125-149 85 1.4 45 0.25 150-174 85 1.4 45 0.2 175-199 85 1.4 45 0.2 200-224 85 1.4 44 0.15 225-249 85 1.4 40 0.15 250-274 85 1.4 36 0.1 275-299 85 0.31 24 0.1 300-324 85 0.28 22 0.1 325-349 85 0.26 20 0.1 350-374 85 0.25 19 0.1 375-399 85 0.22 18 0.1 400-424 85 0.21 17 0.1 425-449 85 0.2 16 0.1 450-475 85 0.19 15 0.1 475+ 85 0.15 14 0.1

In various embodiments, the composite load curve **1530** may be suited to smaller surgical devices and/or smaller tissue bites.

(271) FIG. **81** illustrates a graphical representation of a second example composite load curve **1532**. The composite load curve **1532** may have a maximum pulse power of 45 Watts RMS and a maximum pulse width of 0.5 seconds. In FIG. **81**, the power as a function of tissue impedance is indicated by **1546**, while the pulse width as a function of tissue impedance is indicated by **1548**. Table 2 below illustrates values for the composite load curve **1532** for tissue impedances from 0Ω to 475Ω.

(272) TABLE-US-00002 TABLE 2 Load, V Lim, I Lim, P Lim, PW, Ohms RMS RMS W Sec 0-24 85 3 45 0.5 25-49 85 2 45 0.5 50-74 85 1.4 45 0.5 75-99 85 1.1 45 0.5 100-124 85 0.9 45 0.5 125-149 85 0.7 45 0.5 150-174 85 0.55 45 0.5 175-199 85 0.48 45 0.5 200-224 85 0.42 32 0.5 225-249 85 0.38 28 0.5 250-274 85 0.33 26 0.3 275-299 85 0.31 24 0.3 300-324 85 0.28 22 0.25 325-349 85 0.26 20 0.25 350-374 85 0.25 19 0.25 375-399 85 0.22 18 0.25 400-424 85 0.21 17 0.25 425-449 85 0.2 16 0.25 450-475 85 0.19 15 0.25 475+ 85 0.15 14 0.25

The composite load curve **1532** may be targeted at small, single vessel tissue bites and, according to various embodiments, may be a first composite power curve applied in region two 1480.

(273) FIG. **82** illustrates a graphical representation of a third example composite load curve **1534**. The composite load curve **1534** may have a maximum pulse power of 60 Watts RMS and a maximum pulse width of 2 seconds. In FIG. **82**, the power as a function of tissue impedance is indicated by **1550**, while the pulse width as a function of tissue impedance is indicated by **1552**. Table 3 below illustrates values for the composite load curve **1534** for tissue impedances from 0Ω to 475Ω.

(274) TABLE-US-00003 TABLE 3 Load, V Lim, I Lim, P Lim, PW, Ohms RMS RMS W Sec 0-24 85 3 60 2 25-49 85 3 60 2 50-74 100 3 60 2 75-99 100 3 60 2 100-124 100 3 60 2 125-149 100 3 60 2 150-174 100 3 55 0.5 175-199 100 3 50 0.5 200-224 85 0.42 32 0.3 225-249 85 0.38 28 0.3 250-274 85 0.33 26 0.3 275-299 85 0.31 24 0.3 300-324 85 0.28 22 0.25 325-349 85 0.26 20 0.25 350-374 85 0.25 19 0.25 375-399 85 0.22 18 0.25 400-424 85 0.21 17 0.25 425-449 85 0.2 16 0.25 450-475 85 0.19 15 0.25 475+ 85 0.15 14 0.25

The composite load curve **1534** may be more aggressive than the prior curve **1532** by virtue of its generally higher power. The composite load curve **1534** may also, initially, have higher pulse widths than the prior curve **1532**, although the pulse widths of the composite load curve **1534** may

begin to drop at just 150Ω. According to various embodiments, the composite load curve **1536** may be utilized in the algorithm **1542** as a load curve implemented sequentially after the composite load curve **1532**.

(275) FIG. **83** illustrates a graphical representation of a fourth example composite load curve **1536**. The composite load curve **1536** may have a maximum pulse power of 90 Watts RMS and a maximum pulse width of 2 seconds. In FIG. **83**, the power as a function of tissue impedance is indicated by **1554**, while the pulse width as a function of tissue impedance is indicated by **1556**. Table 4 below illustrates values for the composite load curve **1536** for tissue impedances from 0Ω to 475Ω.

(276) TABLE-US-00004 TABLE 4 V P Load, Lim, I Lim, Lim, PW, Ohms RMS RMS W Sec 0-24 85 3 90 2 25-49 85 3 90 2 50-74 100 3 90 2 75-99 100 3 90 2 100-124 100 3 80 2 125-149 100 3 65 2 150-174 100 3 55 0.5 175-199 100 3 50 0.5 200-224 85 0.42 32 0.3 225-249 85 0.38 28 0.3 250-274 85 0.33 26 0.3 275-299 85 0.31 24 0.3 300-324 85 0.28 22 0.25 325-349 85 0.26 20 0.25 350-374 85 0.25 19 0.25 375-399 85 0.22 18 0.25 400-424 85 0.21 17 0.25 425-449 85 0.2 16 0.25 450-475 85 0.19 15 0.25 475+ 85 0.15 14 0.25

(277) The composite load curve **1536** may, again, be more aggressive than the prior curve **1534** and, therefore, may be implemented sequentially after the curve **1534** in the algorithm **1452**. Also, according to various embodiments, the composite load curve **1536** may be suited to larger tissue bundles.

(278) FIG. **84** illustrates a graphical representation of a fifth example composite load curve **1538**. The composite load curve **1538** may have a maximum pulse power of 135 Watts RMS and a maximum pulse width of 2 seconds. In FIG. **84**, the power as a function of tissue impedance is indicated by **1558**, while the pulse width as a function of tissue impedance is indicated by **1560**. Table 5 below illustrates values for the composite load curve **1538** for tissue impedances from 0Ω to 475Ω.

(279) TABLE-US-00005 TABLE 5 V I P Load, Lim, Lim, Lim, PW, Ohms RMS RMS W Sec 0-24 85 3 135 2 25-49 85 3 135 2 50-74 100 3 135 2 75-99 100 3 100 2 100-124 100 3 80 2 125-149 100 3 65 2 150-174 100 3 55 0.5 175-199 100 3 50 0.5 200-224 85 0.42 32 0.3 225-249 85 0.38 28 0.3 250-274 85 0.33 26 0.3 275-299 85 0.31 24 0.3 300-324 85 0.28 22 0.25 325-349 85 0.26 20 0.25 350-374 85 0.25 19 0.25 375-399 85 0.22 18 0.25 400-424 85 0.21 17 0.25 425-449 85 0.2 16 0.25 450-475 85 0.19 15 0.25 475+ 85 0.15 14 0.25

The composite load curve **1538** may be used sequentially after the prior curve **1536** in the algorithm **1452**.

(280) FIG. **85** illustrates a graphical representation of a sixth example composite load curve **1540**. The composite load curve **1540** may have a maximum pulse power of 90 Watts RMS and a maximum pulse width of 2 seconds. In FIG. **85**, the power as a function of tissue impedance is indicated by **1562**, while the pulse width as a function of tissue impedance is indicated by **1564**. Table 6 below illustrates values for the composite load curve **1540** for tissue impedances from 0Ω to 475Ω.

(281) TABLE-US-00006 TABLE 6 V I P Load, Lim, Lim, Lim, PW, Ohms RMS RMS W Sec 0-24 85 3 90 2 25-49 85 3 90 2 50-74 100 3 90 2 75-99 100 3 90 2 100-124 100 3 80 2 125-149 100 3 65 2 150-174 100 3 55 0.5 175-199 100 3 50 0.5 200-224 85 0.42 32 0.3 225-249 85 0.38 28 0.3 250-274 85 0.33 26 0.3 275-299 85 0.31 24 0.3 300-324 85 0.28 22 0.25 325-349 85 0.26 20 0.25 350-374 85 0.25 19 0.25 375-399 85 0.22 18 0.25 400-424 85 0.21 17 0.25 425-449 85 0.2 16 0.25 450-475 85 0.19 15 0.25 475+ 85 0.15 14 0.25

The composite power curve **1540** is less aggressive than the prior power curve **1538**. According to various embodiments, the composite power curve **1540** may be implemented in the algorithm **1452** sequentially after the curve **1538**. Also, in some embodiments, the composite power curve **1540** may be implemented in the algorithm **1452** as a third or fourth region-specific composite power curve.

(282) As described above, the various composite power curves used in the algorithm **1452** may each be implemented for a predetermined number of pulses. Table 7 below illustrates the number of pulses per composite power curve for an example embodiment utilizing the power curves **1532**, **1534**, **1536**, **1540** sequentially in the algorithm **1452**.

(283) TABLE-US-00007 TABLE 7 Composite Load Curve Number of Pulses 1532 4 1534 2 1536 2 1538 8 1540 n/a

The last composite power curve **1540** is shown without a corresponding number of pulses. For example, the composite power curve **1540** may be implemented until the clinician terminates the operation, until the timeout time is reached, until the threshold tissue impedance is reached, etc.

(284) According to various embodiments, the generator **102** may provide power to a tissue bite in a manner that brings about a desired value of other tissue parameters. FIG. **86** illustrates one embodiment of a block diagram **1570** describing the application of an algorithm **1572** for maintaining a constant tissue impedance rate of change. The algorithm **1572** may be implemented by the generator **102** (e.g., by a digital device of the generator **102**). For example, the algorithm **1572** may be utilized by the generator **102** to modulate the drive signal. Sensors **1574** may sense a tissue condition, such as tissue impedance and/or a rate of change of tissue impedance. The sensors **1574** may be hardware sensors or, in various embodiments may be software implemented sensors.

For example, the sensors **1574** may calculate tissue impedance based on measured drive signal current and voltage. The drive signal may be provided by the generator **102** to the cable/implement/load **1576**, which may be the electrical combination of the tissue, the surgical device **104**, **106** and a cable **112**, **128** electrically coupling the generator **102** to the device **104**, **106**.

(285) The generator **102**, by implementing the algorithm **1572**, may monitor the impedance of the tissue or load including, for example, the rate of change of impedance. The generator **102** may modulate one or more of the voltage, current and/or power provided via the drive signal to maintain the rate of change of tissue impedance at a predetermined constant value. Also, according to various embodiments, the generator **102** may maintain the rate of change of the tissue impedance at above a minimum impedance rate of change.

(286) It will be appreciated that the algorithm **1572** may be implemented in conjunction with various other algorithms described herein. For example, according to various embodiments, the generator **102** may sequentially modulate the tissue impedance to different, increasingly aggressive rates similar to the method **1330** described herein with reference to FIG. **70** herein. For example, a first impedance rate of change may be maintained until the total energy delivered to the tissue exceeds a predetermined energy threshold. At the energy threshold, if tissue conditions have not reached a predetermined level (e.g., a predetermined tissue impedance), then the generator **102** may utilize the drive signal to drive the tissue to a second, higher impedance rate of change. Also, in various embodiments, tissue impedance rates of change may be used in a manner similar to that described above with respect to composite load curves. For example, instead of utilizing plurality of composite load curves, the algorithm **1452** of FIG. **75** may call for applying a plurality of rates of tissue impedance change. Each rate of tissue impedance change may be maintained for a predetermined amount of time and/or a predetermined number of pulses. The rates may be successively applied in order of value (e.g., rates may successively increase). In some embodiments, however, the driven rates of tissue impedance change may peak, and then be reduced.

(287) As previously discussed, in one embodiment, the generator **102** may comprise an ultrasonic generator module **108** to drive an ultrasonic device, such as the ultrasonic device **104** be by providing a drive signal to the transducer **114** in any suitable manner. For example, the generator **102** may comprise a foot switch **120** coupled to the generator **102** via a footswitch cable **122** (FIG. **8**). Control circuits may be employed to detect the presence of the foot switch **120** and the ultrasonic generator module **108** such that the operation of the foot switch **120** is corresponds with the activation of the ultrasonic generator module **108**. Accordingly, a clinician may activate the

transducer **114**, and thereby the transducer **114** and blade **151**, by depressing the foot switch **120**. In one embodiment, the foot switch **120** may comprise multiple pedals, where each pedal may be activated to perform a particular function. In one embodiment, the foot switch **120** may comprise a first pedal, for example, to enable and/or control the ultrasonic mode of the generator **120** and a second pedal, for example, to enable and/or control the electrosurgical mode of the generator **120**. Accordingly, the generator **102** may configure the foot switch **120** pedals depending upon which drive mode of the generator **102** is active.

(288) As previously discussed, the drive mode of the generator **102** may be determined by the device identification. The characteristics of the interrogation signal may uniquely indicate the state or configuration of a control circuit, which can be discerned by the generator **102** and used to control aspects of its operation. Any of the control circuits **280**, **280-1** to **280-9**, **282**, and **282-1** to **282-8** previously described in connection with FIGS. **14-32** may be configured to set or indicate the drive mode of the generator **102**, for example. Such control circuits may be contained, for example, in an ultrasonic surgical device (e.g., in a handpiece of an ultrasonic surgical device) or may be contained in an electrosurgical device (e.g., in a handpiece of an electrosurgical device).

(289) In one embodiment, the generator **102** may be configured to operate in ultrasonic mode using the ultrasonic generator module **108**. Accordingly, the first pedal of the foot switch **120** enables and/or controls the operation of the ultrasonic generator module **108**. During ultrasonic mode operation, the first pedal (e.g., the left pedal) may be configured to activate the ultrasonic output of the ultrasonic generator module **108** to generate a drive signal that corresponds to a minimum (P.sub.Min) ultrasonic power level (e.g., minimum energy setting) and the second pedal (e.g., the right pedal) may be configured to activate the ultrasonic generator module **108** to generate a drive signal that corresponds to a the maximum (P.sub.Max) ultrasonic power level (e.g., maximum energy setting). It will be appreciated that either the first or second pedal may be configured to activate the ultrasonic generator module **108** to generate a drive signal that corresponds to the minimum (P.sub.Min) or maximum (P.sub.Max) ultrasonic power levels, without limitation. Therefore, either the left or right pedal may be assigned as the first pedal, and the other pedal may be assigned as the second pedal. In embodiments where the foot switch **120** comprises more than two pedals, each of the pedals may be assigned a predetermined function. In some embodiments, one or more of the foot pedals may be deactivated or ignored by the generator **102** DSP software or logic (e.g., executable instructions, hardware devices, or combinations thereof).

(290) In one embodiment, the generator **102** is configured in ultrasonic mode and the foot switch **120** comprises two pedals, e.g., first and second or left and right. During ultrasonic foot switch **120** activation mode, when a valid activation of the first pedal (e.g., left pedal) of the foot switch **120** is detected and an ultrasonic surgical device **104** (FIG. **8**) is connected to the ultrasonic generator module **108**, the generator **102** DSP software is configured to drive the ultrasonic transducer **114** of the ultrasonic surgical device **104** at the maximum power level, for example. For ultrasonic transducers **114**, the maximum power (P.sub.Max) level can be obtained at the maximum drive current (I.sub.Max). In embodiments where the foot switch **120** comprises more than two pedals, any two of the multiple pedals may be configured in accordance with the above functionality.

(291) In one embodiment, during ultrasonic foot switch **120** activation mode, when a valid activation of the second pedal (e.g., right pedal) of the foot switch **120** is detected and an ultrasonic surgical device **104** (FIG. **8**) is connected to the ultrasonic generator module **108**, the generator **102** DSP software or logic is configured to drive the ultrasonic transducer **114** of the ultrasonic surgical device **104** at the minimum power level, for example. For ultrasonic transducers **114**, the maximum power level can be obtained at the minimum drive current.

(292) In another embodiment, the generator **102** may comprise an electrosurgery/RF generator module **110** to drive the electrosurgical device **106** in any suitable manner. Control circuits may be employed to detect the presence of the foot switch **120** and the electrosurgery/RF generator module **110** such that the operation of the foot switch **120** is corresponds with the activation of the

electrosurgery/RF generator module **110**. Accordingly, the generator **102** may be configured to operate in electrosurgical mode (e.g., bi-polar RF mode) using the electrosurgery/RF generator module **110** and the foot switch **120** may comprise two pedals, e.g., first and second or left and right. The first pedal of the foot switch **120** enables and/or controls the operation of the electrosurgery/RF generator module **110**. During electrosurgical mode operation, the first pedal (e.g., the left pedal) may be configured to activate the bi-polar RF output of the electrosurgery/RF generator module **110** and the second pedal (e.g., the right pedal) switch may be ignored and may be referred to as an inactive switch. It will be appreciated that the inactive switch (e.g., the right pedal) may be used by the user for input to the generator **102** other than power level. For example, tapping the inactive switch pedal may be a way for the user to acknowledge or clear a fault, among other functions, as well as other inputs from the user to the generator by way of the inactive switch pedal. In embodiments where the foot switch **120** comprises more than two pedals, any two of the multiple pedals may be configured in accordance with the above functionality.

(293) In one embodiment, during electrosurgical foot switch **120** activation mode when an electrosurgical or RF surgical device **106** (FIG. **8**) is connected to the electrosurgery/RF generator module **110**, the generator **102** DSP and user interface (UI) software or logic is configured to ignore the status of the second pedal of the foot switch **120**, which is the maximum activation switch. Accordingly, there is no change in power delivery and no change in visual or audible feedback to the user when the second pedal of the foot switch **120** is activated. Therefore, when the electrosurgery/RF generator module **110** transitions from inactive to active of the Max switch-input on the second pedal of the foot switch **120** while all other activation switch inputs are inactive, the electrosurgery/RF generator module **110** ignores the switch input from the second pedal of the foot switch **120** and does not sound or emit a tone.

(294) In one embodiment, during electrosurgical foot switch **120** activation mode when an electrosurgical or RF surgical device **106** (FIG. **8**) is connected to the electrosurgery/RF generator module **110**, the generator **102** DSP software is configured to drive the electrosurgical or RF device **106** in accordance with a predetermined algorithm specific to the electrosurgical or RF device **106**. Various algorithms are discussed in the present disclosure.

(295) As previously discussed, according to various embodiments, the generator **102** may implement a user interface in conjunction with the algorithm **1452** described in connection with FIG. **75**. One embodiment of a user interface now will be described with reference to FIGS. **75** and **76** where FIG. **75** illustrates a process flow illustrating one embodiment of the algorithm **1452**, as implemented by the generator **102** (e.g., by a digital device of the generator **102**) and FIG. **76** illustrates shows a process flow illustrating one embodiment of the algorithm **1452**, as implemented by the generator **102** (e.g., by a digital device of the generator **102**). The algorithm **1452** may be activated at **1476**. It will be appreciated that the algorithm **1452** may be activated in any suitable manner. For example, the algorithm **1452** may be activated by a clinician upon actuation of the surgical device **104**, **106** (e.g., by pulling or otherwise actuating a jaw closure trigger **138**, **142**, switch, handle, etc.).

(296) Accordingly, in one embodiment, the algorithm **1452** may be implemented to control the operation of the electrosurgery/RF generator module **110** to control the electrosurgical or RF surgical device **106** (FIG. **8**). Accordingly, in one embodiment of the algorithm **1452**, a first activation tone occurs during the activation of the electrosurgical or RF surgical device **106** that transitions to a second activation tone that occurs when the impedance and/or energy conditions/threshold(s) are met and then transitions to a third activation tone that occurs when the impedance and/or energy conditions/threshold(s) and an end effector **132** (e.g., knife) position condition are all met. The tone from the second activation tone to third activation tone is reset upon opening of the jaws as determined by impedance as described with reference to the algorithm **1452** as described and illustrated in connection with FIG. **76** used to drive the electrosurgical or RF surgical device **106**.

(297) Accordingly, turning now to FIG. 76, in one embodiment the algorithm 1452 controls the activation of various audible feedback during certain power delivery regions and transitions between regions. In one embodiment, multiple audible feedback is provided based on operational cycle and region of the electrosurgery/RF generator module 110 of the generator 102. In one embodiment, for example, a first audio tone (Audio Tone-I) is emitted when the electrosurgical (RF) power delivery cycle begins. The first audio tone is emitted, for example, whenever the electrosurgery/RF generator module 110 of the generator 102 enters the first region 1478 and continues in the second region 1480.

(298) A second audio tone (Audio Tone-II) is emitted when the tissue impedance threshold is reached. In one embodiment, the second tone is emitted, for example, upon the electrosurgery/RF generator module 110 of the generator 102 entering the third region 1482. In one aspect, the second tone may be latched such that if a transition occurs from the third region 1482 to the second region 1480, the second tone is continued to be emitted.

(299) A third audio tone (Audio Tone-III) is emitted when the electrosurgical (RF) power delivery cycle is complete. In one embodiment, when the generator 102 DSP software and/or logic determines that a “cycle complete” status has been reached, the third tone is emitted upon completion of a fourth energy pulse in the fourth region 1484.

(300) Still with reference to FIG. 76, in one embodiment, the algorithm 1452 may be employed when marching on tissue, which is defined as opening the jaws of the end effector 132 (FIG. 8) without releasing the energy activation, re-grasping on tissue, and then continuing energy activation. To accommodate the marching on tissue functionality, the algorithm 1452 may be implemented as follows. While in the second region 1480 or third region 1482, if the tissue impedance $|Z_{sub}.mag| > |Z_{sub}.jaws_open|$, the algorithm 1452 re-enters the first region 1478 and implements the first region 1478 applicable Power (P), Current (I), Voltage (V) limits. In one embodiment, while re-entering the first region 1478 T.sub.timeout is not reset and the T.sub.start timer and the energy accumulation parameter is reset. In one embodiment, when re-entering the first region 1478, the algorithm 1452 shall cause the first audio tone (Audio Tone-I) to be emitted and the logic to switch again to the second audio tone (Audio Tone-II) is kept the same as previously discussed.

(301) It will be appreciated that the previously described sequence of tones, first, second, and third audio tones, may be applied to indicate energy delivery to tissue when the generator 102 is configured to operate in ultrasonic mode using the ultrasonic generator module 108. For example, the first audio tone (Audio Tone-I) is emitted when the ultrasonic energy delivery cycle begins, the second audio tone (Audio Tone-II) is emitted when the tissue impedance threshold is reached, and the third audio tone (Audio Tone-III) is emitted when the ultrasonic energy delivery cycle is complete.

(302) FIGS. 87-99 illustrate various additional embodiments of control circuits of surgical devices are provided. In one embodiment, the control circuit may comprise a first circuit portion coupled to at least one switch operable between an open state and a closed state. The first circuit portion communicates with a surgical generator over a conductor pair to receive a control signal to determine a state of the at least one switch. The control circuit also comprises at least one resistor coupled to the at least one switch, where the state of the at least one switch is determinable based on the value of the at least one resistor. In one embodiment, the at least one resistor is coupled in series with the at least one switch. In one embodiment, the at least one resistor is coupled in parallel with the at least one switch. In one embodiment, the control circuit comprises a voltage reference coupled in parallel to the at least one switch. In one embodiment, the control circuit comprises a capacitor coupled to the voltage reference, where the state of the at least one switch is determinable based on a slope of the voltage on the capacitor. In one embodiment, the control circuit comprises an oscillator in communication with the at least one switch, where the state of the at least one switch is determinable based on a frequency of the oscillator, which is determined by the value of

the at least one resistor. In one embodiment, the control circuit comprises a one-wire multi-switch input device coupled to the at least one switch and the at least one resistor, wherein the state of the at least one switch is determinable based on the at least one resistor and is communicated to the generator over a one-wire communication protocol. In one embodiment, the control signal is a differential constant current pulse having a positive phase and a negative phase. In one embodiment, the control circuit comprises a second circuit portion comprising a data circuit element coupled to the first circuit portion. In one embodiment, the data circuit element comprises at least one memory device. In one embodiment, the at least one memory device comprises at least one one-wire electrically erasable programmable read only memory (EEPROM).

(303) In various embodiments, a control circuit of a surgical device is provided. In one embodiment, the control circuit comprises a first circuit portion is coupled to at least one switch operable between an open state and a closed state. The first circuit portion communicates with a surgical generator over a conductor pair to receive a control signal from input terminals to determine a state of the at least one switch. The control signal having a first phase and a second phase. A first transistor coupled between the input terminals, a first capacitor, and a first resistor coupled in series with the first capacitor. During the first phase of the control signal the first transistor is held in cutoff mode while the first capacitor charges to a predetermined voltage and during an initial portion of the second phase of the control signal the first transistor transitions from cutoff mode to saturation mode and is held in saturation mode until the first capacitor discharges through the first resistor. During a final portion of the second phase of the control signal the first transistor transitions from saturation mode to cutoff mode when the first capacitor voltage drops below a predetermined threshold. In one embodiment, during the initial portion of the second phase of the control signal while the first transistor is in saturation mode, a first impedance is presented between the input terminals and during the final portion of the second phase of the control signal while the first transistor is in cutoff mode, a second impedance is presented between the input terminals, where the state of the at least one switch is determinable based on the first and second impedance values. In one embodiment, the control circuit comprises at least one second resistor coupled to the at least one switch, where the second impedance is based at least in part on the at least one second resistor when the at least one switch is in the open state. In one embodiment, the control circuit comprises at least one second resistor coupled to the at least one switch, where the second impedance is based at least in part on the at least one second resistor when the at least one switch is in the closed state. In one embodiment, the control circuit comprises a second transistor coupled to the first capacitor to charge the first capacitor during the first phase of the control signal. In one embodiment, the control circuit comprises a second circuit portion comprising a data circuit element coupled to the first circuit portion, wherein the first capacitor voltage is sufficient to supply voltage to the data element. In one embodiment, the control circuit comprises a voltage reference coupled in parallel to the at least one switch. In one embodiment, the control circuit comprises a second capacitor coupled to the voltage reference, where the state of the at least one switch is determinable based on a slope of the voltage on the second capacitor. In one embodiment, the first phase of the control signal is a positive transition of a current pulse and the second phase of the control signal is a negative transition of a current pulse.

(304) In various embodiments, a method is provided. In one embodiment, the method comprises receiving a control signal at a control circuit of a surgical device and determining the state of the at least one switch based on the value of the resistor. The control circuit comprises a first circuit portion coupled to at least one switch operable between an open state and a closed state, the first circuit portion to communicate with a surgical generator over a conductor pair to receive the control signal, the first circuit portion comprising at least one resistor coupled to the at least one switch. In one embodiment, the method comprises presenting a first impedance during a first phase of the control signal and presenting a second impedance during a second phase of the control signal based on the state of the at least one switch, where determining the state of the at least one switch

comprises comparing the first impedance to the second impedance. In one embodiment, the method comprises generating a first voltage slope during a first phase of the control signal and generating a second voltage slope during a second phase of the control signal, where determining the state of the at least one switch comprises comparing the first voltage slope to the second voltage slope. In one embodiment, the method comprises generating a first frequency during a first phase of the control signal and generating a second frequency during a second phase of the control signal, where determining the state of the at least one switch comprises comparing the first frequency to the second frequency. In one embodiment, the method comprises reading a value of the least one resistor by a one-wire multi-switch input device coupled to the at least one switch and communicating the value of the at least one resistor to the generator over a one-wire communication protocol.

(305) The control circuits disclosed in connection with FIGS. **87-99** are suitable for use with surgical devices, including, for example, disposable surgical devices. The control circuits can overcome challenges such as cost constraints, non-ideal surgical device usage conditions, and compatibility with existing equipment. In addition, the series resistance in the switch lines may increase over time and usage due to residue build up from cleaning agents on the surgical device handpiece contacts.

(306) The configuration and detection methodology of each of the control circuits described in connection with FIGS. **87-99**, may be configured to operate with any one of, any combination of, or all of, the following requirements: compatibility with constant current sources using voltage level state detection as may be provided in the generator **102**; compensation circuits for the cable **112** (e.g., line) impedance (i.e., contaminated handpiece contacts); support functionality for multiple switches provided on the surgical device handpiece (scalable design to support one, two, three, or more switches on the handpiece of the surgical device); provide a circuit presence state; support functionality for multiple simultaneous switch states; support functionality for multiple one-wire memory components (power and communication); low cost for disposable device; maximize use of “off the shelf” components; minimize circuit footprint (e.g., approximately 15 mm×15 mm); provide for mitigation of resultant hazardous conditions due to single point component failure (e.g., open/short circuit); provide electrostatic discharge (ESD) mitigation.

(307) The various embodiments of the control circuits described in connection with FIGS. **87-99** are configured to be located in a handpiece of a surgical instrument. The control circuits receive an interrogation signal transmitted by the generator **102** through a cable **112**. As discussed above in connection with FIG. **10**, a control circuit may modify the characteristics of an interrogation signal transmitted by the generator **102**. The characteristics of the interrogation signal, which may uniquely indicate the state or configuration of the control circuit and/or the handpiece of the surgical device, can be discerned by circuitry located at the generator **102** side and are used to control aspects of its operation. The various embodiments of the control circuits described in connection with FIGS. **87-99** may be contained in an ultrasonic surgical device (e.g., in the handpiece **116** of the ultrasonic surgical device **104**), in an electrosurgical device (e.g., in the handpiece **130** of the electrosurgical device **106**), or in a combination ultrasonic/electrosurgical device (e.g., in the handpiece of a device that incorporates the functional aspects of the ultrasonic device **104** and the electrosurgical device **106** in a single handpiece), as described herein.

(308) FIG. **87** illustrates one embodiment of a control circuit **1602** comprising parallel switched resistance circuit with high speed data communication support and at least one data element comprising at least one memory device. In one embodiment, the control circuit **1602** is operable by a control signal in the form of a differential current pulse. As shown in FIG. **87**, in one embodiment, the control circuit **1602** may be connected or coupled to the generator **102** via the cable **112** (and/or **128**) to receive an interrogation signal (e.g., a bipolar interrogation signal at a predetermined frequency, such as 2 kHz, for example) from a signal conditioning circuit **202** (e.g., from generator terminals HS and SR shown in FIG. **10** via a conductive pair of cable **112** or **128**).

In one embodiment, the control circuit **1602** comprises multiple parallel switches SW1, SW2, SW.sub.n. The embodiment of the control circuit **1602** in FIG. **87** can be employed with any suitable number n of switches in parallel with a fixed resistor R1. Accordingly, any suitable number of switches SW.sub.n, where n is an integer greater than zero, may be located in parallel with a predetermined fixed resistance to provide an indication of the configuration and/or operation of a surgical device **1600** (e.g., an ultrasonic device **104**, an electrosurgical device **106**, or a combined ultrasonic/electrosurgical device **104/106**). The number of switches n determines the number of possible states. For example, a single switch SW1 can indicate up to two states, two switches can indicate up to four states, and n switches can indicate up to 2n states. The switches SW1, SW2, SW.sub.n may be any type of switch. In one embodiment, the switches SW1, SW2, SW.sub.n are pushbutton switches located on the handpiece of the surgical device **1600**. The embodiments, however, are not limited in this context.

(309) Depending on the open or closed state of each of the switches SW1, SW2, SW.sub.n, for example, the signal conditioning circuit **202** (e.g., from generator terminals HS and SR shown in FIG. **10** via a conductive pair of the cable **112**, **128**) would see a different impedance (e.g., resistance). For embodiments where n=2 that employ two switches SW1, SW2, for example, four unique switch configurable impedances can be detected by the signal conditional conditioning circuit **202**, as described in Table 8, below:

(310) TABLE-US-00008 TABLE 8 SW1 SW2 Impedance Open Open R1 Open Closed R1 || R3 Closed Open R1 || R2 Closed Closed R1 || R2 || R3

(311) The interrogation signal may be coupled across the inputs HS/SR of the control circuit **1602** and applied across both branches of switches SW1, SW2. When both SW1 and SW2 are open, the control circuit **280** may define an open circuit corresponding to a first state. When SW1 is open and SW2 is closed, the interrogation signal may define a second state. When SW1 is closed and SW2 is open, the interrogation signal may define a third state. When both SW1 and SW2 are closed, the interrogation signal may define a fourth state. Accordingly, based on the different characteristics of the interrogation signal corresponding to the different states of SW1 and SW2, the state or configuration of the control circuit **280** may be discerned by the generator **102** based on a voltage signal appearing across the inputs of the control circuit **280** (e.g., as measured by an ADC of the signal conditioning circuit **202**).

(312) Still referring to FIG. **87**, in one embodiment, the control circuit **1602** comprises a diode D1 placed across the HS/RS terminals of the control circuit **1602** to clamp and protect the control circuit **1602** from electro-static discharges (ESD) and/or transients coming from the surgical device **1600** and the generator **102** when the surgical device **1600** is connected and disconnected to the generator **102**. In the illustrated embodiment, diode D1 is a bi-directional transient voltage suppressor. Diode D1 may be selected based on a predetermined ESD rating, such as for example 8 kV contact discharge, 5V nominal working voltage, small size, and low capacitance to minimize affecting the pulse waveforms. Any suitable bi-directional transient voltage suppressor may be employed without limitation. In other embodiments, as shown in FIGS. **93-95** and **97**, for example, multiple diodes can be placed across the inputs HS/SR to protect the surgical device **1600** and/or the control circuit **1602** from ESD and other transients.

(313) Diode D2 is provided for unidirectional routing of negative current pulses to the resistor network R1, R2, R3, and R.sub.n+1 configured by the switches SW1, SW2, SW.sub.n. In the illustrated embodiment, diode D2 is a single Schottky diode to provide a low voltage drop and minimize impact to the usable voltage range to the ADC of the generator **102**. Diode D2 may be provided in discrete form, as part of a multi-diode package, or may be formed as part of a custom integrated circuit (e.g., and ASIC). In other embodiments, other suitable diodes may be employed without limitation. During the positive phase (rising edge) of the current pulse, diode D2 is reverse biased and current flows through transistor M1, which charges gate capacitor C2 through diode D3. During the negative phase (falling edge) of the current pulse, diode D2 is forward biased and the

ADC of the generator **102** sees the impedance of the resistor network **R1**, **R2**, **R3**, and **R.sub.n+1** configured by the switches **SW1**, **SW2**, **SW.sub.n**. In other embodiments, a pair of diodes may be employed for bidirectional power as shown in connection with FIGS. **92-95** and **97**.

(314) Transistor **M1** is provided to route a positive current pulse from the generator **102** to gate capacitor **C2**. In the illustrated embodiment, transistor **M1** is a p-channel MOSFET, although in other embodiments, any suitable transistor may be employed, without limitation. In the illustrated embodiment, the drain terminal of transistor **M1** is connected to the HS input terminal and to the cathodes of diodes **D1**, **D2** and the gate terminal is connected to the SR input terminal of the surgical device **1600**. The source of transistor **M1** is connected to the SR input terminal through resistor **R5**. A positive current pulse, as described in more detail with reference to FIG. **88**, received at the HS terminal charges capacitor **C2** through **M1**. Capacitor **C2** is coupled to the gate of transistor **M2** and when charged can turn on transistor **M2** during a negative phase of the current pulse or supply power to the at least one memory device **1604** during a positive phase of the input current pulse. Although shown as a discrete component, transistor **M1** may be provided a multi-transistor package or may be formed as part of a custom integrated circuit (e.g., and ASIC). In other embodiments, other suitable transistors may be employed without limitation.

(315) In one embodiment, the at least one memory device **1604** may comprise, for example, one or more one-wire EEPROM(s), known under the trade name "1-Wire." As previously discussed, the one-wire EEPROM is a one-wire memory device **1604**. Generally, multiple EEPROMs may be employed in the control circuit **1602**. In the illustrated embodiment, up to two EEPROMs may be used in the surgical device **1600**. The EEPROM(s) are powered during the positive phase of the current pulse. Transistor **M1** also provides low forward voltage drop and minimizes impact to the EEPROM high speed communications. Transistor **M1** provides reverse blocking capabilities, which cannot be realized with a simple diode.

(316) A second transistor **M2** provides a substantial low impedance path to short out the resistor network **R1**, **R2**, **R3**, **R.sub.n+1** in order for the ADC to make an initial reference measurement. The drain terminal of transistor **M2** is connected to the anode of diode **D2** and one end of resistors **R1**, **R2**, **R3**, and **R.sub.n+1**. The source terminal of transistor **M2** is connected to the SR terminal of the surgical device **1600**. In the illustrated embodiment, transistor **M2** is also a p-channel MOSFET transistor, although in other embodiments, any suitable transistor can be employed. During a positive phase of the current pulse, transistor **M2** turns on and applies a substantial low impedance path across the HS/RS terminals of the surgical device **1600** for a relatively short duration of time as defined by an RC time constant on the gate terminal of transistor **M2**, where the RC time constant is determined by resistor **R4** and gate capacitor **C2**, each of which is coupled to the gate to transistor **M2**. Resistor **R4** provides a discharge path and timing for the voltage decay of gate capacitor **C2**. Gate capacitor **C2** charges during the positive phase of the current pulse, when transistor **M2** is cut off or open circuit. Eventually, the voltage on gate capacitor **C2** is sufficient to negatively bias the gate-source and turns on the transistor **M2** during the negative phase of the current pulse. A first measurement is then obtained by the ADC of the signal conditioning circuit **202** in the generator **102**, which is referred to herein as the "ADC_NEG1" measurement, as shown in FIG. **90**, for example.

(317) Diode **D3** provides a current path during the positive phase of the current pulse and enables gate capacitor **C2** to charge quickly while bypassing resistor **R4**, which provides the discharge path for gate capacitor **C2**. In one embodiment, transistor **M2** may be selected due to a well defined minimum and maximum gate threshold to provide better characterization of ON/OFF timing for the differential measurement. In one embodiment, transistor **M2** component may be provided as a stand alone discrete component or in the same package as transistor **M1**. Resistor **R4** may be selected based on the 1% tolerance, 100 ppm temperature coefficient, and small package size. The identified component selection criterion is provided merely as a non-limiting example. Any suitable component may be substituted to perform the desired functionality.

(318) As previously discussed, resistor network **R1**, **R2**, **R3**, and **R.sub.n+1** is configured by the switches **SW1**, **SW2**, **SW.sub.n** to present a different resistance (e.g., impedance) to the ADC of the generator **102**. Resistor **R1** sets the maximum resistance used for switch detection, e.g., when all switches **SW1**, **SW2**, **SW.sub.n** are open (“off”). When **SW1** is closed (“on”), resistor **R2** is located in parallel with the other switch detection resistors, as shown in Table 8, for example. When **SW2** is closed (“on”), resistor **R3** is located in parallel with other switch detection resistors, as shown in Table 8, for example. Each of the resistors **R1**, **R2**, **R3**, **R.sub.n+1** can be selected based on a 1% tolerance, 100 ppm temperature coefficient, and small package size, without limitation. Capacitor **C1** may be located in parallel with the resistors **R1**, **R2**, **R3**, and **R.sub.n+1** to provide high frequency filtering from conducted and/or radiated noise, for example. In one embodiment, capacitor **C1** may be a ceramic capacitor, without excluding other capacitor types.

(319) Resistor **R5** provides a relatively low impedance path to rapidly turn on transistor **M2** during the start of the negative phase of the current pulse. Gate capacitor **C2** charges in conjunction with resistor **R4** to increase the gate voltage above the turn on threshold of transistor **M2** for a short duration. The value of resistor **R5** also may be selected to provide a sufficient supply voltage to the one or more EEPROM(s) during the positive phase of the current pulse.

(320) FIG. **88** is a graphical representation of one aspect of a control signal in the form of a constant current pulse waveform **1610** that may be generated by the signal conditioning circuit **202** of the generator **102** as shown in FIG. **10**. Time, in milliseconds (mS), is shown along the horizontal axis and Current, in milliamperes (mA), is shown along the vertical axis. In the embodiment illustrated in FIG. **88**, periodic constant current pulses **1612**, **1614**, each having a first phase shown as positive transitions **1612a**, **1614a** and a second phase shown as negative transitions **1612c**, **1614c**, at ± 15 mA are produced by a current source generator located at the generator **102** side. The current source generator is coupled to the HS/SR input terminals of the surgical device **1600**/control circuit **1602** via the cable **112**, **128**. Other constant current amplitudes may be employed. In the illustrated embodiment, the control circuit **1602** utilizes both pulse polarities to detect up to $2^{sup.n}$ unique combinations of switches **SW1**, **SW2**, **SW.sub.n** and also supports communication to the one-wire EEPROM(s).

(321) FIG. **88A** is a graphical representation of one aspect of a control signal in the form of the constant current pulse waveform **1610** of FIG. **88** showing numerical values of various features of the waveform **1610** according to one example embodiment. In the example embodiment of FIG. **88A**, time on the horizontal axis is shown in milliseconds (mS) while current along the horizontal axis is shown in milliamperes (mA). Also, in the example embodiment of FIG. **88A**, the current pulses **1612**, **1614** may have an amplitude of ± 15 mA.

(322) It will be appreciated that although the current pulse is characterized as having a first phase that is a positive transition and a second phase is a negative transition, the scope of the present disclosure includes embodiments where the current pulse is characterized as having the opposite polarity. For example, the current pulse can be characterized as having a first phase that is a negative transition and a second phase is a positive transition. It will also be appreciated that the selection of the components of any of the control circuits disclosed herein can be suitably adapted and configured to operate using a current pulse characterized as having a first phase that is a negative transition and a second phase is a positive transition. Also, although the present disclosure describes the control signal as a differential current pulse, the scope of the present disclosure includes embodiments where the control signal is characterized as a voltage pulse. Accordingly, the disclosed embodiments of the control circuits are not limited in the context of the differential current pulse described in connection with FIG. **88**.

(323) With reference now to both FIGS. **87** and **88**, in one embodiment, the control circuit **1602** operates on a differential pulse with parallel resistors **R1**, **R2**, **R3**, and **R.sub.n+1** configured by the switches **SW1**, **SW2**, **SW.sub.n**. Additionally, high speed data communication is provided through transistor **M1** to the at least one memory device **1604**. In one embodiment, the control circuit **1602**

uses a variable resistance dictated by the closure of one or more pushbutton switches SW1, SW2, SW.sub.n. Both positive edges **1612a**, **1614a** and negative edges **1612c**, **1614c** of the current pulses **1612**, **1614** are used to derive a measurement that will allow for compensation of an unknown series cable **112**, **128** and contamination resistance of the switches SW1, SW2, SW.sub.n contacts within a predetermined range. Initially a positive current pulse **1612a** charges gate capacitor C2 and subsequently a negative current pulse **1612c** enables transistor M2 to turn for a brief time period defined by the RC time constant of gate capacitor C2 and resistor R4. During the first phase of the negative current pulse **1612c**, transistor M2 is placed in a saturated state creating a very low impedance path through the surgical device **1600** between terminals HS/SR. After a time period defined by the RC time constant pair of R4 and C2, gate capacitor C2 discharges through resistor R4 and the FET channel of transistor M2 opens causing the total current to be steered through the resistor R1, R2, R3, R.sub.n+1 path based on the state of the pushbutton switches SW1, SW2, SW.sub.n. The low impedance state when transistor M2 is in closed (saturation mode) and the state of the resistor network R1, R2, R3, R.sub.n+1 when transistor M2 is in an open (cutoff mode) state are the two impedance states in the time domain that are used for detecting the activation of pushbutton switches SW1, SW2, SW.sub.n. This provides the ability to compensate for an external path resistance (handpiece contacts, cabling, connector resistance) by comparing the difference between the two states.

(324) With reference back to FIG. **88**, each of the combination of switches SW1, SW2, SW.sub.n creates a unique resistance presented to the ADC at the generator **102** during the negative phase of the current pulse. The constant current pulses **1612**, **1624** transmitted from the generator **102** to the surgical device **1600** presents a unique voltage back to an ADC in the generator **102** that is proportional to the resistance selected by the switches SW1, SW2, SW.sub.n. Transistor M1 gate capacitor C2 charges during the positive current pulse phase **1612a**, **1614a**. When the current pulse waveform changes from positive to negative at **1612b**, **1614b**, the voltage on gate capacitor C2 turns on transistor M2, which effectively shorts the surgical device **1600** control circuit **1602** between the terminals HS/SR. The short circuit exists for a short duration that is set by the time constant defined by resistor R4 and gate capacitor C2 coupled to the gate of transistor M2. After gate capacitor C2 has fully discharged through resistor R4, the control circuit **1602** transitions to an impedance that represents the state of pushbutton switches SW1, SW2, SW.sub.n. Using these two measurements for each time period when transistor M2 is in an open circuit state and a short circuit state, allows the control circuit **1602** system to tolerate resistance variations due to cable wiring affects, connector contact resistance, and pushbutton switch resistance. Additionally, as previously discussed, using p-Channel MOSFET transistor M1 rather than a diode is an effective means to facilitate high speed EEPROM communications without adversely affecting the hand switch detection capability.

(325) FIG. **89** is a graphical representation **1620** of various detection regions associated with the aspect of the control signal shown in FIG. **88**. Time in “mS” is shown along the horizontal axis and Impedance in “k Ω ” is shown along the vertical axis. Five detections regions (1-5) are depicted in the graphical representation **1620**. With reference also to FIGS. **87** and **88**, region 1 represents the state of transistor M2 when it is in saturation mode during the negative phase **1612c** of the current pulse **1612** creating a low resistance path through the surgical device **1600**. This is the baseline ADC measurement that represents series resistor R1 except for pushbutton switch SW1, SW2, SW.sub.n detection resistance. Region 2 represents the state of transistor M2 after gate capacitor C2 has fully discharged through resistor R4 over a time constant defined by R4 and C2. In region 2, the surgical device **1600** will present a resistance that represents a unique pushbutton switch SW1, SW2, SW.sub.n state. The amplitude in region 2 changes depending on the states of pushbutton switches SW1, SW2, SW.sub.n. Region 3 represents the ohmic region of transistor M2 when it is transitioning from saturation mode to cutoff mode. Region 4 represents the rising positive edge **1612a** of the current pulse **1612** to charge gate capacitor C2 coupled to the gate of transistor M2.

Diode D3 with its anode also coupled to the gate of transistor M2 allows gate capacitor C2 to be charged quickly in order to be within a predetermined total timing budget. Region 5 represents the falling or negative edge **1612c** of the current pulse **1612** when transistor M2 is transitioning from cutoff mode to saturation mode.

(326) FIG. **90** is a graphical representation of one aspect of a control signal in the form of a current pulse waveform **1630** measured at the generator **102** with SW1 closed and a zero ohm cable/connector **112**, **128** impedance. The current pulse waveform **1630** shows a first current pulse **1632** followed by a second current pulse **1634**. Gate capacitor C2, which is coupled to the gate of transistor M2, charges during the initial positive edge **1632a**. During the negative edge **1632b** of the current pulse **1632**, transistor M2 goes into a saturated (on) state creating a very low impedance external to the surgical device **1600** between terminals HS/SR. During a first sample period **1636** the ADC makes a first “reference” impedance measurement labeled as “ADC_NEG1” on the waveform **1630**. After a time period T defined by the time constant set by R4 and C2, gate capacitor C2 discharges through resistor R4 and the FET channel of transistor M2 opens causing the total current to be steered through the parallel combination of resistors R1 and R3 (R1|R3) since SW1 is in the closed position. Accordingly, with transistor M2 in the off state, the impedance external to the surgical device **1600** is measured during a second sample period **1638** by the ADC and is labeled as “ADC_NEG2” on the waveform **1630**. A third sample period **1639** may be reserved for legacy instrument detection and occurs after a subsequent positive edge **1634a** of the current pulse **1634** and is labeled ADC_POS.

(327) FIG. **90A** is a graphical representation of one aspect of a control signal in the form of the current pulse waveform **1630** of FIG. **90** showing numerical values of various features of the waveform **1630** according to one example embodiment. Time in “mS” is shown along the horizontal axis and ADC Counts is shown along the inner vertical axis and ADC Vin in volts (V) shown along the outer vertical axis.

(328) FIG. **91** is an oscilloscope trace of an actual measured ADC input waveform **1640** using a generator **102** and a control circuit **1602**. Time in “mS” is shown along the horizontal axis and ADC Vin in volts (V) is shown along the vertical axis. The actual waveform **1640** shape shown in FIG. **90** is very similar to the simulated waveform shape **1630** shown in FIG. **90**.

(329) FIG. **92** illustrates another embodiment of a control circuit **1702** comprising parallel switched resistance circuit and at least one data element comprising at least one memory device memory device. In one embodiment, the control circuit **1702** is operable by a control signal in the form of a differential current pulse. As previously discussed with respect to the control circuit **1602** shown in FIG. **87**, the control circuit **1702** interfaces with a current source (generator), which outputs periodic positive and negative constant current pulses at +/-15 mA, as shown in FIG. **88**, for example. The control circuit **1702** design utilizes both pulse polarities to detect up to 2.sup.n unique switch combinations and also supports communication to one or more memory devices **1704**, such as, for example a one-wire EEPROM(s).

(330) In one embodiment, the control circuit **1702** may be connected to the generator **102** to receive an interrogation signal (e.g., a bipolar interrogation signal at a predetermined frequency, such as 2 kHz, for example) from the signal conditioning circuit **202** (e.g., from generator terminals HS and SR shown in FIG. **10** via a conductive pair of cable **112**, **128**). The control circuit **1702** may be coupled to a predetermined number of n switches to provide an indication of the configuration and/or operation of a surgical device **1700** (e.g., an ultrasonic device **104**, an electrosurgical device **106**, or a combined ultrasonic/electrosurgical device **104/106**). The control circuit **1702** shown in FIG. **92** is substantially similar in structure and operation to the control circuit **1602** described in connection with FIGS. **87-91**, except that transistor M1 is replaced with diode D4. Otherwise, the two control circuits **1602** and **1702** function in a substantially similar manner.

(331) As shown in FIG. **92**, during the positive phase of the current pulse, as shown in FIG. **88**, diode D2 is reversed biased and diode D4 is forward biased to enable gate capacitor C2 to charge

through diode D3. During the negative phase of the current pulse, diode D4 is reversed biased and diode D2 is forward biased to connect resistors R1, R2, R3, R.sub.n+1 in parallel in a configuration determined by the state of switches SW1, SW2, SW.sub.n. During an initial period of the negative phase of the current pulse, transistor M2 is in saturation mode due to the voltage on charged gate capacitor C2 and provides a low impedance path to the external generator 102 for a period equal to the time constant defined by R4 and C2. During this period, the ADC makes an initial impedance measurement "ADC_NEG1" as shown in the waveform 1630 of FIG. 90. Once gate capacitor C2 discharges through resistor R4, transistor M2 turns off and the ADC sees the impedance configured by the parallel combination of resistors R1 and any of the other resistors R2, R3, R.sub.n+1 that may be switched by corresponding switches SW1, SW2, SW.sub.n, as previously described in connection with FIGS. 87-91. The detection region is similar to that shown in FIG. 89.

(332) FIG. 93 illustrates one embodiment of a control circuit 1802 comprising a serial switched resistance circuit and at least one data element comprising at least one memory device. In one embodiment, the control circuit 1802 is operable by a control signal in the form of a differential current pulse. As previously discussed with respect to the control circuits 1602, 1702 shown in FIGS. 87, 92, the control circuit 1802 interfaces with a current source (generator), which outputs periodic positive and negative constant current pulses at ± 15 mA, as shown in FIG. 88, for example. The control circuit 1802 utilizes both pulse polarities to detect up to 2.sup.n unique switch combinations and also supports communication to one or more memory devices 1804, 1806, such as, for example, two one-wire EEPROM(s).

(333) In one embodiment, the control circuit 1802 may be connected to the generator 102 to receive an interrogation signal (e.g., a bipolar interrogation signal at a predetermined frequency, such as 2 kHz, for example) from the signal conditioning circuit 202 (e.g., from generator terminals HS and SR shown in FIG. 10 via a conductive pair of cable 112, 128). The control circuit 1802 may be coupled to a predetermined number of n switches to provide an indication of the configuration and/or operation of a surgical device 1800 (e.g., an ultrasonic device 104, an electrosurgical device 106, or a combined ultrasonic/electrosurgical device 104/106).

(334) In one embodiment, the control circuit 1802 employs a series variable resistance configurable by up to n pushbutton switches SW1, SW2, SW.sub.n. Both positive and negative current pulses can be used to derive a measurement that will allow for compensation of an unknown series cable and contamination resistance within a given range. Initially a positive phase of a current pulse reverse biases diode D2 and forward biases diode D4 to charge gate capacitor C2 through diode D3. During a subsequent negative phase of the current pulse, diode D2 is forward biased and diode D4 is reversed biased. Accordingly, diode D2 now couples series resistors R1, R2, R3, R.sub.n+1, in a configuration determined by the state of pushbutton switches SW1, SW2, SW.sub.n, to the external ADC in the generator 102. During the first portion of the negative phase of the current pulse, transistor M2 is in saturation mode due to the voltage on charged gate capacitor C2, thus creating a very low impedance path through the surgical device 1800. After a time period defined by the time constant set by resistor R4 and C2, gate capacitor C2 discharges and the FET channel of transistor M2 opens to steer the total current through the series resistors path defined by resistors R1, R2, R3, R.sub.n+1, in a configuration determined by the state of pushbutton switches SW1, SW2, SW.sub.n. The initial shorted state of transistor M2 and the subsequent impedance state of the series resistor path are the only two impedance states in the time domain that are used for detecting the state of pushbutton switches SW1, SW2, SW.sub.n. This provides the ability to compensate for an external path resistance (handpiece contacts, cabling, connector resistance) by comparing the difference between these two states. The detection region is similar to that shown in FIG. 89.

(335) Depending on the open or closed state of switches SW1, SW2, SW.sub.n for example, the signal conditioning circuit 202 (e.g., from generator terminals HS and SR shown in FIG. 10 via a conductive pair of the cable 112, 128) would see a different impedance (e.g., resistance). For the

embodiment where $n=3$, and the control circuit **1802** includes three switches SW1, SW2, SW3, for example, there are eight unique switch configurable impedances that can be detected by the signal conditioning circuit **202** based on the states of switches SW1, SW2, SW3, as described in Table 9, below:

(336) TABLE-US-00009 TABLE 9 SW1 SW2 SW3 Impedance Open Open Open $R1 + R2 + R3 + R4$ Open Open Closed $R1 + R2 + R3$ Open Closed Open $R1 + R2 + R4$ Open Closed Closed $R1 + R2$ Closed Open Open $R1 + R3 + R4$ Closed Open Closed $R1 + R3$ Closed Closed Open $R1 + R4$ Closed Closed Closed $R1$

(337) The dual protection diodes D1 and D5 placed across the HS/RS terminals of the control circuit **1802** are used to clamp ESD and/or transients coming from the surgical device **1800** and the generator **102** when the surgical device **1800** is connected and disconnected to the generator **102**.

(338) FIG. **94** illustrates one embodiment of a control circuit **1902** comprising a serial switched resistance circuit with a precision voltage reference and at least one data element comprising at least one memory device. In one embodiment, the control circuit **1902** is operable by a control signal in the form of a differential current pulse. As previously discussed with respect to the control circuits **1602**, **1702**, **1802** shown in FIGS. **87**, **92**, **93**, the control circuit **1902** interfaces with a current source (generator), which outputs periodic positive and negative constant current pulses at ± 15 mA, as shown in FIG. **88**, for example. The control circuit **1902** utilizes both pulse polarities to detect up to 2^n unique switch combinations and also supports communication to one or more memory devices **1904**, **1906**, such as, for example, two one-wire EEPROM(s).

(339) In one embodiment, the control circuit **1902** may be connected to the generator **102** to receive an interrogation signal (e.g., a bipolar interrogation signal at a predetermined frequency, such as 2 kHz, for example) from the signal conditioning circuit **202** (e.g., from generator terminals HS and SR shown in FIG. **10** via a conductive pair of cable **112**, **128**). The control circuit **1902** may be coupled to a predetermined number of n switches to provide an indication of the configuration and/or operation of a surgical device **1900** (e.g., an ultrasonic device **104**, an electrosurgical device **106**, or a combined ultrasonic/electrosurgical device **104/106**).

(340) In one embodiment, the control circuit **1902** employs a series variable resistance configurable by up to n pushbutton switches SW1, SW2, SW_{sub.n}. The control circuit **1902** in FIG. **94**, however, comprises an additional shunt voltage reference implemented with zener diode D6, which allows higher resistance values to be used for switch detection. Higher resistance values are better suited when a low pushbutton contact resistance cannot be maintained (i.e., silver ink flex circuits). In other respects, the control circuit **1902** is substantially similar in operation to the control circuit **1802** described in connection with FIG. **93**, except that the switch configurable impedance is in parallel with resistor R6.

(341) Depending on the open or closed state of switches SW1, SW2, SW_{sub.n} for example, the signal conditioning circuit **202** (e.g., from generator terminals HS and SR shown in FIG. **10** via a conductive pair of the cable **112**, **128**) would see a different impedance (e.g., resistance). For the embodiment where $n=3$ and the control circuit **1902** includes three switches SW1, SW2, SW3, for example, there are eight unique switch configurable impedances that can be detected by the signal conditioning circuit **202** based on the states of switches SW1, SW2, SW3, as described in Table 10, below:

(342) TABLE-US-00010 TABLE 10 SW1 SW2 SW3 Impedance Open Open Open $R6 \parallel (R1 + R2 + R3 + R4)$ Open Open Closed $R6 \parallel (R1 + R2 + R3)$ Open Closed Open $R6 \parallel (R1 + R2 + R4)$ Open Closed Closed $R6 \parallel (R1 + R2)$ Closed Open Open $R6 \parallel (R1 + R3 + R4)$ Closed Open Closed $R6 \parallel (R1 + R3)$ Closed Closed Open $R6 \parallel (R1 + R4)$ Closed Closed Closed $R6 \parallel R1$

(343) FIG. **95** illustrates one embodiment of a control circuit **2002** comprising a variable frequency switched resistance circuit and at least one data element comprising at least one memory device. As previously discussed with respect to the control circuits **1602**, **1702**, **1802**, **1902** shown in FIGS. **87**, **92**, **93**, **94**, the control circuit **2002** interfaces with a current source (generator), which outputs

periodic positive and negative constant current pulses at ± 15 mA as shown in FIG. 88, for example. The control circuit **2002** utilizes both pulse polarities to detect up to $2^{\text{sup.n}}$ unique switch combinations and also supports communication to one or more memory devices **2004**, **2006**, such as, for example, two one-wire EEPROM(s).

(344) In one embodiment, the control circuit **2002** may be connected to the generator **102** to receive an interrogation signal (e.g., a bipolar interrogation signal at a predetermined frequency, such as 2 kHz, for example) from the signal conditioning circuit **202** (e.g., from generator terminals HS and SR shown in FIG. 10 via a conductive pair of cable **112**, **128**). The control circuit **2002** may be coupled to a predetermined number of n switches to provide an indication of the configuration and/or operation of a surgical device **2000** (e.g., an ultrasonic device **104**, an electrosurgical device **106**, or a combined ultrasonic/electrosurgical device **104/106**).

(345) The switched resistance configuration control circuit **2002** employs variable pulse frequency to determine the states of pushbutton switches SW1, SW2, SW.sub.n. The control circuit **2002** utilizes both pulse polarities to detect up to $2^{\text{sup.n}}$ unique switch combinations and also supports communication to one or more memory devices **2004**, **2006**, such as, for example, two one-wire EEPROM(s). The use of variable frequency allows the detection method employed by the control circuit **2002** to be independent of external variable series resistance. A Schmitt trigger circuit **2008** is configured as an RC oscillator, where the frequency of oscillation is determined by the RC time constant set by the pushbutton switch configured resistors R1, R2, R3, R.sub.n+1 and capacitor C3. An initial negative current pulse charges bulk capacitor C4 to be used as a power supply in order to freewheel between current source pulses. Then the current source generator is configured for zero current during frequency measurement. The output pulses are capacitive coupled by C5 back to the generator comparator. From here the oscillator frequency can be processed to determine the states of the switches SW1, SW2, SW.sub.n.

(346) Depending on the open or closed state of switches SW1, SW2, SW.sub.n for example, the signal conditioning circuit **202** (e.g., from generator terminals HS and SR shown in FIG. 10 via a conductive pair of the cable **112**, **128**) would see a different frequency ($f_{\text{sub.o}}$) based on the switch selectable resistance. For an embodiment where $n=3$ and the control circuit **2002** includes three switches SW1, SW2, SW3, for example, there are eight unique switch configurable frequencies determined by the RC time constant for the Schmitt trigger **2008** based oscillator set by the by the pushbutton switch configured resistors R1, R2, R3, R4 and capacitor C3. The frequency ($f_{\text{sub.o}}$) seen by the signal conditional conditioning circuit **202** based on the states of switches SW1, SW2, SW3, as described in Table 11, below:

(347) TABLE-US-00011

SW1	SW2	SW3	RC Time Constant	Frequency (Hz)
Open	Open	Open	$(R1 + R2 + R3 + R4)C3$	$f_{\text{sub.1}}$
Open	Open	Closed	$(R1 + R2 + R3)C3$	$f_{\text{sub.2}}$
Open	Closed	Open	$(R1 + R2 + R4)C3$	$f_{\text{sub.3}}$
Open	Closed	Closed	$(R1 + R2)C3$	$f_{\text{sub.4}}$
Closed	Open	Open	$(R1 + R3 + R4)C3$	$f_{\text{sub.5}}$
Closed	Open	Closed	$(R1 + R3)C3$	$f_{\text{sub.6}}$
Closed	Closed	Open	$(R1 + R4)C3$	$f_{\text{sub.7}}$
Closed	Closed	Closed	$R1C3$	$f_{\text{sub.8}}$

(348) FIG. 96, is a graphical representation **2010** of one embodiment of a detection method showing detection regions for the control circuit **2002** comprising a variable frequency switched resistance circuit and a memory device, as described in connection with FIG. 95. Time (mS) is shown along the horizontal axis and Voltage (V) is shown along the vertical axis. As shown in FIG. 96, region 1 shows a pulse output of the generator **102** comparator circuit derived from the RC based oscillator circuit **2008** of control circuit **2001**. As shown in Table 11, for $n=3$, there would be eight unique frequencies representing that state of each pushbutton switch SW1, SW2, SW3.

Region 2 shows bulk supply capacitor C4 for the oscillator circuit **2008** being charged by a negative current pulse. Region 3 shows a pulse waveform on the generator **102** comparator input.

(349) FIG. 97 illustrates one embodiment of a control circuit **2102** comprising a parallel switched resistance circuit with a precision voltage reference and at least one data element comprising at least one memory device employing a variable slope waveform to determine switch states. In one

embodiment, the control circuit **2102** is operable by a control signal in the form of a differential current pulse. As previously discussed with respect to the control circuits **1602**, **1702**, **1802**, **1902**, **2002** shown in FIGS. **87**, **92**, **93**, **94**, and **95**, the control circuit **2102** interfaces with a current source (generator), which outputs periodic positive and negative constant current pulses at ± 15 mA as shown in FIG. **88**, for example. The control circuit **2102** utilizes both pulse polarities to detect up to 2.sup.n unique switch combinations and also supports communication to one or more memory devices **2104**, **2106**, such as, for example, two one-wire EEPROM(s).

(350) In one embodiment, the control circuit **2102** may be connected to the generator **102** to receive an interrogation signal (e.g., a bipolar interrogation signal at a predetermined frequency, such as 2 kHz, for example) from the signal conditioning circuit **202** (e.g., from generator terminals HS and SR shown in FIG. **10** via a conductive pair of cable **112**, **128**). The control circuit **2102** may be coupled to a predetermined number of n switches to provide an indication of the configuration and/or operation of a surgical device **2100** (e.g., an ultrasonic device **104**, an electrosurgical device **106**, or a combined ultrasonic/electrosurgical device **104/106**).

(351) The control circuit **2102** employs a variable slope waveform to determine the states of pushbutton switches SW1, SW2, SW.sub.n. A constant current pulse source drives a current into a capacitive load, e.g., capacitor C6 to generate a slope dependent on the capacitance value of capacitor C6. Using a shunt regulator D6 and charged capacitor combination creates an active circuit that varies the slope by varying the switch configurable resistance R1, R2, R3, R.sub.n+1 rather than varying the capacitance C6. Each pushbutton switch SW1, SW2, SW.sub.n state creates a unique ramp slope as shown in FIG. **98** based on a unique time constant determined by a combination of capacitor C6 and resistance R1, R2, R3, R.sub.n+1 as selected by switches SW1, SW2, SW.sub.n.

(352) Depending on the open or closed state of switches SW1, SW2, SW.sub.n for example, the signal conditioning circuit **202** (e.g., from generator terminals HS and SR shown in FIG. **10** via a conductive pair of the cable **112**, **128**) would see a different ramp slope based on a unique time constant based on the switch selectable resistance. For an embodiment where n=3 and the control circuit **2102** includes three switches SW1, SW2, SW3, for example, there are eight unique switch configurable ramp slopes determined by the RC time constant determined by the pushbutton switch configured resistors R1, R2, R3, R4 and capacitor C6. The ramp slope seen by the signal conditional conditioning circuit **202** based on the state of each switch combination SW1, SW2, SW3, as described in Table 12, below:

(353) TABLE-US-00012

SW1	SW2	SW3	RC Time Constant	Ramp Slope (V/t)
Open	Open	Open	R1C6	Ramp/Slope.sub.1
Open	Open	Closed	(R1 R4)C6	Ramp/Slope.sub.2
Closed	Open	Open	(R1 R3)C6	Ramp/Slope.sub.3
Open	Closed	Closed	(R1 R3 R4)C6	Ramp/Slope.sub.4
Closed	Open	Open	(R1 R2)C6	Ramp/Slope.sub.5
Closed	Open	Closed	(R1 R2 R4)C6	Ramp/Slope.sub.6
Closed	Closed	Open	(R1 R2 R3)C6	Ramp/Slope.sub.7
Closed	Closed	Closed	(R1 R2 R3 R4)C6	Ramp/Slope.sub.8

(354) FIG. **98** is a graphical representation **2110** of one embodiment of a detection method showing detection regions for the control circuit **2102** comprising a variable ramp/slope switched resistance circuit and a memory device, as described in connection with FIG. **97**. Time (mS) is shown along the horizontal axis and ADC counts is shown along the vertical axis. Region **1** shows the variable slope controlled by the time constant determined by capacitor C6 and switch SW1, SW2, SW3 configurable resistance R1, R2, R3, R4. Each switch SW1, SW2, SW3 state provides a unique slope. Region **2** is the time region to reset the RC time constant. Simulation modeling confirmed immunity to very high contact corrosion resistance. Nevertheless, time based measurement may be susceptible to cable inductance and capacitance due to fast transient edges of signaling.

Accordingly, compensation circuits for cable inductance and capacitance due to fast transient edges of signaling are contemplated within this disclosure. Slope detection would minimize risk of false switch state detection. The current pulse rising and falling edges may need to be slower than the

lowest resonance frequency of the cable, for example. The embodiments, however, are not limited in this context.

(355) FIG. **99** illustrates one embodiment of a control circuit **2202** comprising a one-wire multi-switch input device. As previously discussed with respect to the control circuits **1602**, **1702**, **1802**, **1902**, **2002**, **2102** shown in FIGS. **87**, **92**, **93**, **94**, **95**, and **97**, the control circuit **2202** interfaces with a current source (generator), which outputs periodic positive and negative constant current pulses at ± 15 mA as shown in FIG. **88**, for example. The control circuit **2202** utilizes both pulse polarities to detect up to 2^{sup.n} unique switch combinations and also supports communication to one or more memory devices, such as, for example, two one-wire EEPROM(s), which are not shown in FIG. **99** for conciseness and clarity, but may be included in various aspects of this embodiment. Also, diode pair D2/D4 for bidirectional power are not shown in FIG. **99** for conciseness and clarity, but may be included in various aspects of this embodiment.

(356) In one embodiment, the control circuit **2202** may be connected to the generator **102** to receive an interrogation signal (e.g., a bipolar interrogation signal at a predetermined frequency, such as 2 kHz, for example) from the signal conditioning circuit **202** (e.g., from generator terminals HS and SR shown in FIG. **10** via a conductive pair of cable **112**, **128**). The control circuit **2202** may be coupled to a predetermined number of n switches to provide an indication of the configuration and/or operation of a surgical device **2200** (e.g., an ultrasonic device **104**, an electrosurgical device **106**, or a combined ultrasonic/electrosurgical device **104/106**).

(357) In one embodiment, the one-wire multi-switch input device of the control circuit **2202** employs a one-wire port expansion device **2204** such as a Maxim DS2408 8-channel, programmable I/O one-wire integrated circuit available from Maxim Integrated Products, Inc., Sunnyvale, CA, or equivalents thereof, known under the trade name “1-Wire.” The port expansion device **2204** may be powered from the negative current pulse of the generator **102**. The port expansion device **2204** complies with the one-wire communication protocol to accommodate up to eight switch inputs **2208**. The port expansion device **2204** outputs are configured as open-drain and provide an on resistance of **1000** max. A robust channel-access communication protocol ensures that the output-setting changes occur error-free. A data-valid strobe output can be used to latch port expansion device **2204** logic states into external circuitry such as a D/A converter (DAC) or microcontroller data bus. The port expansion device **2204** selects one of eight switches SW1-SW8 to select one of eight corresponding resistors **1-8** in resistor bank **2210**. In one embodiment, for example, a detection method comprises reading any one of switches SW1-SW8 state using the one-wire communication protocol of the control circuit **2202** utilizing the same software functionality used to read the surgical device **2200** EEPROMs. In other embodiments, the control circuit **2202** can be expanded to read additional switch inputs. In one embodiment, cyclic redundancy code (CRC) error checking may be employed to eliminate or minimize uncertainty of switch states. It may be possible that one-wire communications may be susceptible to activation noise from other electrosurgical instruments, accordingly, compensation circuits and techniques to minimize interference from such electrosurgical instruments are contemplated within this disclosure.

(358) Although the various embodiments of the devices have been described herein in connection with certain disclosed embodiments, many modifications and variations to those embodiments may be implemented. For example, different types of end effectors may be employed. Also, where materials are disclosed for certain components, other materials may be used. The foregoing description and following claims are intended to cover all such modification and variations.

(359) Any patent, publication, or other disclosure material, in whole or in part, that is said to be incorporated by reference herein is incorporated herein only to the extent that the incorporated materials does not conflict with existing definitions, statements, or other disclosure material set forth in this disclosure. As such, and to the extent necessary, the disclosure as explicitly set forth herein supersedes any conflicting material incorporated herein by reference. Any material, or portion thereof, that is said to be incorporated by reference herein, but which conflicts with existing

definitions, statements, or other disclosure material set forth herein will only be incorporated to the extent that no conflict arises between that incorporated material and the existing disclosure material.

Claims

1. A control circuit to receive a control signal from a surgical generator, the control signal defining a first phase and a second phase, the control circuit comprising: a resistor network comprising resistors arranged in parallel to each other; a capacitor coupled in series with the resistor network; and a plurality of switches, each of the switches configured to transition between an open state and a closed state; wherein the control circuit is to transition between a plurality of states corresponding to a plurality of operational modes of a surgical instrument; wherein, in the first phase of the control signal, the control circuit is to communicate surgical instrument information to the surgical generator; wherein, in the second phase of the control signal, the control circuit is to provide an output corresponding to one of the plurality of states of the control circuit; wherein the plurality of switches are configured to transition the control circuit between the plurality of states; and wherein a state of the plurality of switches is determinable based on a slope of a voltage of the capacitor.
 2. The control circuit of claim 1, wherein each of the switches is coupled to a pushbutton disposed on the surgical instrument.
 3. The control circuit of claim 1, further comprising: a memory device storing the surgical instrument information; wherein the memory device is powered in the first phase of the control signal to communicate the surgical instrument information to and from the surgical generator.
 4. The control circuit of claim 3, wherein the memory device comprises an electrically erasable programmable read-only memory.
 5. The control circuit of claim 3, wherein: the control signal comprises a differential constant current pulse; the first phase defines a positive phase of the differential constant current pulse; and the second phase defines a negative phase of the differential constant current pulse.
 6. The control circuit of claim 1, wherein the surgical instrument information is selected from the group consisting of a number of uses of the surgical instrument, a date of use of the surgical instrument, a time of use of the surgical instrument, a model number, a serial number, and any combination thereof.
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